

A New Auxiliary Model Approach To Fermionic Criticality: Insights on Mottness in Strongly Correlated Systems

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by

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Chapter 1

Introduction

In recent years, the term *quantum materials* has come to describe a class of materials that, loosely speaking, display quantum mechanical properties and many-body effects over large length and energy scales. These effects manifest in the form of exotic phenomena, examples of which include unconventional superconductivity and the fractional quantum hall effect. Such materials often involve strong inter-electron interactions, low dimensions and topological properties, making them highly sensitive to small changes in parameters or temperature. This typically means that such materials cannot be described using single-particle or weak-coupling approaches, making the study of such materials more challenging and interesting. Examples of quantum materials include the copper oxide superconductors, heavy fermion materials, Moire materials and quantum spin liquids. The exotic phases that emerge from the interplay of competing tendencies in these materials are often referred to as *quantum matter*. Such phases are usually marked by non-trivial patterns of topology and entanglement. There are several examples of phases of quantum matter, some were already provided: unconventional superconducting states, fractional quantum hall states, strange metals and topologically ordered phases, among others. In this thesis, we are concerned with the study of a specific aspect of quantum matter – Mott metal-insulator transition. In doing so, we will deal with a number of correlated phases such as Mott insulators, strange metals and the pseudogap, all of which will be explained in due course. The phenomenon of Mott insulation is also intimately tied to other exotic effects such as high- T_c superconductivity and spin liquid behaviour; these will not be explored in this thesis.

1.1 Mott metal-insulator transition: An Introduction

The rich physics of Mott metal-insulator transitions in strongly correlated systems has been an active subject of study for quite some time [1–4]. Their appearance in various materials such as the nickelates, cuprates, iridates and manganites, among others, is marked by complex phase diagrams that display dramatic changes in properties across changes in filling and/or bandwidth, even at low temperatures. The physics of Mott transitions is driven by the competition between two energy scales in the system: (i) the scale of itineracy, quantified by the bandwidth, and (ii) the scale of electron-electron repulsion. A Mott insulator arises when the electronic correlations are strong enough that

low-energy electronic excitations become gapped, and the metallic valence band gets split into lower and upper bands; for an initially half-filled valence band, the lower band is filled at $T = 0$ while the upper band is vacant. Because of strong electronic repulsion, the electrons in the lower band form local moments which can show a variety of behaviour. This includes various symmetry-broken ground states such as an antiferromagnet [5], or symmetry-preserved states such as quantum spin liquids [6]. The physics of Mottness has been proposed as being responsible for several very interesting phenomena, such as high-temperature superconductivity [5], non-Fermi liquids [7] and the various phases in magic-angle twisted bilayer graphene [8, 9].

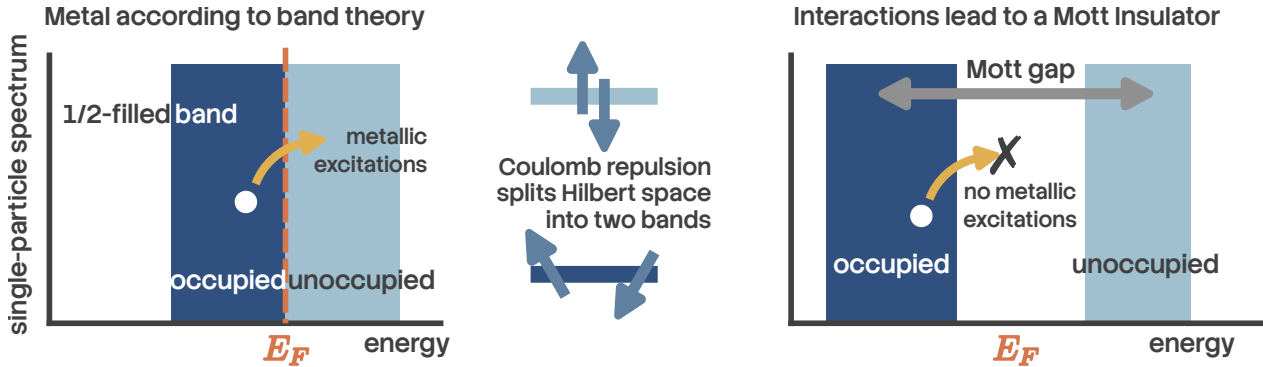


Figure 1.1: A Mott insulator is an example of strongly correlated phenomena, where inter-electron repulsion leads to insulating behaviour. In the absence of interactions, a half-filled band (left) would result in metallic behaviour according to band theory. Upon including sufficiently strong interactions, the set of states within the band get split into two sub-bands, separated by a spectral gap of the order of the interaction strength. This makes the system metallic, because the lower sub-band remains completely filled while the upper sub-band is empty.

Initial attempts at explaining metal-insulator transitions were based on a non-interacting picture of electrons (through the work of Bethe, Sommerfeld, Bloch and Wilson among others); in such a picture, electronic bands can arise only from the scattering of electrons off the periodic lattice potential. One then distinguishes between metals and insulators at zero temperature by the position of the Fermi level E_F with respect to the bands: if E_F lies within a band gap, an integer number of bands will be completely filled and the system will be an insulator, whereas if E_F lies within one of the bands, the system will be a metal because there are vacant gapless excitations within the band.

It was first pointed out by Boer and Verwey [10] that certain insulators and semiconductors (such as nickel oxides) have a particle filling that, purely from band theory, would lead to partially filled bands. This poses a challenge because a partially filled band necessarily leads to a metallic phase (see Fig. 1.1 (left)). The insulating behaviour of these materials therefore signals a fundamental limitation of single-particle band theory and points to a qualitatively different mechanism for electron localization—one driven by strong electron–electron interactions rather than by scattering from the lattice potential alone. This realization led to the concept of a Mott insulator, in which charge localization emerges as a genuine many-body effect despite the absence of a band gap in the non-interacting electronic structure. A very simple mechanism for this was put forth by Mott [11–13]: in the presence of a large local Coulomb repulsion cost U on a lattice site, the quantum states $|\Psi_1\rangle$

in which the site is singly occupied get separated in energy from the states $|\Psi_2\rangle$ where the site is doubly occupied by a gap of size U ; the former class of states $|\Psi_1\rangle$ form the lower band (LB) while the latter $|\Psi_2\rangle$ form the upper band (UB). If we now have a half-filled system (one electron per site), this electron will completely fill LB and leave UB vacant, leading to an insulating state (see Fig. 1.1 (right)).

Given the above discussion, it is clear that a Mott insulator must involve a gap to charge excitations (excitations that add or remove an electron). It does not however determine the fate of the spin, orbital or other degrees of freedom. One scenario is when the spin or orbital degree of freedom undergoes spontaneous symmetry breaking into an ordered state; in the case of spin, this usually results in a superexchange interaction being generated and leads to a long-range ordered antiferromagnetic Neel state coexisting with the Mott state. Examples of such systems include vanadium sesquioxide V_2O_3 and its derivatives and the high- T_c cuprates (such as La_2CuO_4) [4]. The other scenario is one where the spin degree of freedom retains its symmetry, and we end up with a quantum mechanically non-trivial entangled state (in order to quench the entropy). Examples of such states include quantum spin liquids [14]. Mott insulators without long-range order are of particular interest because of the possibility of topological order, fractional excitations, and novel metallic parent states.

A prototypical model for gaining theoretical understanding of Mott metal-insulator transitions on a lattice is the single-band Hubbard model [1, 15–17]

$$H = -t \sum_{\langle i,j \rangle, \sigma} (c_{i,\sigma}^\dagger c_{j,\sigma} + \text{h.c.}) - \frac{U}{2} \sum_i (n_{i,\uparrow} - n_{i,\downarrow})^2 - \mu N. \quad (1.1)$$

The first term incorporates nearest-neighbour hopping on the lattice, while the second term introduces inter-electron repulsion when two electrons site on the same lattice site; in this way, the model incorporates the two competing tendencies of delocalisation and localisation that are responsible for the Mott MIT. The third term fixes the chemical potential and determines the filling of the lattice. Many of the materials that are studied for Mott MIT involve d -electrons that are five-fold degenerate, and these orbitals can hybridise with ligand atoms. The Hubbard model avoids these complexities by focusing on only a single band that lies closest to the Fermi energy, and it is the states of this band that are described by the Hubbard model.

This leaves us with two channels for observing Mott MIT [4]: (i) tuning the bandwidth, and (ii) tuning the filling. The former can be done, in experiments, by applying pressure. Filling is tuned by doping specific elements of a compound with elements of a different valence. The former is referred to as bandwidth-control metal-insulator transition, while the latter is called filling-control metal-insulator transition.

1.2 Features of Mott metal-insulator transition

1.2.1 Two complementary pictures of the Mott transition: Brinkman-Rice vs Mott-Hubbard.

Initial insights into the Mott MIT were obtained by Hubbard [1], thinking along lines similar to Mott. Within this picture, one starts from the insulating large U limit of the Hubbard model (eq. 1.1) ($U \gg t$). We stay at half-filling for simplicity ($\mu = 0$). The large on-site repulsion leads to the formation of two bands at each site, the lower Hubbard band (LHB) associated with singly-occupied states and the upper Hubbard band (UHB) associated with doubles and holes (states that incur a cost U). In the large U limit, these bands are well-separated, and the dynamics within the bands is a result of the motion of tightly bound holons and doubles (more on this later) that are generated because of the small but non-zero hopping probability t . As U/t is lowered, the bands come closer and a metal-insulator transition occurs at the critical value of U at which the gap vanishes. For lower U/t , the bands overlap and we get a metal. This picture fails to provide an accurate description of the metal resulting from the melting of such a Mott insulator [18].

An alternative approach to the Mott MIT is due to Brinkman and Rice [19], where the authors applied Gutzwiller's variational method [20] to the half-filled Hubbard model. Gutzwiller's approach involves modifying, through variational factors, the hopping matrix elements that start from doubly occupied states, in order to reflect the electronic correlation at each site. By requiring that the number of doubly-occupied states vanish in the Mott insulating ground state, they obtained a critical value of the interaction strength. In the Brinkman-Rice scenario, therefore, the transition from the metal to the insulating phase occurs when the hopping matrix elements from the doubly-occupied to the singly-occupied states strictly vanishes. Note that in contrast to the Mott-Hubbard picture, this approach starts from a correlated Fermi liquid (renormalised due to the Gutzwiller factors). While it gives a better low-energy description of the correlated metal compared to the Mott-Hubbard picture, it fails to account for the high-energy physics, and treats the Mott insulator as a direct product state of local moments at each state (on account of the strictly zero hopping). This of course means that this method cannot explain antiferromagnetic superexchange processes in the Mott insulator [18].

1.2.2 Holon-doublon unbinding picture of the Mott MIT.

In 1949, Mott proposed a zero temperature explanation for a correlation-driven insulator and a mechanism for its transition into a metal in terms of the motion of holes and doubles in the background of a half-filled lattice (see Fig. 1.2). For $U \ll t$ (strong correlation), the ground state predominantly involves lattice sites with a single electron. Hopping processes between two nearest-neighbour sites transfer an electron between them and create a pair of hole and double, but the pair remains tightly bound to each other. This is because moving the hole independently involves scattering processes that cross the Mott gap, and these processes are not allowed at $T = 0$. But without the independent motion of the hole (or double), there is no current, and the system is an insulator (left, Fig. 1.2).

As U/t is lowered, the gap closes and processes that transport the hole without transporting the double become gapless. This essentially means that the pair becomes deconfined, and the hole can be moved far away from the double (right, Fig. 1.2). Current can then be passed by having the hole

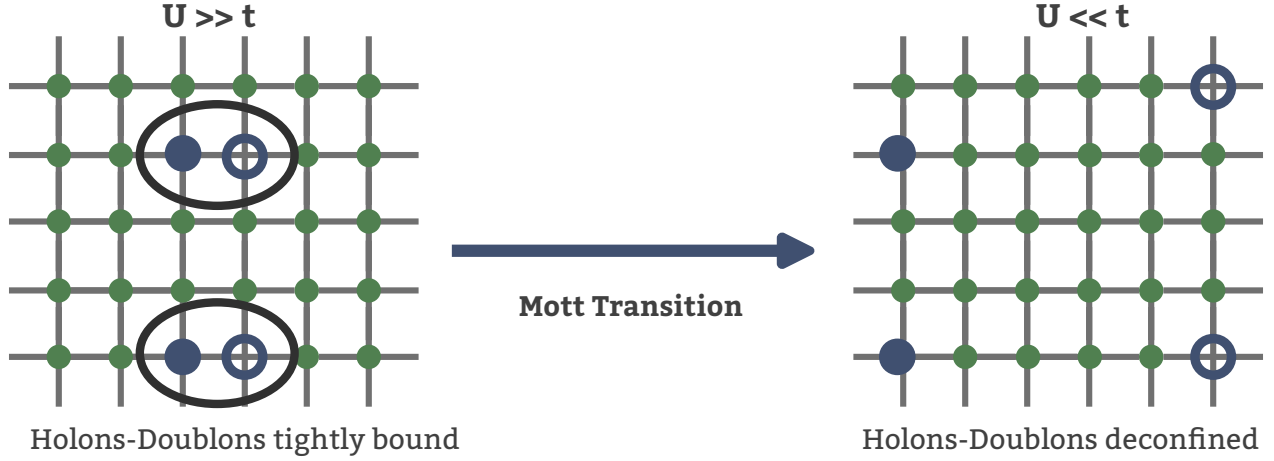


Figure 1.2: A schematic depiction of Mott's picture of correlation-driven metal-insulator transition. In the insulating state (left), each site is mostly comprised of single spins because of the strong repulsion, but virtual hopping processes can create locally bound pairs of holes and doubles. Because of the gap to charge excitations, these pairs cannot unbind and delocalise. In contrast, in the metallic phase, the presence of gapless excitations means that the hole can move independently of the double, leading to charge transport and metallicity. This frames the Mott MIT as a confinement-deconfinement transition of the holes and doubles.

move through the lattice, and we get a metallic state. This leads to a picture of the Mott transition into a metal in terms of an unbinding of the hole-double bound states (excitons) that were present in the insulator.

There have been more quantitative studies of this mechanism. I list some of the relatively recent ones here. In [21], Prelovšek et al. studied the motion of holon-doublon pairs in a spin polarised background by rewriting the repulsive Hubbard model in terms of electrons as an attractive Hubbard model of holes and doubles. The model was treated using a BCS-like treatment, and the Mott transition was obtained when the binding energy for the hole-double condensate vanishes. To mark the deconfinement, they show that the approach to the transition from the insulating side is concomitant with an increase in the size of the pairs. This approach however fails to capture the transition for a system away from half-filling. Following a slave-boson treatment of holon-doublon unbinding, Zhou et al. [22] show that the Mott transition from a paramagnetic metal to a Mott insulator on the half-filled honeycomb and 2D square lattices occur through an intervening Slater insulator with coherent quasiparticles. In particular, their approach sheds light on the nature of the incoherent excitations in the Hubbard sidebands of the local spectral function. Finally, an older work by Watanabe et al. [23] studies the half-filled Hubbard model on an anisotropic triangular lattice using a Monte carlo method with a variational binding parameter between holons and doublons. Their approach reveals a first order Mott transition, along with robust d -wave superconductivity and short-range magnetic order.

1.2.3 Spectral features of the Mott transition.

One of the important diagnostics of the Mott metal-insulator transition is the spectrum for one-particle electron-like excitations, as expressed through the one-particle retarded Greens function $G(\mathbf{k}, \sigma; t) = -i\theta(t)\langle\{c_{\mathbf{k},\sigma}(t), c_{\mathbf{k},\sigma}^\dagger\}\rangle$. In general, the presence of poles in the Green function at the Fermi energy and in its neighbourhood imply the existence of gapless excitations and hence a metallic phase. It turns out that this statement is equivalent to Luttinger's theorem [24, 25],

$$\frac{N}{V} = 2 \int_{G(0,\mathbf{k})>0} \frac{d\mathbf{k}}{(2\pi)^3}, \quad (2.1)$$

which states that the particle density in a metal is equal to the number of states in the Fermi volume. As clarified in the equation, the Fermi volume is defined as the region of k -space where the Greens function is positive at $\omega = 0$.

In a Mott insulator, strong correlations lead to the destruction of electronic excitations around $\omega = 0$. This results in the Greens function poles transmuting into zeros. This is also tracked by analogous changes in the electronic self-energy $\Sigma(\mathbf{k}; \omega)$; Greens function zeros leads to a Mott pole at $\omega = 0$ in $\Sigma(\mathbf{k}; \omega)$. The points in k -space where $\Sigma(\mathbf{k}; 0)$ vanishes is called a Luttinger surface [26]. The Mott metal-insulator transition is therefore about the topological change of a Fermi surface (surface of Greens function poles) into a Luttinger surface (surface of self-energy poles).

As discussed earlier, the appearance of Greens function zeros at $\omega = 0$ leads to the formation of a charge gap that freezes one-particle low-energy excitations. The spectrum gets redistributed into two bands on either side of the gap, the upper and lower Hubbard bands. In the ground state, the lower band is obviously filled while the upper band is empty. Bands of a Mott insulator formed from electronic correlation differ from those formed due to one-particle scattering in a particular way: dynamical spectral weight transfer. Unlike a band insulator, the bands of a Mott insulator are not rigid; tuning the lattice filling results in a continuous transfer of states from one band to another.

The non-rigidity of the bands can be seen most easily from the atomic limit ($t = 0$) of the Hubbard model (eq. 1.1). The local retarded Greens function at an arbitrary site i can be obtained using the equation of motion approach. The final result is [27]

$$G_R(i; \omega) = \frac{1+x}{\omega + \mu + U/2} + \frac{1-x}{\omega + \mu - U/2}, \quad (2.2)$$

where $x = 1 - \sum_\sigma \langle n_{i\sigma} \rangle$ is the average number of holes at any site. The spectral weight for the two poles makes it clear that the lower band has $1+x$ number of states while the upper band has $1-x$ states. Changing the number of states in the lower band (by tuning μ and hence x) also changes those in the upper band. For example, by adding 0.1 holes per site to the lower band, the spectral weight in the upper band decreases by the same amount. This is the non-rigidity of the bands mentioned previously.

1.2.4 Preliminary Insights into Mottness From a Toy Model

The Baskaran-Hatsugai-Kohmoto model [28, 29] introduced independently in 1991 and 1992 is an exactly solvable model of electronic correlation that displays a transition from a metallic phase to a

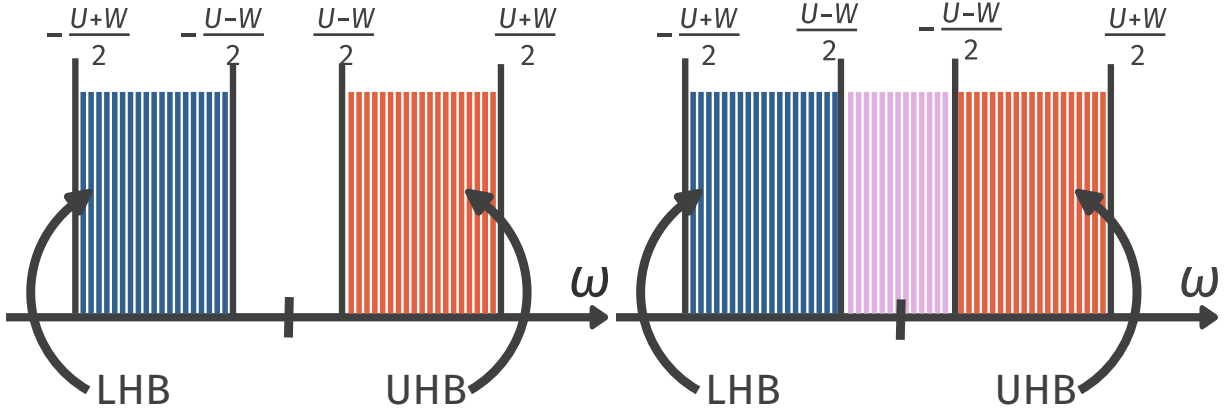


Figure 1.3: The Mott insulator (left) and non-Fermi liquid (right) phases of the Hatsugai-Kohmoto model. In the Mott insulating phase, each momentum state is singly occupied and the lower (blue) is completely filled, while the upper band consisting of the doubly occupied and empty states is completely empty. There is, thus, a spectral gap to the excitations. In the metallic phase (right), the bands overlap and one can have correlated excitations from singly occupied states to hole and double states. These excitations are non-electronic, and the metal is a non-Fermi liquid.

Mott insulator. While its solvability offers an instructive method for studying Mott transition, the infinitely long-ranged nature of its interaction makes it tricky to apply these conclusions to realistic materials.

At half-filling, the model is defined by the following Hamiltonian:

$$H_{\text{BHK}} = \sum_{\mathbf{k}, \sigma} \epsilon(\mathbf{k}) n_{\mathbf{k}, \sigma} - \frac{U}{2} \sum_{\mathbf{k}} (n_{\mathbf{k}, \uparrow} - n_{\mathbf{k}, \downarrow})^2. \quad (2.3)$$

Clearly the model is diagonal in momentum-space; the Hamiltonian splits up into commuting 4×4 blocks for every point $\mathbf{k}$:

$$H_{\text{BHK}} = \sum_{\mathbf{k}} H(\mathbf{k}), \quad H(\mathbf{k}) = \epsilon(\mathbf{k}) \sum_{\sigma} n_{\mathbf{k}, \sigma} - \frac{U}{2} (n_{\mathbf{k}, \uparrow} - n_{\mathbf{k}, \downarrow})^2. \quad (2.4)$$

The vacant eigenstate $|0\rangle$ of $H(\mathbf{k})$ has zero energy ($E_0 = 0$), the two singly-occupied states $|\uparrow\rangle$ and $|\downarrow\rangle$ have energy $E_1 = \epsilon(\mathbf{k}) - U/2$ and the doubly-occupied state $|\uparrow, \downarrow\rangle$ has energy $E_2 = 2\epsilon(\mathbf{k})$. This leads to two possible single-particle excitations – one for adding an electron on the vacant state: ($\Delta_1 = E_1 - E_0 = \epsilon(\mathbf{k}) - U/2$), the other for adding an additional electron on a singly-occupied state: ($\Delta_2 = E_2 - E_1 = \epsilon(\mathbf{k}) + U/2$).

Upon combining states from other k -modes, these isolated excitations broaden into bands around $\pm U/2$ because of the continuum of values available to $\epsilon(\mathbf{k})$. For large enough U (compared to the non-interacting bandwidth W), these bands remain well-separated in energy (Fig. 1.3 left), forming lower and upper Hubbard bands. The half-filled system results in the lower band being completely filled and the upper band empty, leading to a Mott insulator. The states within this band are all singly-occupied (per k -state). Upon lowering the correlation strength U , the bands touch

each other at the critical value of $U_c = W$ (obtained by equating the top of the lower band to the bottom of the upper band). This mechanism is qualitatively similar to the Mott-Hubbard picture of Mott transition. For lower values of U , the system is a metal due to the presence of gapless excitations within the combined band (Fig. 1.3 right). This therefore provides a simple demonstration of a correlation-driven Mott metal-insulator transition.

1.3 Models and materials for strong local correlation

1.3.1 Introduction to the auxiliary model approach

One of the main results obtained within this thesis is the development of a formalism to study the more realistic case of finite-dimensional interacting models by using appropriately chosen impurity models as auxiliary models. The key to an *auxiliary model method* is to construct an appropriate auxiliary model that while being tractable captures some essential property of the full lattice model [30]. This involves separating the complete degrees of freedom into a system part (H_S) which one treats exactly and the rest of the system (H_R) that one might simplify in order to be able to solve the problem. The non-trivial nature of the problem arises of course from the hybridisation H_{SR} between the system and the bath:

$$H = \begin{bmatrix} H_R & H_{RS} \\ H_{RS}^* & H_S \end{bmatrix}. \quad (3.1)$$

This separation can always be done exactly, if one does not care about tractability. For example, one can take the Hubbard model that describes electrons hopping on a lattice and interacting when two electrons reside on the same lattice site:

$$H_{\text{hub}} = -t \sum_{\langle i,j \rangle, \sigma} (c_{i,\sigma}^\dagger c_{j,\sigma} + \text{h.c.}) + U \sum_i n_{i\uparrow} n_{i\downarrow}, \quad (3.2)$$

and separate this into parts, the system being the local orbitals of a specific site l and the rest of the system being all the other sites:

$$H_S = U n_{l\uparrow} n_{l\downarrow}, \quad H_R = -t \sum'_{\langle i,j \rangle, \sigma} (c_{i,\sigma}^\dagger c_{j,\sigma} + \text{h.c.}) + U \sum_i n_{i\uparrow} n_{i\downarrow}, \quad H_{SR} + H_{RS} = -t \sum_{i,\sigma} (c_{l,\sigma}^\dagger c_{i,\sigma} + \text{h.c.}), \quad (3.3)$$

where the primed summation is carried out over all nearest-neighbour pairs that do not involve l and the summation in $H_{SR} + H_{RS}$ is over all sites adjacent to l .

The smaller system H_S (often called the *cluster*) is typically chosen such that its eigenstates are known exactly. Progress is then made by choosing a simpler form for H_R and its coupling H_{RS} with the smaller system. This combination of the cluster and the simpler bath is then called the *auxiliary system*. A tractable auxiliary system for the Hubbard model is the single-impurity Anderson model (SIAM); the correlated impurity site represents the set of local orbitals of any given site, while the conduction bath represents the rest of the lattice sites in an approximate fashion. Such a construction is shown in fig. 1.4.

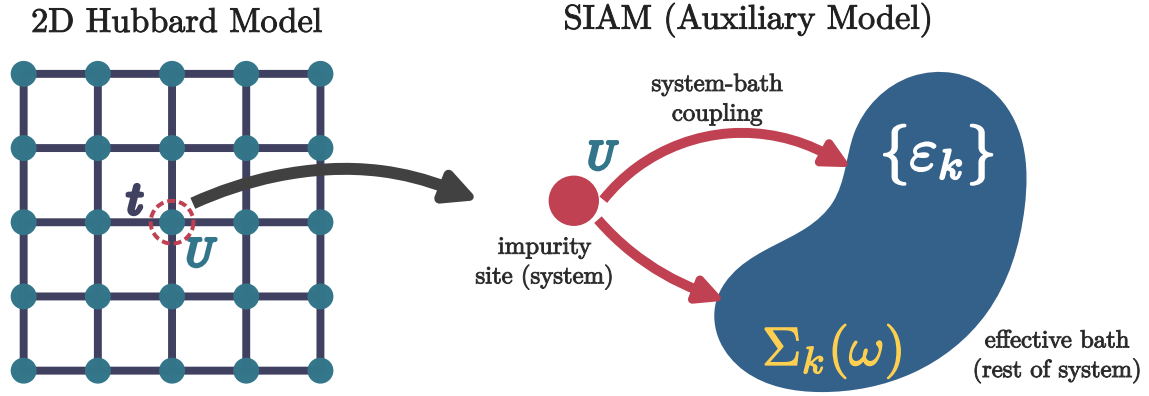


Figure 1.4: Left: Full Hubbard model lattice with onsite repulsion U^H on all sites and hopping between nearest neighbour sites with strength t^H . Right: Extraction of the auxiliary (cluster+bath) system from the full lattice. The central site on left becomes the impurity site (red) on the right (with an onsite repulsion ϵ_d), while the rest of the $N - 1$ sites on the left form a conduction bath (green circles) (with dispersion ϵ_k and correlation modelled by the self-energy $\Sigma_k(\omega)$) that hybridizes with the impurity through the coupling V .

It should be noted that any reasonable choice of the cluster and bath would break the translational symmetry of the full model: To allow computing quantities, one would need to make the bath (which is a much larger system) simpler than the cluster (which is a single site). This distinction breaks the translational symmetry of the Hubbard model. As a result, calculations that make use of the translation symmetry of the full model (such as k -space objects) will require a periodisation operation of an auxiliary model quantity.

1.3.2 Correspondence between impurity and lattice models in $d = \infty$

Dynamical Mean-Field Theory (DMFT) provides a nonperturbative framework for treating strong local electronic correlations in lattice models by mapping them onto a self-consistent quantum impurity problem [18, 31]. The central idea of DMFT is that in the limit of infinite spatial dimension, or equivalently infinite coordination number, the electronic self-energy becomes purely local while retaining its full dynamical (frequency-dependent) structure. As a result, the many-body lattice problem reduces to an effective single-site problem embedded in a self-consistently determined bath, capturing local quantum fluctuations exactly while neglecting nonlocal correlations. DMFT has become a cornerstone of modern correlated-electron theory, successfully describing Mott metal-insulator transitions, quasiparticle formation, and spectral weight transfer in Hubbard-like models [18, 32].

In order to motivate the impurity model-lattice model mapping inherited by DMFT, we first recall the textbook example of classical mean-field theory. Mean-field theory provides one of the earliest and most instructive examples of controlled approximations in many-body physics. In its

classical incarnation, exemplified by the Ising model,

$$H_{\text{Ising}} = - \sum_{i \neq j} J_{ij} S_i S_j - h \sum_i S_i = - \sum_i S_i (h + \sum_{j \neq i} J_{ij} S_j), \quad (3.4)$$

mean-field theory proceeds by replacing the interaction between a given spin and its neighbors by an *effective static field* proportional to the average magnetization. Writing $S_j = \langle S_j \rangle + \delta S_j$ and neglecting quadratic fluctuations and constant parts, one obtains an *auxiliary problem* in the form of a mean-field Hamiltonian

$$H_{\text{Ising}}^{(\text{MF})} = - \sum_i (h + h_{\text{MF}}(i)) S_i, \quad (3.5)$$

where the mean-field h_{MF} comes out to be

$$h_{\text{MF}}(i) = \sum_{j \neq i} 2 \langle S_j \rangle J_{ij}. \quad (3.6)$$

The above approach assumes two things: (i) the presence of a non-zero *order parameter* $m_j = \langle S_j \rangle$ (in this case, a non-zero magnetisation), and (ii) the smallness of the fluctuations $S_j - \langle S_j \rangle$. The mean-field Hamiltonian can be used to compute the partition function (at inverse temperature β) and hence the magnetisation at a site i of the lattice:

$$m_i = \tanh(\beta(h_{\text{MF}}(i) + h)) = \tanh(\beta \sum_{j \neq i} 2m_j J_{ij} + \beta h). \quad (3.7)$$

The problem hasn't yet been solved of course; one needs to determine the appropriate value of m_j . We therefore need an additional equation; this is provided by the so-called *self-consistency condition*. The translation invariance of the lattice ensures that the magnetisation must be uniform across sites i and j . This allows us to replace m_j with m_i in the above equation:

$$m_i = \tanh(\beta m_i \sum_{j \neq i} 2J_{ij} + \beta h). \quad (3.8)$$

This equation can be solved numerically to obtain the uniform magnetisation $m_i = m$. The mean-field approximation becomes exact in the limit of large coordination number, where fluctuations are suppressed and spatial correlations beyond a single site vanish.

An equivalent way of framing the self-consistency equation is to equate the mean-fields on different lattice sites. Combining eqs. 3.7 and 3.6, we have

$$h_{\text{MF}}(i) = \sum_{j \neq i} 2m_j J_{ij} = \sum_{j \neq i} 2 \tanh(\beta(h_{\text{MF}}(j) + h)) J_{ij}. \quad (3.9)$$

The equation in terms of $h_{\text{MF}}(i)$ and $h_{\text{MF}}(j)$ can again be solved numerically by claiming that the mean-field has to be uniform across all the sites owing to translation invariance. This form of the self-consistency requirement (in terms of the field) will be useful to us in the upcoming discussion.

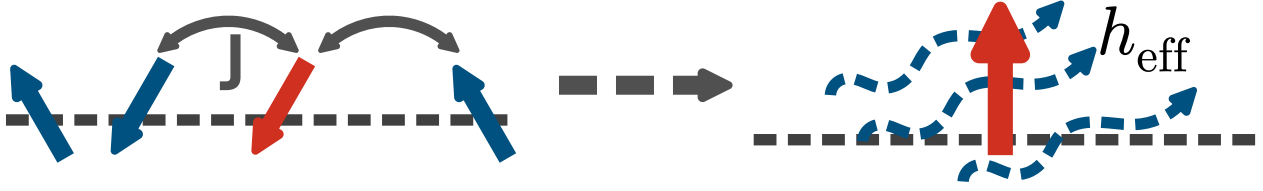


Figure 1.5: The Ising mode mean-field treatment as an auxiliary model approach. Through the mean-field decoupling followed by the identification of a self-consistent field, the effect of the neighbouring spins is approximated by a magnetic field. This self-consistently determined model of a spin in a magnetic field acts as an auxiliary model for the complete Ising model, representing the local properties of the Ising spins through the properties of the spin subjected to the field.

The essential feature of classical mean-field theory is therefore the replacement of a many-body problem by an effective *single-site* problem embedded in a self-consistently determined environment. Quantum lattice models admit a closely analogous construction, but with an important conceptual difference: while the classical mean field is static, quantum fluctuations necessarily introduce a tower of excited states that allow for temporal dynamics. This observation underlies the formulation of dynamical mean-field theory (DMFT), which may be viewed as the natural quantum generalization of classical mean-field theory [18, 31].

Similar to the previous case of the Ising model, we will now derive the self-consistency equation for a quantum model. Consider an interacting fermionic lattice Hamiltonian, such as the Hubbard model:

$$H = - \sum_{ij\sigma} t_{ij} c_{i\sigma}^\dagger c_{j\sigma} + U \sum_i n_{i\uparrow} n_{i\downarrow}, \quad (3.10)$$

defined on a lattice with coordination number z .

The point of this exercise is to calculate the interacting self-energy $\Sigma_k(\omega)$ for the fermions. This therefore acts as an “order parameter” and plays the same role as the local magnetisation in the previous exercise. The self-energy defines the interacting k -space Green’s function for a translation-invariant problem:

$$G_{\mathbf{k}}(\omega) = \frac{1}{\omega - \varepsilon_{\mathbf{k}} - \Sigma_{\mathbf{k}}(\omega)}, \quad (3.11)$$

where $\varepsilon_{\mathbf{k}}$ denotes the noninteracting dispersion, and a real-space local Green’s function, obtained by summing over momentum:

$$G_{\text{loc}}(\omega) = \sum_{\mathbf{k}} \frac{1}{\omega - \varepsilon_{\mathbf{k}} - \Sigma_{\mathbf{k}}(\omega)}. \quad (3.12)$$

Similar to the earlier exercise, we need to apply a mean-field decoupling to drop certain fluctuations and obtain a similar effective problem. In the context of interacting electrons, such a decoupling is obtained in the limit of infinite dimensionality or coordination number, where the self-energy becomes k -independent [31]:

$$\Sigma_{\mathbf{k}}(\omega) = \Sigma_{\text{loc}}(\omega), \text{ or (equivalently) } \Sigma_{ij}(\omega) = \delta_{ij} \Sigma_{\text{loc}}(\omega). \quad (3.13)$$

The mean-field decoupling here therefore corresponds to ignoring spatial fluctuations of the one-particle self-energy, and leads to a local Greens function that is directly connected to the local self-energy:

$$G_{\text{loc}}(\omega) = \sum_{\mathbf{k}} \frac{1}{\omega - \varepsilon_{\mathbf{k}} - \Sigma_{\text{loc}}(\omega)}. \quad (3.14)$$

It turns out that a local Greens function connected to a local self-energy also defines a quantum impurity problem describing the dynamics of a single correlated site with the other environment sites integrated out:

$$S_{\text{imp}} = - \int_0^\beta d\tau d\tau' c^\dagger(\tau) G_{\text{MF}}^{-1}(\tau - \tau') c(\tau') + U \int_0^\beta d\tau n_\uparrow(\tau) n_\downarrow(\tau). \quad (3.15)$$

This acts as an auxiliary problem for the more complete problem of locally interacting electrons on a lattice. The non-interacting Greens function G_{MF} is as of now unknown, and defines the environment of the impurity site and comprises the dynamics of the sites that have been integrated out. Such an impurity problem is solvable, and the impurity Green's function is given by

$$G_{\text{imp}}(\omega) = \frac{1}{G_{\text{MF}}^{-1}(\omega) - \Sigma_{\text{imp}}(\omega)}, \quad (3.16)$$

The first term in eq. 3.15 is therefore what makes this problem dynamical – it represents the probability of electrons hopping into and out of the impurity site, and leads to fluctuations of its occupancy. We can make the identification of the Greens function G_{MF} as the *mean-field* encountered in the previous example. More specifically, G_{MF} acts as a “field” because it is obtained by integrating out the other sites on the lattice, and it acts on the impurity site by affecting its dynamics.

The complete set of equations for the solving such a dynamical mean-field problem is finally obtained by equating the impurity local Greens function G_{imp} to the lattice local Greens function G_{loc} . As discussed earlier, such a solution exists in an exact fashion only in the limit of infinite dimensions, as long as the correct G_{MF} can be found. In order to find the appropriate G_{MF} that allows equating the impurity model Greens function to a lattice mode Greens function, we require an additional equation: the self-consistency condition. Similar to the case of the Ising model, we require that G_{MF} be equal to G_{loc} , owing to translation invariance. The full set of coupled DMFT equations is

$$\begin{aligned} G_{\text{MF}} &= G_{\text{loc}}, \\ \Sigma_{\text{imp}}(\omega) &= G_{\text{MF}}^{-1}(\omega) - G_{\text{imp}}^{-1}(\omega), \\ \Sigma_{\text{loc}}(\omega) &= \Sigma_{\text{imp}}(\omega), \\ G_{\text{loc}}(\omega) &= \sum_{\mathbf{k}} \frac{1}{\omega - \varepsilon_{\mathbf{k}} - \Sigma_{\text{loc}}(\omega)}. \end{aligned} \quad (3.17)$$

One can map various features of this self-consistent program with the set of equations obtained above for the Ising model:

- The first equation is equivalent to the feature in the previous exercise where we equated the mean-fields on i and j .

- The second equation is equivalent to computing, in the previous exercise, the magnetisation on the j^{th} site from the partition function.
- The third equation is the essential mean-field approximation, where we equate the self-energy of the simpler decoupled problem to that of the more complete lattice problem and is justified in the limit of infinite dimensions.
- The fourth equation is equivalent, in the previous exercise, to using the computed “magnetisation” of site j in the second step to further compute the mean-field at site i .

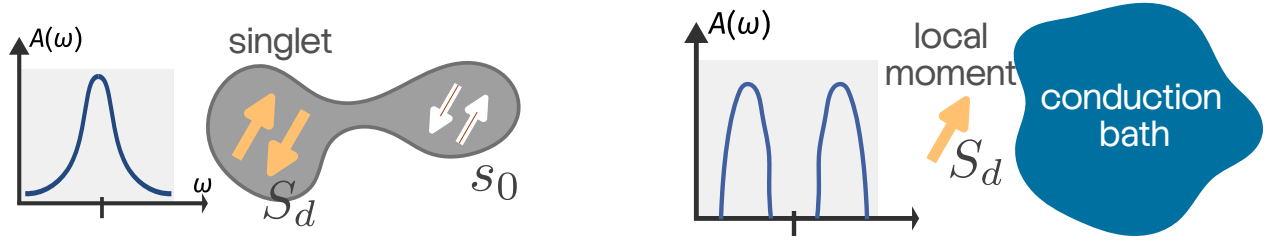


Figure 1.6: The auxiliary model mapping between interacting lattice models and quantum impurity models involves a dictionary for relating phenomena on either side. For example, Kondo screening and the associated local Fermi liquid excitations on the impurity model (left) map to metallic Fermi liquid behaviour for the lattice model. On the other hand, breakdown in Kondo screening and the presence of a residual unscreened moment at low-energies in the impurity model maps to insulating behaviour in the lattice model.

In direct analogy with classical mean-field theory, DMFT replaces a lattice model by an effective single-site problem embedded in a self-consistent environment. The crucial distinction is that the mean field is now dynamical rather than static, encoding quantum fluctuations exactly while neglecting spatial correlations. The fact that the correlated self-energy of the impurity site feeds into the impurity-bath hybridisation during the convergence towards self-consistency means that the bath must get correlated in the process. Importantly, this demonstrates the importance of quantum impurity models in investigating correlated lattice models, particularly in the presence of strong local interactions. This is because the effects of local correlations are captured exactly by such an auxiliary model mapping.

1.3.3 Generator of strong local correlations: Quantum impurity models

Within this thesis, we have developed an auxiliary model approach that combines the physics of quantum impurity models with a manybody version of Bloch’s theorem to investigate interacting fermionic lattice models. This is motivated by the relative ease in solving impurity models and transparency afforded by such solutions. We now review the phenomenology of some of these impurity models and point out how the various impurity features affect the physics of the lattice models.

The single-impurity Anderson model describes the dynamics of a system of non-interacting itinerant electrons ($c_{k,\sigma}$) hybridising with a localised correlated impurity orbital (d_σ) [15]:

$$H_{\text{AIM}} = \sum_{k,\sigma} \epsilon_k n_{k,\sigma} + \sum_{k,\sigma} V_k (c_{k,\sigma}^\dagger d_\sigma + \text{h.c.}) + \epsilon_d \sum_{\sigma} n_{d\sigma} + U n_{d\uparrow} n_{d\downarrow} . \quad (3.18)$$

V_k controls the hybridisation strength, and ϵ_d and U control the onsite energy and local correlation respectively of the impurity site. In the limit of large U , the impurity dynamics gets projected onto its spin states, and the model then describes the scattering of non-interacting electrons off magnetic impurities, through the Kondo model [33]. Formally, the Kondo model is obtained by carrying out a Schrieffer-Wolff transformation on the Anderson impurity model and then using the condition to $V \ll U$ to drop all contributions beyond second order in the ratio V^2/U [34]:

$$H_{\text{Kondo}} = \sum_{k,\sigma} \epsilon_k n_{k,\sigma} + J \sum_{k_1,k_2} \sum_{\alpha,\beta} \vec{S}_d \cdot \vec{\sigma}_{\alpha,\beta} c_{k_1,\alpha}^\dagger c_{k_2,\beta} , \quad (3.19)$$

where $\vec{S}_d$ is the impurity spin formed by projecting the Hilbert space of the impurity to the singly-occupied states.

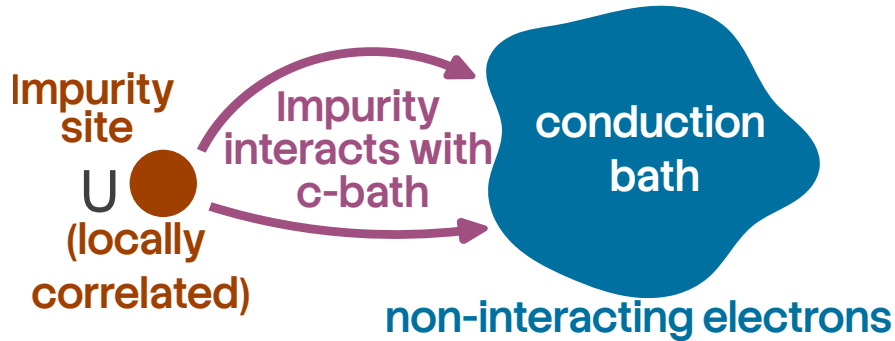


Figure 1.7: Schematic of a quantum impurity model. The impurity is essentially a locally correlated site that presents a magnetic scattering center for the non-interacting itinerant electrons of the conduction bath. The impurity-bath scattering can appear in multiple forms, such as a one-particle hybridisation through which an electron simply hops between the two regions, or a spin-exchange Kondo interaction that emerges when the impurity site has been reduced to a local moment that can undergo spin-flip scattering with the conduction bath electrons.

Even though the conduction bath is non-interacting, the presence of local repulsion on the impurity site makes the problem *many-electron* in nature: that is, the "U" term makes it impossible to study the electrons in isolation. This can be seen as follows: imagine two conduction electrons $|k_1 \uparrow\rangle$ and $|k_2 \downarrow\rangle$ trying to hybridise with an empty impurity site. The first electron moves onto the impurity site through a scattering process of the form $c_{d\uparrow}^\dagger c_{k_1\uparrow}$ and makes the impurity site single occupied ($|\uparrow\rangle_d$), but now the second electron k_2 *must pay a cost* of the order of U in order to hybridise with this singly-occupied impurity site through a process of the kind $c_{d\downarrow}^\dagger c_{k_1\downarrow}$. This shows that the conduction electrons *are aware* of the other electrons, and all of them must be treated together.

Another way of seeing this collective behaviour is from the Kondo model. Again imagine two spin-up electrons k_1, k_2 that are about to scatter off the impurity spin (which is presently down) via a *spin-flip* process. The first electron scatters and leaves the impurity spin up while it itself becomes down, but now the second conduction electron cannot spin-flip scatter because the spins cannot be exchanged (both are down!). This is another example of how the conduction electrons can *feel* the presence of the other electrons, and this is what makes the problem many-electron in nature.

Kondo screening Despite the interacting nature of the problem, the problem can be solved in an effectively exact fashion, via Wilson’s numerical renormalisation group approach [35, 36], density-matrix renormalisation group approach [37] or the method of *Bethe ansatz* [38, 39]. The former solution is more transparent: for any non-zero value of the hybridisation V (or Kondo coupling J), the low energy physics of the impurity is described by a strong-coupling fixed point ($V = \infty$, or $J = \infty$), where the impurity forms a maximally-entangled singlet state with the local spin density of the conduction bath. The low-energy excitations about such a fixed point can be calculated from renormalised perturbation theory, and describe a local Fermi liquid – one particle excitations in the neighbourhood of the singlet renormalised by a local repulsion term [40, 41]. The crossover from the free local moment at high temperatures to the screened non-magnetic state at low temperatures occurs at an emergent energy scale T_K , the Kondo temperature, which depends exponentially on the strength of the Kondo coupling, and is the only energy scale remaining at low temperatures, so that all thermodynamic quantities must go as a function of T/T_K [35]. Unitary renormalisation group studies show that this crossover is marked by the crystallisation of entanglement between the impurity and the Kondo screening cloud, as well as among the electrons forming the cloud [42]. This entanglement also leads to a non-trivial low-energy theory for the excitations of the Kondo cloud, and it is these excitations that are responsible for the screening mechanism [42].

At half-filling (attained for $\epsilon_d = -U/2$), the only other fixed points are the free orbital fixed point at $V = 0, U = \epsilon_d = 0$ where the impurity site is freely occupied by two independent spins, and the local moment fixed point at $V = 0, U = \infty$, where the impurity site is occupied by a local moment that is decoupled from the bath [36, 43]. Away from half-filling, two other fixed points become available [36, 43]: a valence fluctuation fixed point at $V = 0, \epsilon_d = 0, U = \infty$ where the doubly-occupied state has been projected out from the symmetric free orbital fixed point, and a frozen impurity fixed point at $V = U = 0, \epsilon_d = \infty$ where the impurity site is left completely empty.

The crossover from local moment physics to Kondo screening as the system is tuned from high to low temperatures can also be seen from the impurity spectral function [44–47]. At low frequencies, the spectral function exhibits a sharp resonance around $\omega = 0$ that indicates the presence of gapless single-particle local Fermi liquid excitations (see Fig. 1.6). At higher frequencies, one sees satellite peaks that involve impurity excitations that cost energy of the order of U . The height of the central Kondo resonance is a topological invariant; it is fixed by the impurity occupancy owing to an application of the Friedel sum rule to impurity models [48], and remains unchanged as the interaction U is increased adiabatically while remaining at particle-hole symmetry. The topological nature can also be ascertained by identifying such a sum rule as the impurity model analogue of Luttinger’s theorem [24, 25, 49] that relates the total number density of electrons to the number of Greens function poles within the Fermi volume, and which is itself topological [50, 51].

Breakdown of Kondo screening While the Anderson model (or the Kondo model derived from it) only display a strongly-coupled screened phase, it is possible to enhance it in order to introduce Kondo frustration which can eventually lead to a breakdown in the Kondo screening process. Such models are interesting because they provide a means of studying zero temperature *continuous* phase transitions and address experimental signatures of quantum criticality in correlated systems. The canonical example of this is the multichannel Kondo model, involving a single impurity hybridising with multiple independent conduction channels [52]:

$$H_{\text{MCK}} = \sum_{k,\sigma,\alpha} \epsilon_{k,\lambda} n_{k,\sigma,\lambda} + \sum_{k_1,k_2,\lambda} J_\lambda \sum_{\alpha,\beta} \vec{S}_d \cdot \vec{\sigma}_{\alpha,\beta} c_{k_1,\alpha,\lambda}^\dagger c_{k_2,\beta,\lambda}, \quad (3.20)$$

here λ is the channel index. In the case of an isotropic Kondo coupling across the channels ($J_\lambda = J$), the strong-coupling $J = \infty$ fixed-point of the single-channel Kondo model is replaced by an intermediate-coupling fixed-point at a finite value J^* [52, 53]. Powerful methods like boundary conformal field theory [54–59], bosonization [60–64], numerical renormalization group [56, 65, 66] and Bethe ansatz [67–71] have shown that the low energy theory of such models exhibit non-Fermi liquid excitations. Further, the system displays anomalous behaviour in thermodynamic properties near $T = 0$ and a fractional zero temperature entropy in the thermodynamic limit (upon ensuring that temperature is taken to zero after the system size is taken to infinity [63, 72]). This anomalous divergent behaviour is a signature of the fact that the MCK system with a single channel-symmetric exchange coupling is quantum critical: the ground state is susceptible to perturbations that introduce channel anisotropy in the exchange couplings [52, 56, 62, 70, 73]. The criticality arises from a degeneracy of the impurity states, and it can be shown that the anomalous features – non-Fermi liquid physics, diverging impurity contributions to thermodynamic quantities and the presence of fractional excitations – that appear at the critical point can be tied to this degeneracy [74].

Another class of models that shows Kondo breakdown is that of pseudogap Anderson and Kondo models, where the impurity hybridises with a conduction bath whose density of states (DOS) varies as $\rho(\omega) \sim |\omega|^r$ with a paramter $r > 0$ that controls how fast the DOS vanishes as the Fermi level is approached. Such behaviour can arise in semi-metals and in long-range ordered systems where the order parameter vanishes at certain points on the Fermi surface (such as d -wave superconductors). The pseudogap impurity models exhibit a boundary quantum phase transition between a Kondo-screened strong-coupling phase and an unscreened local-moment phase as a function of interaction strength or exchange coupling [75–78]. At the critical Kondo coupling, the impurity spectral function and spin susceptibility exhibit nontrivial power-law scaling with anomalous exponents determined by r [76, 79–82]. For $0 < r < 1/2$ (in the particle-hole symmetric case), both the local-moment and strong-coupling phases are stable, separated by a critical fixed point characterized by fractional residual entropy and nontrivial hyperscaling relations [76, 83]. For larger exponents, the strong-coupling phase may disappear entirely, and the system remains in the local-moment phase for all couplings [78].

The multi-impurity Anderson and Kondo models furnish another canonical realization of Kondo breakdown, driven by the competition between Kondo screening and inter-impurity interaction. The paradigmatic example is the two-impurity Kondo model (TIKM), in which two localized spins couple both to conduction electrons and to each other via an antiferromagnetic exchange K [84, 85]. In the limit of large $K > 0$, the impurities form a non-magnetic singlet decoupled from the

conduction electrons and suppressing the Kondo resonance. This fixed point is separated from the strong-coupling Kondo screened fixed point through a quantum critical point [84–86] displaying logarithmic or power-law scaling in thermodynamic quantities and fractional residual entropy [87]. A modified version of this model that uses conduction bath-mediated inter-impurity interaction leads to a metallic spin liquid displaying nonlocal impurity-spin entanglement and fractionalized charge excitations [88]. More intricate geometries can be constructed by arranging antiferromagnetically coupled Kondo impurities along a triangle, each hybridising with its own conduction bath; such a model displays a topological phase transition from a symmetry-preserved gapless spin liquid phase with “topological order” to a screened Fermi liquid phase [89]. Indeed, recent experiments have validated such phase diagrams for the case of two impurities by studying pairs of copper atoms in the presence of strong magnetic fields [90].

1.3.4 Examples of materials with strong local correlations

As discussed in the previous section, the route to studying lattice problems via quantum impurity models makes most sense for materials with strong local correlations. Such systems are often referred to as strongly correlated quantum materials, and their phenomenology often lies beyond conventional approaches like density functional theory or perturbative weak-coupling methods. In practise, such materials often involve transition metal elements or rare earth/actinide elements, due to the presence of partially filled $3d$ – or $4f$ –orbitals in such elements. This is because these orbitals are more localised around the nucleus than the s – or p –orbitals of comparable energy, and hence lead to a reduced itinerant energy scale as well as amplify the effects of local correlations [32].

The localised nature of these orbitals can be understood as follows. For the $3s$ –orbital, the $1s$ and $2s$ orbitals have same orbital angular momentum l but lower principal quantum number n ; since all such wavefunctions must be orthogonal to each other, the $3s$ –orbital must develop nodes in its radial form and extend farther away from the nucleus in order to be orthogonal to these other orbitals. However, for the $3d$ –orbital, there is no orbital with same l but lower n ; this means that the orthogonality is already enforced by the behaviour of the angular part (the spherical harmonics), and the radial part can remain localised around the nucleus without having to spread out. A similar argument holds for the localised nature of the $4f$ –orbital.

We now discuss the phenomenology of some of these quantum materials in order to identify the challenges in trying to explain them and the kind of models that are likely to be useful in this endeavour.

Transition Metal Oxides (TMOs) Transition Metal Oxides (TMOs) are defined chemically by the presence of partially filled d -shells in transition metals (e.g., Cu^{2+} , Mn^{3+} , V^{4+}) coordinated by oxygen ligands. The physics of TMOs is governed by the competition between the kinetic energy hopping amplitude t and the on-site Coulomb repulsion U ; due to the small overlap between the localised d –orbitals, the d –electrons can delocalise only by hopping onto the ligand atoms. According to the Zaanen-Sawatzky-Allen (ZSA) classification scheme [91], these materials are categorized as either Mott-Hubbard insulators, where the charge gap is determined by the d – d Coulomb interaction ($E_g \approx U$), or Charge-Transfer insulators, where the gap is dictated by the energy difference between

the oxygen p -band and the metal d -band. These derive from considerations of several factors, such as crystal field splitting, arrangement of atoms and lattice symmetries.

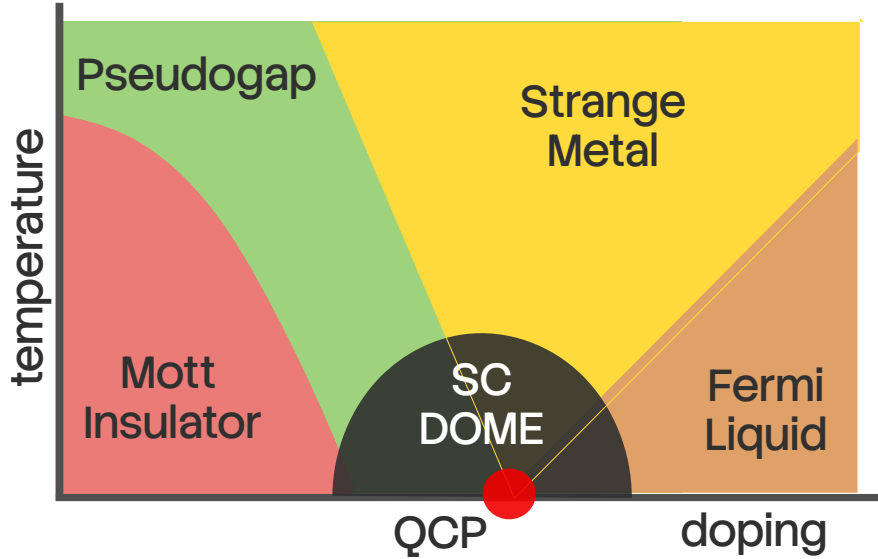


Figure 1.8: Experimental phase diagram of the high- T_c copper oxide materials. The normal state at half-filling is an antiferromagnetic Mott insulator; doping it leads to a plethora of fascinating phases. Upon (here, hole) doping at low temperatures, the Mott insulator melts and gives rise to a d -wave superconductor with transition temperatures ranging upto 130K at ambient pressures. Further doping the system leads to a correlated Fermi liquid. At higher temperatures, two other regions are observed: the strange metal region with linear-in-temperature resistivity, and the pseudogap with partial gapping of the Fermi surface.

For example, in the case of octahedral symmetry, repulsion from the surrounding atoms leads to the raising of energy of the $d_{x^2-y^2}$ and $d_{3z^2-r^2}$ orbitals forming the e_g doublet, while the other three orbitals (d_{xy} , d_{yz} and d_{zx}) form the t_{2g} multiplet. For the heavier TMOs (Ti, V, Cr...), the Fermi level lies within e_g . The increased positive nuclear charge (compared to the lighter elements) lowers the energy of the d -orbitals and brings them closer to the p -orbitals, reducing the charge transfer energy $\Delta = \epsilon_d - \epsilon_p$ of transporting an electron from a d -state to a p -state, compared to the d -level repulsion U ($|\Delta| < U$). Geometric reasons (the e_g orbitals point towards the p -orbitals) also enhance the hybridisation between d - and p -orbitals in such materials. If Δ becomes larger than the bandwidth, we obtain a *charge-transfer insulator*, where the charge gap is decided by Δ . In the lighter TMOs, the Fermi level passes through t_{2g} , and because they point away from the p -orbitals, the hybridisation is weaker. Moreover, the reduced positive nuclear charge leads to an increased $\Delta > U$; a U greater than the bandwidth leads to what's called a *Mott-Hubbard insulator*.

Experimental studies of transition metal oxides (TMOs) have established metal-insulator transitions (MITs) as a defining consequence of strong electronic correlations in systems with partially filled (d)-electron shells [92, 93]. Early experiments on vanadates such as V_2O_3 characterised pressure-driven MITs through the pressure- and temperature-dependent conductivity as well as an anomalous enhancement of the optical conductivity [94–96]. Copper oxides constitute the most

extensively studied class of correlated TMOs and provide a canonical example of a doping-driven MIT. Transport measurements on parent compounds such as La_2CuO_4 establish a robust insulating state, while carrier doping leads to anomalous metallic behavior characterized by linear-in-temperature resistivity and unconventional superconductivity [97–99]. Such metal-insulator transition is also reported from angle-resolved photoemission spectroscopy (ARPES) measurements in doped La_2CuO_4 and $Bi_2Sr_2CaCu_2O_8$, along with the emergence of pseudogap near the antinodes at high temperatures and a hole-to-electronlike Fermi surface reconstruction [100–102]. Nickelates ($LaNiO_2$, $La_3Ni_2O_7$, $La_4Ni_3O_{10}$, etc) exhibit related but distinct experimental signatures. With Ni^{2+} sharing the same $3d^9$ valence configuration as Cu^{2+} in the cuprate superconductors, these materials also display superconductivity, although the mechanism in the nickelates is different due to the strong coupling between the layers [103, 104]. The optimal doped regime in thin films shows strange metal behaviour, while infinite-layer nickelates show the absence of long-range antiferromagnetic order [105].

Transition Metal Dichalcogenides Transition Metal Dichalcogenides (TMDs) are layered van der Waals materials with the stoichiometry MX_2 , where M is a transition metal (e.g., Mo, W) and X is a chalcogen (e.g. S, Se, Te) [106, 107]. Recent interest has focused on the interplay between strong spin-orbit coupling (SOC), electronic correlations and direct bandgap – properties that make them particularly useful for electronic applications, spintronics and DNA sequencing. Layered TMDs have the structure $X - M - X$, with the transition metal sandwiched between two hexagonal planes of chalcogenides. Different combinations of the metals with the chalcogens lead to widely varying structures and electronic properties in TMDs. Experimental studies show that metal–insulator transitions (MITs) in TMDs are commonly intertwined with charge density wave (CDW) formation rather than arising from purely on-site Coulomb localization [108, 109]. In prototypical compounds such as 1T-TaS₂, ARPES and transport measurements reveal fluctuating magnetic order parameters in the Mott state and a transition to superconductivity driven by pressure [110–112]. Scanning tunnelling microscope (STM) pulses have been successfully used to manipulate the metal–insulator transition of 1T-TaS₂, demonstrating the macroscopic controllability of the Mott phase [113].

ARPES measurements in 2H-NbSe₂ and 2H-TaSe₂ exhibit partial gapping of the Fermi surface restricted to nested momentum regions and a temperature-dependent pseudogap similar to the cuprates [114, 115]. This is further supported by optical conductivity experiments that show spectral weight in the midinfrared region along with a low-frequency Drude contribution [116]. Transport measurements on single crystals of TaSe₂ reveal that disorder enhances superconductivity by suppressing other competing orders such as CDW [117]. Collectively, these experimental results establish TMDs as a distinct class of correlated materials in which reduced dimensionality, band narrowing, and lattice reconstruction cooperate to produce metal–insulator behavior beyond conventional band theory.

Iron-Based Superconductors The iron-based superconductors (FeSCs), discovered in 2008, constitute a broad family of layered materials that share a common structure: a layer of iron atoms joined by tetrahedrally coordinated pnictogen (P, As) or chalcogen (S, Se, Te) anions arranged in a stacked

sequence; the layers are separated by alkali, alkaline-earth or rare-earth and oxygen/fluorine ‘blocking layers’ [118–121]. Doping or applying external pressure reveals coupled antiferromagnetic and structural transitions along with superconducting and nematic orders [122–124]. Electrical resistivity measurements in the neighbourhood of the quantum critical point show a crossover from a bad-metal behavior (T –linear resistivity) to a Fermi liquid behaviour (T^2 –resistivity) [125]. Optical conductivity measurements indicate point to intermediate-to-strong electronic correlations [126].

ARPES measurements reveal a k –dependent superconducting gap and de Haas-van Alphen measurements indicate substantial quasiparticle mass enhancements, shedding more light on the correlated nature of the material [127, 128]. Strain-tuning measurements of the nematic susceptibility, and differential elastoresistance measurements have provided compelling evidence for the nematic transition and its surrounding fluctuations being the cause of the structural transition as well as the superconductivity [129, 130]. Neutron scattering and NMR measurements further reveal persistent low-energy spin fluctuations across wide regions of the phase diagram, supporting scenarios in which unconventional superconductivity emerges from the interplay of magnetic, orbital, and nematic degrees of freedom [131, 132].

Heavy-Fermion / Kondo-Lattice Systems Heavy-fermion materials are a class of strongly correlated electron systems typically realized in intermetallic compounds of rare earth metals (such as Ce, Sm and Yb) and actinides (such as U, Np and Pu), where localized f electrons hybridize with itinerant conduction bands [133, 134]. At high temperatures, these systems exhibit local-moment behavior governed by Curie–Weiss susceptibilities and incoherent transport, while at low temperatures the Kondo effect leads to the formation of a coherent many-body state with extremely large quasiparticle effective masses [43]. Experimentally, this crossover is most clearly manifested in transport and thermodynamic measurements: the electrical resistivity shows a characteristic maximum followed by a rapid drop signaling the onset of coherence, while the linear specific heat coefficient $\gamma = C/T$ reaches values orders of magnitude larger than in conventional metals, directly reflecting the heavy quasiparticle mass [135].

A defining feature of heavy-fermion materials is their proximity to magnetic quantum phase transitions, which can be accessed experimentally by pressure, chemical substitution, or magnetic field [136]. Near such quantum critical points (QCPs), transport measurements reveal pronounced non-Fermi-liquid behavior (T –linear resistivity) and a vanishing Hall response, along with a divergent effective quasiparticle mass as the transition is approached [137, 138]; these are all indications of the breakdown of electronic excitations near the transition. Neutron scattering experiments provide direct evidence for spin fluctuations down to very low temperatures, interplaying with the itinerant electrons at the QCP and potentially driving the superconducting transition [139, 140]. Quantum oscillation measurements through de Haas–van Alphen experiments further demonstrate Fermi surface reconstruction across the quantum critical point, highlighting the drastic change in Fermi volume [141]. Together, these experimental observations establish heavy-fermion compounds as paradigmatic systems in which strong local correlations, hybridization-driven coherence, and quantum critical fluctuations conspire to produce emergent low-energy quasiparticles and unconventional metallic behavior [142].

1.4 Phenomenology of Mott transition in lattice models

1.4.1 The Infinite-Dimensional Hubbard Model

The Hubbard model (introduced in eq. 1.1) presents one of the most fruitful playgrounds for investigating correlation-driven metal-insulator transitions and its various facets. This model, despite its conceptual simplicity, exhibits a rich variety of phenomena, including antiferromagnetism, superconductivity, and, most notably for our purposes, the correlation-driven Mott metal–insulator transition (MIT) [4, 143–145]. While exact solutions exist in the limits of $d = 1$ [146] and $d = \infty$ [147], the problem remains largely open in more realistic situations such as two and three spatial dimensions.

As discussed earlier, the limit of infinite dimensions presents a drastic simplification for the manybody scattering processes within the problem: the one-particle k -space self-energy becomes momentum-independent due to the vanishing of all non-local contributions (they scale as negative powers of the coordination number, which diverges in the limit of $d = \infty$). This simplification lies at the heart of dynamical mean-field theory and its ability to capture the Mott metal–insulator transition at both zero and finite temperatures for this limiting case [18, 147–151]. At half-filling and zero temperature, DMFT predicts a continuous transition as a function of U wherein the quasiparticle weight Z tends to zero at a critical interaction strength U_{c2} , signaling the vanishing of coherent Fermi liquid behavior and the opening of a Mott gap in the single-particle spectral function [18, 149].

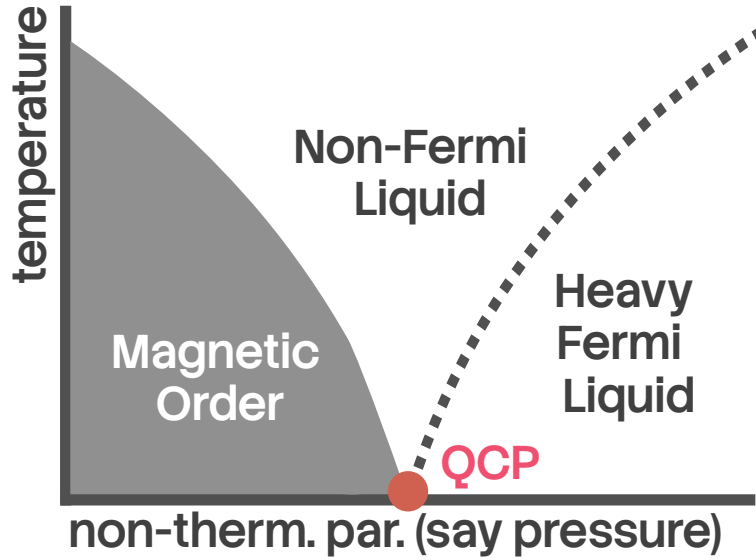


Figure 1.9: Experimental phase diagram of the Kondo-breakdown class of heavy fermions compounds, as a function of the inter-layer hybridisation. For small hybridisation, the correlated f -layer remains decoupled from the itinerant layer in the ground state and undergoes magnetic ordering. For larger values, the f -layer undergoes Kondo screening by the itinerant layer, forming inter-layer entangled states. Lying in between these states is a quantum critical point with strange metallic behaviour arising from it at finite temperatures. The transition across the QCP at zero temperature is characterised by a change from a small to a large Fermi volume.

The three-peak structure of the interacting spectral function, with a central quasiparticle resonance flanked by incoherent lower and upper Hubbard bands, indicates the presence of highly renormalised Landau quasiparticles [149]. A doping-driven continuous metal-insulator transition also exists for this model at zero temperature, marked by the vanishing of electronic compressibility at a critical value of the chemical potential [152]. The quantum critical point itself (at $T = 0$) is likely to be an exotic state, and important questions about it remain unanswered. *Is it possible to obtain a low-energy theory for the local gapless excitations precisely at the MIT, where the metal is on the brink of destruction? How do these excitations compare with those of the local Fermi liquid, e.g., in terms of self-energies and two-particle correlation functions?*

While the DMFT solution captures the zero temperature transition accurately, the self-consistently determined nature of the conduction bath renders the solution somewhat opaque and makes it difficult to write down an effective Hamiltonian for the low-energy theories. Since the simplest version Anderson impurity model does not contain the mechanism to stabilise a localised phase over a finite-sized parameter regime, an important question arising from the DMFT solution is: *what are the minimal changes that one must implement into the Anderson impurity model in order to capture such a metal-insulator transition (1/2-filled Hubbard model on the Bethe lattice in $d = \infty$), in sufficient accuracy and precision?* Finding such an impurity model would also reduce the considerable computational effort that is presently required in self-consistent approaches.

At finite temperatures, the interaction-driven as well as doping-driven Mott transition become first order, and is characterized by a region of phase coexistence and hysteresis [150, 153]. Two spinodal lines, $U_{c1}(T)$ and $U_{c2}(T)$, emerge; within this region metallic and insulating DMFT solutions coexist, and responses such as resistivity show a significant discontinuity [154]. These lines meet at a finite-temperature critical endpoint (U_c, T_c), above which there is no sharp distinction between metallic and insulating solutions and only a crossover exists between the Mott insulator and the Fermi liquid through a bad-metal regime, as the chemical potential is tuned [150, 152, 155–157]. The coexistence of metallic and insulating phases shows that the insulating solution is present within the many-body spectrum of the metallic phase. This naturally leads to the following question: *since a quantum impurity model lies at the heart of the self-consistency procedure, what does the presence of coexistence region mean for the Hamiltonian dynamics and the spectrum of this impurity model? Can the impurity model Hamiltonian display the emergence of the insulating state prior to the transition, through changes in the eigenstates?* Analytic insight of this kind is particularly important because approaching a coexistence region numerically is often fairly tricky.

The bad metal regime displays anomalous linear-in-temperature resistivity which corresponds to quantum critical ω/T scaling, along the quantum widom line [158, 159]. This is accompanied by the disappearance of the Drude peak in optical conductivity, signalling the loss of quasiparticle transport [158]. Similar quantum critical scaling is also obtained from DMFT solutions at very low temperatures, hinting at a “hidden” quantum criticality that is continuously connected to and responsible for the critical behaviour above the finite temperature endpoint [160]. Luttinger’s theorem, which relates the poles of the Greens function within the Fermi volume to the number of electrons in the system at zero temperature, is satisfied within the Fermi liquid phase but is violated in the particle-hole asymmetric Mott insulator by a universal magnitude [161]. The emergence of a quantum critical regime at high temperatures and its connection to similar signatures at low temperatures suggests the presence of quantum fluctuations in the underlying impurity model that modify

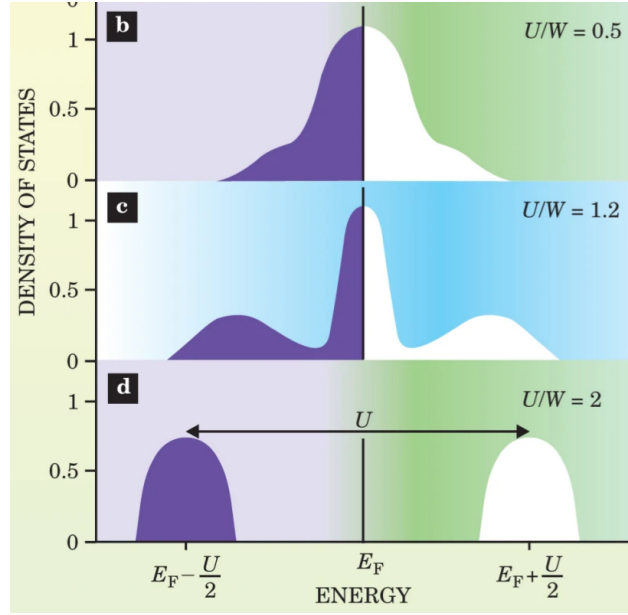


Figure 1.10: Schematic depiction of the evolution of the local spectral function of the infinite-dimensional Hubbard model as obtained from dynamical mean-field theory (DMFT). At small values of the Hubbard interaction U , the spectral function has a central resonance, indicating good metallic behaviour. For larger values, the central resonance splits and satellite peaks emerge, indicating loss of some spectral weight from zero frequency. Beyond a critical value, the central peak disappears and only the satellite peaks remain, leading to a gapped Mott insulating state.

the impurity-bath scattering processes and the associated impurity responses. One can therefore ask: *What are the fluctuations that destroy the metal and lead to the insulating phase? Can we obtain a universal theory for these competing tendencies?*

The above studies of single band models describe the Mott transition to be of the Mott-Hubbard kind (described in previous sections), where the size of the gap is determined by the double-occupancy cost U . In order to obtain transition of the charge-transfer kind, one needs to consider at least a two-band model. Indeed, in a non-degenerate two-band Hubbard model with a Charge-Transfer energy Δ for hopping from a d -orbital to a p -orbital, one sees a Mott-Hubbard transition when $U < \Delta$ and a charge-transfer transition when $U > \Delta$ [162]. The degenerate two-band model with inter-band local correlation shows successive Mott-Hubbard transitions at low temperatures whenever the filling passes through integer values and the interaction strength is intermediate [163].

In the later chapters of this thesis, we will be addressing several of these questions.

1.4.2 The two-dimensional Hubbard model

The 2D Hubbard model is one of the most well-studied models in condensed matter physics, particularly because of its paradigmatic role in Mott metal-insulator transition and its connections to the physics of copper oxide materials and d -wave superconductivity. Photoemission experiments reveal the presence of p and d bands close to the Fermi surface in such materials [100]; further work by Zhang and Rice showed that a specific combination of these orbitals governs the low-energy physics, leading to an effective description in terms of a single band Hubbard model [164]. The discovery of high-temperature superconductivity in nickelates and organic charge transfer salts has provided more platforms where the Hubbard model and its extensions can prove to be useful [165, 166].

While simple in appearance, the model has proved to be notoriously hard to solve except in highly specific parameter regimes, despite several decades of computational effort. This is due to several reasons [145]: (i) the presence of closely-lying energy scales arising from competing tendencies, (ii) the inability of several methods to either reach zero temperature or extrapolate to the thermodynamic limit, (iii) the likelihood of approximate/finite-size methods to overestimate tendencies towards spontaneous symmetry breaking, and (iv) the presence of differing phases separated by slow crossovers without any sharp boundaries.

At half-filling, the 2D Hubbard model has a spectral gap to charge excitations for any non-zero interaction strength [167–171]. The behaviour below the charge gap is described by a nonlinear sigma model with parameters that depend on the interaction strength U [172]. Upon tuning to finite temperatures, the system goes through a sequence of crossovers from the insulating ordered state to a Fermi liquid state with coherent quasiparticles into a non-Fermi liquid state with partial gapping of the Fermi surface [169, 171, 173]. The excitations of the non-Fermi liquid are described by a marginal Fermi liquid theory, and the Mott insulator displays topological order in the absence of spontaneous symmetry breaking [174]. *Capturing the momentum dependence of static and dynamical quantities in sufficient detail remains an open challenge in such problems, since results from cluster methods suffer from sensitivity to periodisation schemes* [175].

Away from half-filling but at weak coupling, the Neel order is accompanied by incommensurate magnetic orders (also called spiral magnetic state) with wave vectors away from (π, π) , as seen from mean-field studies [176, 177]. Functional RG studies indicate that such solutions can exist up

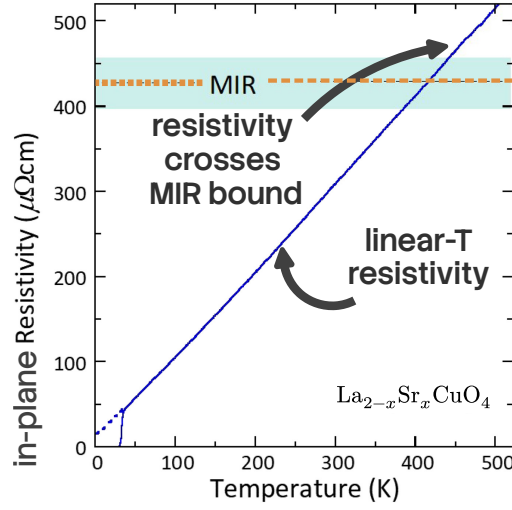


Figure 1.11: Resistivity measurement in the strange metal phase of the overdoped cuprates. There are two striking features. The first is the linear dependence of the resistivity on temperature, which cannot be explained from Landau’s Fermi liquid theory. The second is the unbounded growth of the resistivity beyond the Mott-Ioffe-Regel (MIR) limit as temperature is increased. For a system of electrons on a lattice, the minimum possible mean-free path for propagating electrons is the lattice spacing, and this sets an upper bound for the resistivity, which is the MIR bound. Resistivity crossing the MIR bound implies that electrons are not the propagating degrees of freedom in such a system.

to fairly high doping if the superconducting instability is suppressed [178, 179]. Superconductivity with d -wave pairing has been achieved from renormalised perturbation studies as well as from diagrammatic Monte Carlo (diagMC), functional RG (fRG) calculations and unitary RG calculations [180–183]. More intricate competition between the superconducting and stripe states is found in three-band Hubbard models from variational Monte Carlo, cluster DMFT and DMRG calculations [184–186]. *Further work is necessary to resolve the closely-lying phases in such models.*

The phenomena of pseudogap (PG), where the Fermi surface acquires a momentum-dependent suppression of spectral weight, has also been observed above a certain temperature from methods like diagMC and fRG [171, 187]. The gap is seen to open first near the antinodal point $(\pi, 0)$ and spread across the Fermi surface until it reaches the node $(\pi/2, \pi/2)$. The mechanism of the pseudogap at weak to intermediate coupling is found to be long-range antiferromagnetic (AF) correlations [188]. At underdoping, the pseudogap is marked by a dominant charge gapping of the antinodes accompanied by strong spin-nematic fluctuations and sub-dominant superconducting phase fluctuations [183]. The pseudogap disappears at large doping when the Fermi surface undergoes a topological transition from hole-like to electron-like. At strong coupling, the competition between various phases such as d -wave superconductivity, antiferromagnetic order and stripe states (spin/charge order modulating in real space) is an active subject of study. Recent studies using various methods like DMRG and iPEPS found that the period 8 charge stripe state is lower in energy than the superconducting state at strong coupling $U/t = 8$ and no next-nearest-neighbour hopping [189]. This was later corroborated by studies from variational Monte Carlo and other DMRG studies [190, 191]. *It is not yet clear whether the stripe period gets lowered as the interaction strength is increased.*

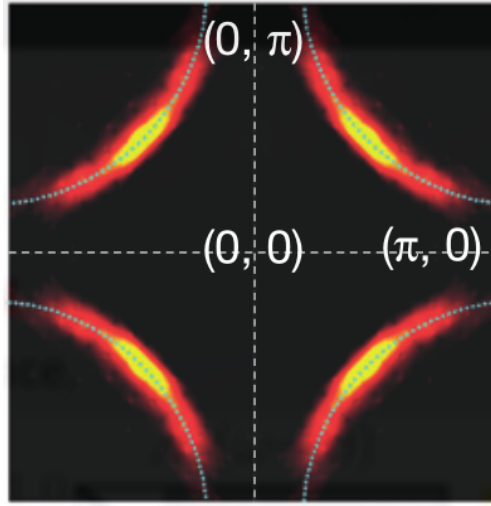


Figure 1.12: ARPES measurement of the Fermi surface in the pseudogap of underdoped cuprates. The Fermi surface shows partial gapping around the antinodes and gapless behaviour around the nodes. Further doping leads to a reconstruction of the Fermi surface into the strange metal.

At weaker coupling, the above methods report a reduced stability for the strip state and increased tendency to realise a d -wave SC state [181, 192]. Further investigation is necessary to map the crossovers between stripe and superconducting phases as doping is tuned. Moreover, the precise parameter regime in which the d -wave superconducting state becomes more stable than the stripe states is still an open question.

At strong-coupling, the pseudogap is obtained below optimal doping and has been studied using dynamical methods such as cluster DMFT (DCA, dynamical cluster approximation) and cellular DMFT (CDMFT) [193–196]; in contrast to the weak-coupling feature (where the PG was borne from long-range AF fluctuations), it is believed that the momentum-space differentiation in this regime is brought about by the proximity to the Mott insulator [195]. Conflicting indications are however provided by a fluctuation analysis that finds that long-lived long-ranged spin-fluctuations at the antinode are responsible for the opening of the PG [197]. The PG is found to be accompanied by non-Fermi liquid behaviour in the one-particle self-energy [194]. The doping value at which the non-Fermi liquid first appears was found to be lower for electron-doping than hole-doping [198]. At half-filling, DCA calculations reveal that the extent of the PG decreases as the next-nearest-neighbour hopping strength is increased [199]. Moreover, while DCA+continuous-time auxiliary field methods do not show any effect of van Hove singularities on the generation of the PG [199], DCA+determinantal Monte Carlo approaches identify the proximity to van Hoves as favouring the formation of a PG [200]. *Methods with better momentum-space resolution is necessary in order to identify the role of nesting and van Hoves on the pseudogap.*

The question of whether superconductivity can arise from repulsion was one of the primary motivations for studying the Hubbard model. DiagMC calculations reveal superconductivity in the ground state at weak-coupling and moderate doping, and with multiple symmetries of the gap [181, 201]. For higher couplings (which is relevant to the cuprates) and moderate doping, stripe states take

over. Quantum Monte Carlo and DCA calculations suggest a $d_{x^2-y^2}$ pairing symmetry [168, 202]. DCA calculations on the 2D Hubbard model suggest low-frequency spin fluctuations as the pairing glue for the superconductivity [203]. Unitary RG studies show the presence of preformed Cooper pairs near the antinodes at finite temperature states above the superconducting dome [183]. *The description of how various parent phases such as the pseudogap and strange metal precipitate superconductivity, in terms of the nature of fluctuations and the appropriate degrees of freedom, is still an open question.* Apart from superconductivity, several strongly correlated materials also exhibit strange metallicity, with transport properties that defy Fermi liquid theory, such as a linear-in-temperature resistivity that extends across large regimes of temperature. Within the Hubbard model, evidence for such T -linear resistivity and robustness across temperatures is available from CDMFT and determinantal Monte Carlo calculations [204, 205]. Studies using the unitary renormalisation group method reveal a two electron-one hole composite as the dominant excitation within the strange metal [183]. *More reliable methods are needed to access and understand the intermediate temperature regime of strange metals* [145, 206].

In the later chapters of this thesis, we will be addressing several of these questions.

1.4.3 Theories of heavy fermion materials

Heavy fermion materials represent the canonical environment for studying quantum phase transitions in correlated systems, owing to the explicit observation of quantum critical points at the boundary of antiferromagnetic order [207]. These systems involve the coexistence of magnetic moments (arising from localised $4f$ - and $5f$ -electrons) and itinerant conduction electrons. The ground state of such a system is dictated by the competition between two kinds of interactions: internal spin-spin interaction J_{RKKY} between localised moments in the f -layer, and interaction J_K between f -electrons and conduction electrons [208].

Analogous to the screening of a single moment by a conduction bath, the lattice of f -moments are prone to undergoing Kondo screening by the conduction electrons through the inter-orbital hybridisation J_K . The local Fermi liquid excitations associated with the spins overlap with each other and become delocalised, leading to a paramagnetic heavy Fermi liquid with highly renormalised quasiparticle mass [209]. Because of the delocalisation of the excitations of the f -spin through Kondo screening, the number of carriers increases to $1 + x$ compared to x coming purely from the conduction electrons; this is commonly referred to as the large Fermi surface phase [210]. Indeed, a perturbation theoretic calculation in the on-site interaction strength of the f -layer allows one to rewrite the two-orbital model as two decoupled quasiparticle bands with dispersions renormalised by phenomenological parameters like the quasiparticle residue [43]. These phenomenological parameters can be estimated from mean-field approaches [211, 212]. While the temperature T^* at which coherent Landau quasiparticles can proliferate was found to be generically lower than the single impurity Kondo temperature T_K [213], T^* was found to be higher than T_K near half-filling of the conduction layer [214]. At half-filling in both layers, the chemical potential passes through the middle of the gap between the two quasiparticle bands, and the system is a Kondo insulator [215]; for bipartite lattices, the ground state can be generally shown to be unique and a singlet ($S = 0$) [216–218].

Tuning the f -layer J_{RKKY} to large values compared to the inter-layer hybridisation can bring forth a quantum phase transition into a different ground state [219]. Experimental evidence sug-

gests that this transition usually falls into one of two classes, depending on the behaviour of the Kondo singlet binding energy E^* in the paramagnetic phase. If E^* remains non-zero as the QCP is approached from the heavy Fermi liquid into the ordered state, the transition is said to fall in the SDW type, because it can be described as an instability of the heavy quasiparticles into a spin-density-wave ordered phase [220–222], in the form of a quantum action that involves the spatial and temporal fluctuations of an order parameter. CeCu₂Si₂ is a prototypical material that exhibits quantum criticality of the SDW type [134]. Because of the non-zero binding energy of the singlet state at the QCP, heavy quasiparticles survive across the transition and the Fermi surface is expected to evolve continuously, along with the order parameter [223, 224].

On the other hand, if E^* vanishes at the QCP, the transition belongs to the Kondo breakdown type, which is not compatible with the spin-density-wave scenario [225]. This happens when gapless spin fluctuations within the f -layer compete with the Kondo spin-flip processes and lead to the destruction of Kondo screening on the Fermi surface [136]. This is tracked by the vanishing of the quasiparticle residue of the heavy Fermi liquid as one approaches the QCP from the paramagnetic phase [82, 224]; this in also implies the critical non-Fermi liquid nature of the large Fermi surface at the transition [226]. *A complete understanding of the Kondo breakdown class of QCP is still missing in the literature.*

Due to the destruction of Kondo screening across the QCP in this class of materials, the localised f -spins cease to contribute to the Fermi surface away from the paramagnetic phase, and the Fermi surface shrinks discontinuously from large $(1 + x)$ to small (x) across the QCP [227]. *Such a change in the Fermi volume violates Luttinger's theorem which equates the particles density to the Fermi surface volume, and Oshikawa's topological flux-tuning argument suggests the presence of additional degrees of freedom that respond to the flux while not being part of the Fermi surface. Additional work is needed to identify these degrees of freedom.* Materials like YbRh₂Si₂ and CuCu_{6-x}Au_x, among others, appear to exhibit behaviour that falls in this class [228, 229]. As mentioned before, several features of this class of experiments are not compatible with the SDW instability transition of the heavy Fermi liquid; these include the jump discontinuous change in Fermi volume mentioned previously (and an associated discontinuous change in the hall conductivity) [230], vanishing of the scale T_{FL} (associated with the onset of Fermi liquid behaviour) and the divergence of the quasiparticle mass upon approaching the QCP from the paramagnet [225], a universal linear-in-temperature strange metal behaviour of the resistivity in an intermediate finite temperature quantum critical regime above the QCP [228, 231] and the possible lack of magnetic ordering immediately away from the QCP [232, 233]. *At present, the connection of the physics of heavy fermion models like the periodic Anderson model to strange metal behaviour is an open question; the mechanism for generating such behaviour from the QCP remains to be completely understood.*

A fairly comprehensive model for describing the phase transition in heavy fermion systems is the periodic Anderson model (PAM):

$$H_{\text{PAM}} = \varepsilon_f \sum_{i,\sigma} n_{i,f,\sigma} + U_f \sum_i n_{i,f,\uparrow} n_{i,f,\downarrow} + \sum_{k,\sigma} \epsilon_k n_{k,c,\sigma} + V_\perp \sum_{i,\sigma} \left(c_{i,f,\sigma}^\dagger c_{i,c,\sigma} + \text{h.c.} \right), \quad (4.1)$$

where the subscripts f and c in the operators indicate whether they belong to the f -layer or the conduction electron layer. At the Kondo breakdown QCP and in the RKKY-dominated phase, the

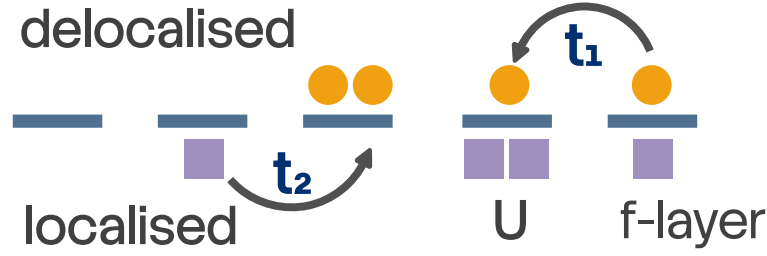


Figure 1.13: Schematic model of heavy fermions. There are two layers: one uncorrelated delocalised layer (top), and a correlated localised layer (bottom). The delocalised layer has very weak correlation and a large bandwidth, leading to small masses for the carriers. The localised layer has very strong local repulsion leading to extremely narrow bands and highly correlated hopping for the carriers. These two layers hybridise with each other through hopping or spin-exchange processes. This results in a modification of the properties of the weakly correlated layer by the strongly-correlated layer.

inter-layer hybridisation is expected to be irrelevant and the f – and conduction electrons fail to bind into a Kondo singlet in the ground state [234, 235]. Several numerical and analytical methods have been applied to investigate various parts of the phase diagram and probe varying aspects of the phenomenology. However, many of these methods suffer from a lack of predictions of various transport measures like strange metallicity, and they often suffer from insufficient resolution in frequency or momentum space.

One of the major approaches applied to this problem is that of extended DMFT (EMDFT), in which one maps the PAM to a self-consistently determined Bose-Fermi-Kondo model [82, 236]. One of the smoking gun signatures of quantum criticality is the scaling of dynamical properties as $\omega/k_B T$ due to temperature being the only remaining energy scale in the system; this is captured from EMDFT calculations [224]. Numerically exact unbiased methods like DMRG and auxiliary field quantum Monte Carlo demonstrate the abrupt change in Fermi surface volume across the transition [237, 238]. Parton methods that implement a slave-particle representation followed by a $1/N$ expansion reveal jump in the conductivity across the transition, with the small Fermi surface arising from a spin-charge separation of the f – electrons into a fractionalised Fermi liquid phase [239, 240]. Currently, a complete description of the Kondo breakdown driven transition that captures the behaviour of both the heavy Fermi liquid and the RKKY-dominated phase is missing.

In the later chapters of this thesis, we will be addressing several of these questions.

1.5 An alternative perspective on metal-insulator transitions: Entanglement renormalisation and holography

Role of entanglement in fermionic systems The present millennium has seen a rapid surge in the use of quantum entanglement towards the study of quantum and condensed matter systems [241–246]. Being non-local by nature, it is helpful in tracking correlations among particles across both small and large distances, giving valuable insight into the nature of the ground state

and the low-energy excitations. Entanglement, in the form of various measures like entanglement entropy and mutual information, is the engine that gives rise to new low-energy degrees of freedom from a renormalisation group (RG) perspective, and is therefore key to understanding and characterising phase transitions and criticality in quantum matter. For instance, entanglement entropy is at the heart of understanding the so-called topological states of matter that can be described in terms of a purely topological effective Lagrangian and have a robust ground state degeneracy and gapless edge states [247–267]. In contrast to systems with local interactions whose subsystem entanglement entropy scales with the area of the subsystem [243, 268, 269], topologically ordered systems have an additional topological term dubbed the topological entanglement entropy [270–288] that quantifies the long-ranged nature of the entanglement in such systems. The non-local nature arises from the fact that the ground state of such a system cannot be made separable (non-entangled) by the application of purely local unitary transformations [245, 289, 290]. Although the topological piece of the entanglement entropy is subdominant compared to the non-topological piece that scales with the subsystem area, it can be isolated by considering the *tripartite information* I_3 of a specific arrangement of subsystems [277, 280]. More generally, all multipartite information I_N beyond mutual information ($N \geq 3$) can be expressed as the product of the topological entanglement entropy and the Euler characteristic of the chosen arrangement of subsystems [291].

More pertinent to the work at hand, however, is the entanglement in free fermionic systems. In this thesis, we will consider relativistic Dirac fermions in two dimensions, both massless and massive. They emerge in several condensed matter systems [292], appearing at the surfaces of three-dimensional topological insulators [293–296], quasi-2D organic materials [297–299] and in artificial materials like cold atoms, molecular and microwave crystals [300–306]. For massless critical systems (or conformal field theories (CFTs)) living in one spatial dimension, the entanglement entropy of a single subsystem of length l scales as $\sim \frac{c}{3} \ln l$ [307–316], where c is the conformal central charge of the CFT. The case of multiple subsystems is more involved, but has been solved [314, 317, 318]. For massive fermions, the entanglement entropy (or rather the entropic c -function, which is the spatial derivative of the entropy) can be expressed in the form of differential equations which need to be solved numerically [317, 319–321]. The solutions lend themselves to small and large mass expansions, reducing to $\frac{1}{3} \ln \frac{l}{\epsilon} - \frac{1}{6} (ml \ln ml)^2 + O((ml)^2)$ for the case of small ml , and to $\sim (ml)^{-1/2} e^{-2ml}$ for that of large ml , both for a single interval. The $\ln l$ behaviour is a specific case of the more general $l^{d-1} \ln l$ violation of the area law, as observed in d -dimensional fermionic critical systems [317, 322–336]. This violation arises from the non-local nature of the real-space quantum fluctuations that exist in quantum critical fermionic systems.

While there is no geometry-independent subdominant term in the entanglement entropy of free fermionic systems [337, 338], it is possible to generate such a term by placing the system on a manifold with a non-trivial topology, say a cylinder, and inserting a magnetic flux through the cylinder [337, 339–342]. For a massless Dirac fermion, if there are ϕ number of electronic flux quanta piercing the system, the additional term that is generated in the entanglement entropy of a subsystem of sufficiently (as defined in Ref. [337]) large length is of the form $\frac{1}{6} \ln |2 \sin \frac{\phi}{2}|$ [337, 340]. There is evidence from numerical computations on discrete models that such a term is a signature of the specific field theory and encodes universality [342–344]. In this thesis, we put this idea on a stronger analytical footing by relating the flux-dependent term to a topological invariant of the field theory, its Luttinger volume [24, 25, 50, 51, 345–347]. Given by the number of k -states enclosed

by the Fermi surface, Luttinger volume V_L has proved to be one of the most important indicators that distinguish metals from insulating states of matter. In metals, V_L is constrained by the number of particles in the system, following Luttinger's theorem [24, 25, 50]. As shown most prominently in Ref. [50], Luttinger's theorem depends crucially on the presence of extended states in the metal that can respond to flux insertion experiments. We demonstrate in Chapter 7 that indeed it is this subdominant term in the entanglement entropy that carries this dependence and hence encodes Luttinger's theorem. It is known that for strongly-interacting states like Mott insulators or quantum hall states, V_L is modified due to changes in the topology of the Fermi surface [51, 345–347]. In this context, we relate the topological invariant on the metallic side (Luttinger volume) to the topological invariants (the Chern numbers) in the integer quantum hall system that the metal would lead to when placed in a perpendicular magnetic field. Such a relation makes explicit the transmutation of the extended states into the localised states of the insulating phase.

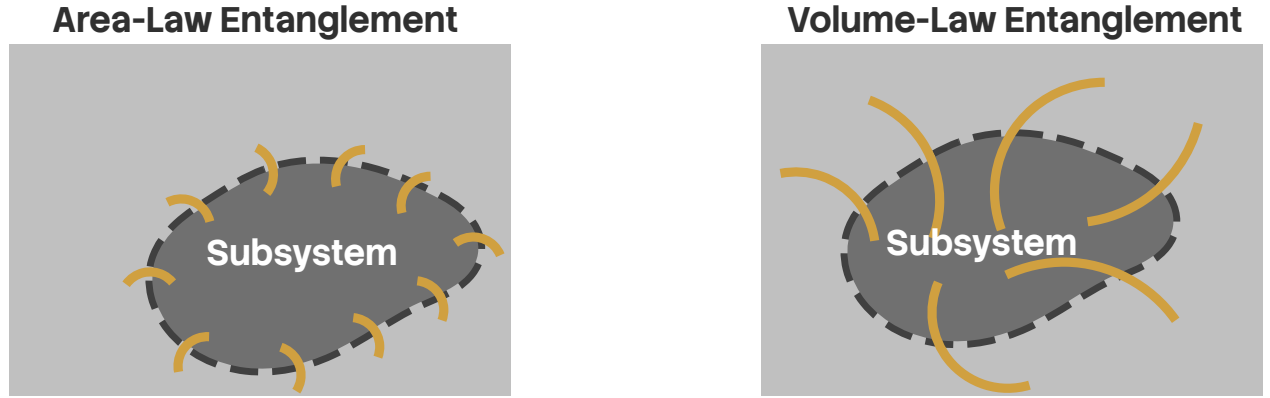


Figure 1.14: Two classes of entanglement found in quantum systems, grouped by how entanglement measures scale with subsystem size. In systems with a spectral gap (left), correlations decay exponentially with distance, and only the degrees of freedom on the boundary of the subsystem entangle appreciably with the rest of the system. As a result, the entanglement entropy of the subsystem scales with the area L^{d-1} of the subsystem: $S_{EE} \propto L^{d-1}$. In contrast, for gapless systems, correlations are more long-ranged and may even span the entire system (for critical states). In such cases, degrees of freedom beyond the proximity of the boundary can entangle with the rest of the system, and such systems can achieve modified area-law $S_{EE} \propto L^{d-1} \ln L$ or even volume law $S_{EE} \propto L^d$.

Entanglement hierarchy and the holographic principle The leading geometry-dependent piece is also found to have intriguing properties of its own. The fact that the free fermion Lagrangian is diagonal in momentum space means that the entanglement arises purely from real space quantum fluctuations, and this allows us to study the entanglement of subsystems of varying sizes in momentum space. We find that the entanglement entropy hence obtained has a Russian doll-like nested structure, such that increasing the size of the subsystem in momentum space gradually reveals layers of entanglement. In fact, we argue that such transformations between the subsystems amount to a renormalisation group (RG) flow in the space of the field theories with varying spectral gaps, and

the above-mentioned hierarchy in the entanglement entropy then stretches across the RG transformations. The precise prescription for choosing the subsystems is described in Chapter 7. We also show, simultaneously, that the hierarchy holds across multipartite entanglement measures, leading to an onion-like structure where increasing the number of parties in the entanglement measure (mutual information, tripartite information, 4-partite information and so on) leads to additional entanglement that did not exist before. More importantly, we interpret the hierarchy of entanglement along the RG flow as the emergence of an additional dimension [348–350], allowing us to define a measure of distance using some non-negative measure of entanglement, like the mutual information [351–354]: the more the entanglement, the less will be the distance between those points. Generation of an additional dimension is, of course, the essence of the AdS/CFT conjecture or, more generally, the holographic principle.

The holographic principle [355–363], also known as the gauge-gravity duality, posits that there exists a duality relation between a quantum field theory with a gauge field, and a gravity theory having one additional spatial dimension [364–367]. The additional dimension can be visualised as a renormalisation group flow that starts from the boundary field theory and extends into the bulk [368–377]. The most popular realisation of the holographic principle is the AdS/CFT correspondence that links a super-symmetric gauge theory with a string theory [378–380]. The essence of holography is that all the information about the bulk gravity theory is already contained within the boundary gauge theory; this idea is very similar to the discovery by Bekenstein and Hawking [381–383] that the entropy of a black hole is determined by its surface area. In fact, Ryu and Takayanagi proposed a generalisation of this result where the entanglement entropy of subsystem region a CFT is determined by the area of the minimal surface that bounds the region in the bulk [384, 385]. The AdS/CFT correspondence happens to be a strong-weak duality - it relates a strongly coupled CFT to a theory of weak classical gravity. This allows physicists to use the holographic principle and convert a problem of strong quantum correlation to one of classical gravity which can be solved using well-known methods [358, 359]. This has led to new insights in various topics of high energy and condensed matter physics, such as the viscosity of the quark-gluon plasma [386], the response functions of the Bose-Hubbard model [387, 388], and the phenomenology of non-Fermi liquids [389–392], holographic superconductors [393–398], striped phases [399–401] and holographic Fermi liquids [402–406]. An overview of the metallic states (phases that satisfy Luttinger’s theorem) and their connections to the gauge-gravity duality have been provided in Refs. [407–409].

Of course, on the flip side, the holographic principle also implies that studying the boundary conformal theory can provide insights into the nature of the bulk quantum gravity theory. This however requires constructing, *ab initio*, the emergent dimension by starting from the field theory, and has proved to be more difficult than the converse, mostly because the field theories are often strongly coupled. Nevertheless, there exist some examples of constructive attempts towards a bulk gravity theory [346, 347, 353, 354, 410–443]. Some well-known approaches include the momentum shell renormalisation approach of Sung-Sik Lee [423–426] and the multi-scale entanglement renormalisation ansatz (MERA) of Vidal [427–431]; both use certain forms of RG flow to generate the extra spatial dimension. The former integrates out high momentum degrees of freedom in favour of generating dynamics for the boundary gauge field, with the RG trajectory variable giving rise to the additional spatial dimension [357]. On the other hand, the MERA works by disentan-

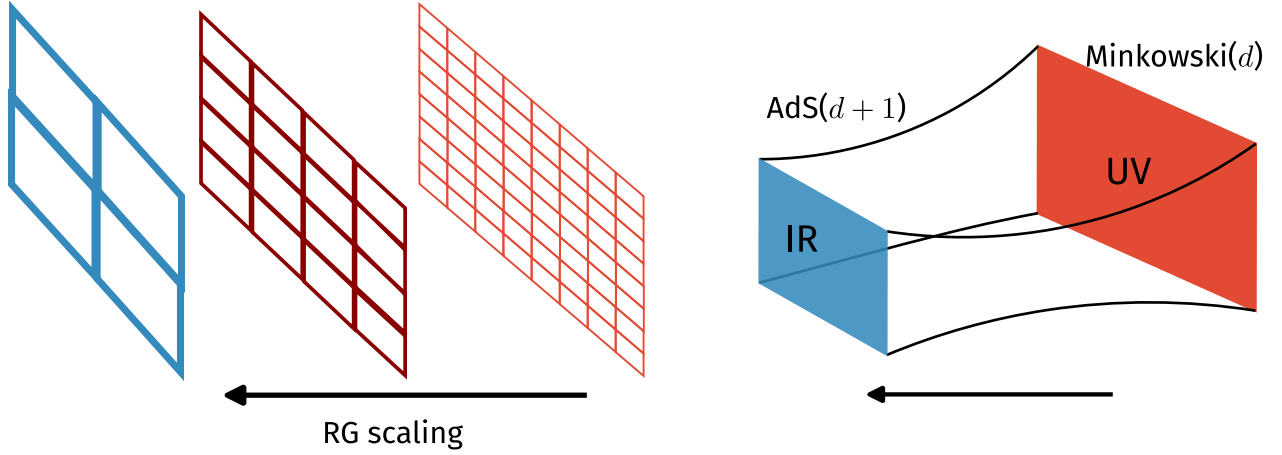


Figure 1.15: The AdS-CFT correspondence (right), or more generally, the holographic principle maps a quantum field theory in d -dimensions to a gravity theory in one extra dimension. The quantum field theory is said to reside at the boundary of this $d+1$ -dimensional space. The additional dimension in the gravity theory arises from the hierarchy of quantum fluctuation scales present within the electronic interactions of the quantum field theory, and can be revealed by considering renormalisation group flows (left) of the boundary theory.

gling degrees of freedom in real space, creating a tensor network in the process [432–434] and leading to a renormalisation in the entanglement. The renormalised entanglement is interpreted as the additional dimension [429]. There exist other approaches that also proceed via entanglement renormalisation, such as the exact holographic mapping [353, 354] (EHM) and the unitary renormalisation group [346, 347, 435] (URG) that operate purely through unitary transformations and have been explicitly shown to lead to a holographic tensor network along the RG flow. Apart from these, there is a body of work due largely to Ki-Seok Kim that provides a geometric description of Wilson’s numerical renormalisation group [36, 436–438, 444], and creates a more general RG framework in terms of an order-parameter field that extends the Ads/CFT correspondence to non-critical theories [439–443].

Questions on entanglement and holography become particularly interesting in systems of critical fermions. As mentioned earlier, such systems differ from bosonic and gapped systems through the presence of longer-ranged entanglement that follows a modified area-law [328, 330] as well as topological features arising from the incompressibility of the Fermi volume [50, 51]. Fermi surface gapping phase transitions in such systems are often driven by changes in topological quantum numbers and the nature of the inter-particle entanglement [136, 346, 347]. Constructing a framework for fermionic criticality is therefore an active area of research with consequences for important systems such as unconventional superconductors, strange metals and heavy fermions [137, 174, 183, 445]. Still less is known about the holographic implications of such systems. We extend this body of work by providing an explicit demonstration of the holographic principle in the form of a simple and tractable exact holographic mapping (EHM) for a free fermion system with and without a mass gap. Our program yields analytic relations between the RG flow parameters and their dual bulk quantities such as distances and curvature. This also allows us to study the consequences of a critical Fermi

surface (lying precisely at the quantum phase transition) on the nature of the holographic space.

1.6 A New Auxiliary Model Approach Towards Fermionic Criticality

To motivate the construction of a new auxiliary model approach in this thesis, we now write down some important outstanding questions that we will address using our approach.

1. *Can we construct an impurity-based auxiliary model approach that describes interaction-driven electronic phase transitions without imposing self-consistency?* As discussed in the preceding sections, one of the most prominent examples of auxiliary model approaches in the study of correlated fermionic models is dynamical mean-field theory (DMFT). However, solutions arising from DMFT and its extensions such as cellular DMFT and dynamical cluster approximation suffer from a lack of transparency owing to the numerical approach towards self-consistency. To go beyond such approaches, can we explicitly construct an impurity model that maps to the local physics of some interacting lattice model, and recover the low energy physics of the latter by solving the former? Apart from being much more transparent due to the explicit nature of the construction, such an approach would also be more tractable – because of the simplicity of impurity models – than studying the full lattice model.
2. *Can such an approach map the renormalisation group flows of an auxiliary impurity model to those of a correlated lattice model?* Both impurity and lattice models, due to the presence of many-particle interactions, typically involve a hierarchy of energy scales that are coupled to one another. The framework of renormalisation group (RG) is a powerful approach to identify the “important” physics at various energy scales. In particular, it can be used to identify low-energy effective Hamiltonians that describe the essential physics in the ground state and at low temperatures, and the microscopic Hamiltonian is said to “flow” into such low-energy fixed-point Hamiltonians. Can our auxiliary model approach map the renormalisation group flows between the impurity and lattice models, at least in terms of the local physics? This would make the underlying mechanism for the RG flows as well as the various fixed-point theories much more transparent.
3. *Does such an auxiliary model approach preserve symmetries of the microscopic/effective theories of the lattice model, and obey various conservation principles such as Luttinger’s theorem?* Importantly, an auxiliary model mapping for a lattice model should preserve the symmetries of the full Hamiltonian. It should also, ideally, show the emergence of conservation principles and spectral sum rules. Luttinger’s theorem, as an example of such a sum rule, equates the electron density in a system with the Fermi volume in the presence of interactions, as long as the excitations are electronic. Violations in Luttinger’s theorem often signal interaction-driven transitions from Fermi liquids to Mott insulators or non-Fermi liquids. Can the above auxiliary model approach capture changes in Luttinger’s theorem? An associated question is whether such an approach can show the emergence of Luttinger surfaces through the appearance of zeros in the one-particle Greens function. Such signatures would

have topological consequences because the Luttinger sum rule is protected by topological invariants.

4. *Is it possible to describe topological transitions using such an approach?* It is becoming increasingly clear that various materials exhibit phase transitions that are not necessarily tied to the spontaneous breaking of a local symmetry and fall outside the Ginzburg-Landau-Wilson paradigm. Examples include (i) the Mott transition driven by a charge gap but without necessarily a spin gap, (ii) the Kondo breakdown QCP scenario in heavy fermion materials where the large-to-small Fermi surface transition is observed to be separate from magnetic ordering and (iii) transition from a trivial insulator to a topological insulator driven by changes in band topology. Such transitions do not exhibit the appearance of non-zero local order parameters, and are instead often described by changes in topological invariants and entanglement patterns. One can therefore ask: is such an auxiliary model approach capable of tracking such generalised order parameters?
5. *Can such an auxiliary model approach capture the effects of multiple orbitals and momentum-space differentiation?* One of the primary difficulties with many of the numerical methods typically employed in such problems is the high computational complexity (in time as well as memory) and the inability to resolve momentum-space features to sufficient precision. The increased complexity becomes particularly important in several classes of problems such as multi-band models, which are pertinent to investigations into orbital-selective transitions, heavy fermion physics and twisted bilayer graphene, among many others. The lack of momentum-space resolution is relevant to studies of the pseudogap (PG) phase that is observed in the underdoped high- T_c superconductors; the PG is known to display partial gapping of the Fermi surface around the antinodes while remaining gapless around the nodes. The approach proposed here can benefit in this regard in two ways: the explicit construction might be helpful to avoid the computational cost of a self-consistency loop whilst providing access to a large number of momentum states, and (ii) the mapping to an impurity model makes the analysis much more tractable.

Within the thesis, we have developed a new approach towards studying correlated models with strong local interactions. The strategy goes as follows: We first identify a quantum impurity model that represents faithfully the local physics of the lattice model we are targeting. This is done by solving the impurity and studying its phenomenology. In order to capture specific aspects of the physics, the impurity model may need to be enhanced by, say, allowing for multiple impurities, or embedding it onto a lattice instead of a continuum, etc. Having identified the appropriate impurity model, we apply manybody translation operators to the impurity model Hamiltonian in order to restore translation invariance and map it to the target lattice Hamiltonian in the process.

For single-particle wavefunctions $\psi_i(r_i)$, the one-particle momentum operator p_i generates a one-particle translation operator $T_i(\mathbf{a})$ that translates the real-space operators associated with the particle i by the vector $\mathbf{a}$. By applying them to the dynamics of local potential wells at each site, these translation operators can be used to construct a translation-invariant Hamiltonians such as the tight-binding model; a similar application on the local wavefunctions generates delocalised translation-invariant Bloch states. Analogous to the single-particle case, one can construct a many-

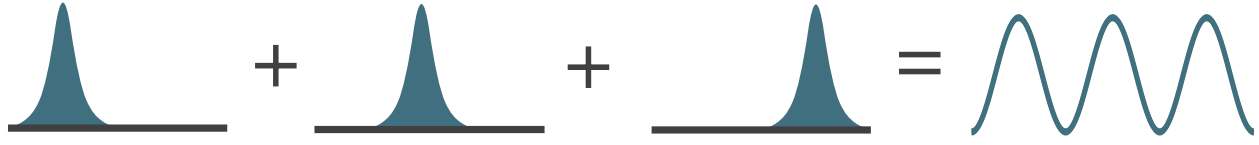


Figure 1.16: For the problem of a single electron, one can construct the full delocalised translation-invariant eigenstate by taking a superposition of the localised wavefunctions at each site. This is equivalent to repeatedly applying translation operations of varying distances on a single localised wavefunction associated with an arbitrarily chosen lattice site. A similar construction also works for the Hamiltonian, where by applying translation operators on the nearest-neighbour hybridisation processes of a single site, one can reconstruct the full tight-binding model.

particle translation operator T [446] that act on an N -particle state $\Psi(r_1, r_2, \dots, r_N)$: $T(\mathbf{a}) = e^{-i\mathbf{a} \cdot \mathbf{P}} = \prod_{i=1}^N T_i(\mathbf{a})$, where $\mathbf{P}$ is the total momentum operator. The action of T is to shift all local operators by $\mathbf{a}$ and to multiply all k -space operators with a phase shift $e^{i\mathbf{a} \cdot \mathbf{k}}$.

Using such a manybody translation operator, we are able to map both the Hamiltonian as well as eigenstates between the "local" auxiliary impurity model and the translation-invariant lattice model. This leads to specific relations between various observables such as correlations and spectral functions, as well as between entanglement measures. This set of mappings can be used to reconstruct the low-energy physics of the lattice model by starting from that of the auxiliary model.

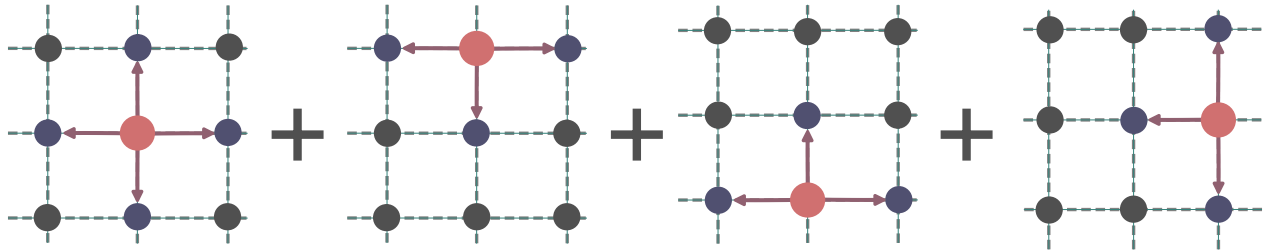


Figure 1.17: Schematic depiction of the "tiling" step in our auxiliary model approach. In order to restore translation invariance and map our impurity model to a lattice model, we apply manybody translation operators on our impurity model Hamiltonian for all lattice sites; this reconstructs the full lattice model, and leads to relations between the auxiliary model couplings and the lattice model couplings.

1.6.1 Important features of the "tiling" auxiliary model approach

Local transition as a diagnostic of lattice transition As mentioned in the preceding paragraphs, the approach developed within this thesis maps the low-energy physics of the lattice model to that of an impurity model with local correlation, hence rendering the problem much more tractable. This, for example, means that the presence of local Fermi liquid excitations on the impurity site marked by a broad central Kondo resonance in the local density of states corresponds to a Landau Fermi liquid phase for the lattice model. By inducing a gap in the impurity site density of states and

stabilising the local moment states, one can also model a Mott insulator on the lattice model. More exotic phases such as non-Fermi liquids can be engineered by enhancing the impurity model with competing interactions that lead to a pseudogap in the impurity density of states. Taken together, this maps the renormalised groups flows of the impurity model to that of the lattice model.

Restoring translation symmetry using a many-body Bloch's theorem At the heart of the impurity-lattice model mapping are manybody translation operators T . Indeed, they serve to restore translation invariance lost by the introduction of the impurity site into the lattice: $H_{\text{lat}} = \sum_i T^\dagger(\mathbf{r}_i) H_{\text{aux}} T(\mathbf{r}_i)$. In a similar fashion, the translation operators serve to "extend" the auxiliary model eigenstates into those of the lattice model: $\Psi = \sum_i T^\dagger(\mathbf{r}_i) \psi_{\text{aux}}$. This constitutes a manybody version of Bloch's theorem [447], where local impurity model states are superposed to form delocalised wavefunctions that span the lattice. This step plays the role of the self-consistent loop in DMFT by restoring the discrete translational invariance of the lattice model. This is also seen from the fact that translation operation leads to effective renormalised relations for the lattice model couplings in terms of the auxiliary model quantities. These translation operators also introduce non-local inter-auxiliary model contributions into various correlation functions.

Mapping across Greens functions and entanglement measures One of the main results of our tiling approach is the set of mappings between real-space and momentum-space Greens functions of the auxiliary model and the lattice model. This makes it much more tractable to characterise phases on the lattice model. The mappings take into account correlations within a single impurity model as well as contributions from inter-auxiliary model connections, through the translation operators. In the absence of long-range critical fluctuations, these expressions can be truncated to terms up to a few sites away from the impurity complex, and this makes the practical computation possible. Near the transition, it becomes necessary to work with much larger number of sites.

Lattice-embedding as a route to encoding lattice geometry and band topology Another important aspect of the tiling approach is the ability to incorporate lattice effects such as momentum-space differentiation and topological features. We accomplish this by considering auxiliary model directly on the appropriate lattice. This ensures that the Hamiltonian couplings of the auxiliary model inherit the symmetries of the lattice, and the auxiliary model quantities are then sensitive to lattice effects. This, for example, has allowed us to study a momentum-space anisotropic pseudogap phase in the context of an extended Hubbard model on a 2D square lattice.

1.7 Summary Of Chapters

For the readers' convenience, we now briefly describe the content and layout of the following chapters.

After the introduction, the *Methods and Preliminaries* chapter describes the various techniques and some preliminary ideas employed throughout the thesis. The chapter begins by deriving the unitary renormalisation group formalism and describes its various properties and a practical protocol for implementing it. That is followed by demonstrations of the unitary RG on three simple models

- a star graph model of spins in a magnetic field, the two-channel Kondo model, and the problem of a particle in a periodic potential. Through these problems, various aspects of the method are clarified. In order to provide more insights, the chapter also details how the unitary RG method is related to other RG-based methods that rely on canonical transformations, such as Poor Man's scaling, Schrieffer-Wolff transformation and CUT (continuous unitary transformation) RG. Apart from the URG, the chapter also describes an iterative diagonalisation method that has been used within this thesis to compute various correlation functions and entanglement measures from the hierarchy of Hamiltonians obtained from the URG. Since various entanglement measures have been used throughout the later chapters, we have also added a section on various entanglement measures here - what they mean, what are their inter-relations, why they are important and how one can compute some of them. The chapter concludes by describing some of the other strategies that have been employed to reduce computational costs in various projects.

In Chapter 3, *Kondo Frustration via Charge Fluctuations: A Route to Mott Localisation in Infinite Dimensions*, we study our first impurity model, the extended Anderson model with local correlation U_b in the conduction bath. We demonstrate that this model hosts a local version of the Mott metal-insulator transition on the Bethe lattice in infinite dimensions, as seen from DMFT. At a critical value of the ratio of U_b and the impurity-bath hybridisation, the effective impurity model shows a transition from a Kondo screened phase into an unscreened local moment phase. For the case of attractive local bath correlations, the model sheds new light on several aspects of the DMFT phase diagram. For example, the $T = 0$ metal-to-insulator quantum phase transition (QPT) is preceded by an excited state QPT (ESQPT) where the local moment eigenstates are emergent in the low-lying spectrum. Long-ranged fluctuations are observed near both the QPT and ESQPT, suggesting that they are the origin of the quantum critical scaling observed recently at high temperatures in DMFT simulations. The $T = 0$ gapless excitations at the quantum critical point display particle-hole interconversion processes, and exhibit power-law behaviour in self-energies and two-particle correlations. These are signatures of non-Fermi liquid behaviour that emerge from the partial breakdown of the Kondo screening. A many-body perturbation theoretic treatment of the Hubbard sidebands reveals that they are comprised of holon-doublon excitations created by the hybridisation of the impurity site with the conduction bath. These excitations consist of decoupled local Fermi liquids for the holons and doublons at the lowest order, which, at higher orders, become coupled via correlated holon-doublon scattering between impurity and bath.

In Chapter 4, *A New Auxiliary Model Approach for Interacting Electrons: Mapping Impurity Models To Lattice Models*, we lay out the details of the auxiliary model method developed to map impurity models to lattice models. We first motivate the general idea behind auxiliary model methods. This is followed by a construction of the manybody translation operators. We use this, along with a manybody Bloch's theorem, to construct lattice models from quantum impurity models and relate their eigenstates. This then allows us to map correlations functions, self-energies and entanglement measures across models. In particular, we demonstrate how the presence of the conduction bath introduces k -space resolution in various quantities.

In Chapter 5, *Mott Criticality as the Confinement Transition of a Pseudogap-Mott Metal: Insights on Mott Transition and Pseudogap in Two Dimensions*, we apply the auxiliary model tiling method to a lattice-embedded extended SIAM. Applying a many-body tiling scheme to the fixed-point impurity model uncovers a lattice model with electron interactions and Kondo physics. At

half-filling, the interplay between Kondo screening and bath charge fluctuations in the impurity model leads to Fermi liquid breakdown. This reveals a pseudogap phase characterized by a non-Fermi liquid (the Mott metal) residing on nodal arcs, gapped antinodal regions of the Fermi surface (Luttinger surfaces), and an anomalous scaling of the electronic scattering rate with frequency. The eventual confinement of holon-doublon excitations of this exotic metal obtains a continuous transition into the Mott insulator. Our results identify the pseudogap as a distinct long-range entangled quantum phase, and offer a new route to Mott criticality beyond the paradigm of local quantum criticality. The phase appears to encode multipartite entanglement, as seen from a large value of the quantum Fisher information. The critical point at the pseudogap-Mott insulator boundary is found to be described by the exactly-solvable Hatsugai-Kohmoto model. The emergence of Luttinger surfaces alters the anomaly structure associated with the generalized symmetry of the FS [44–46]. The corresponding topological charge is defined via the Luttinger–Ward functional. This charge is modified by Green’s function zeros, signalling a shift in the underlying anomaly. Within our framework, this reorganization grants topological protection to the RG flows terminating in the PG regime, ensuring the stability of its NFL excitations.

In Chapter 6, *Competing tendencies In Heavy Fermion Systems: Kondo screening vs. Mott localisation*, we study a periodic Anderson model where we treat both the c - and f -layers as extended Hubbard models, and analyse it using the tiling approach developed and implemented in the earlier chapters. We first study the case of half-filling and obtain a Kondo insulator at large values of the inter-band hybridisation. We also study the more interesting case of a Mott metal in the f -layer hybridising with the c -electrons.

In Chapter 7, *Holography of Entanglement in 2D Free Fermions: Emergent Geometry and Fermionic Criticality*, we provide an explicit demonstration of the holographic principle in the form of a simple and tractable exact holographic mapping (EHM) for a free fermion system with and without a mass gap. Our program yields analytic relations between the RG flow parameters and their dual bulk quantities such as distances and curvature. This also allows us to study the consequences of a critical Fermi surface (lying precisely at the quantum phase transition) on the nature of the holographic space.

We conclude in the final chapter by summarising the main results obtained within the thesis, in terms of the methods developed as well as insights obtained into the behaviour of various models. We describe the impact of these results on the field, the broad conclusions emerging from them and some open questions resulting from the work.

Chapter 2

Kondo Frustration via Charge Fluctuations: A Route to Mott Localisation in Infinite Dimensions

2.1 Introduction

The rich physics of metal-insulator transitions in strongly correlated systems has been an active subject of study for quite some time [2–4], yet much still remains to be understood. It involves diverse aspects such as spin and charge fluctuations, quasiparticle renormalisation effects, anomalous metallic phases and unconventional superconductivity, and has been studied using an equally diverse array of methods like mean-field theory, renormalisation group approaches, numerical techniques like exact diagonalisation, quantum Monte Carlo and dynamical mean-field theory, and many others. In particular, dynamical mean-field theory (DMFT) [18, 31, 196, 448–453] obtains an exact solution of the Mott metal-insulator transition (MIT) for the 1/2-filled Hubbard model [1, 11, 16, 17, 19] on the Bethe lattice with infinite coordination number, in terms of an Anderson impurity model with a self-consistently determined bath obtained by requiring, in an iterative manner, that its local Greens function be equal to that of the impurity site. The above-mentioned transition can be captured by the local spectral function through (i) the continuous appearance of a Mott gap, followed by (ii) the sharpening and vanishing of the central Kondo resonance. These two features are often referred to, respectively, as the Mott-Hubbard [1] and Brinkman-Rice [19] scenarios of the Mott MIT. The exact nature of the solution arises from the fact that all non-local contributions to the lattice self-energy are observed to vanish upon taking the limit of an infinite coordination number for the lattice model. Thus, the simplification is that the dynamics of any local site on the lattice is determined completely by a quantum impurity problem [18, 161]. It must also be noted that this exact solution precludes any long-range order in the system, and corresponds to the case of a maximally frustrated Hubbard model involving long-range and frustrating inter-site hopping such that both the metallic and insulating phases remain paramagnetic [454]. Due to the exact and non-perturbative nature of the DMFT solution for the MIT in $d = \infty$, the method has been extended to models of strongly correlated electrons in finite spatial dimensions [451, 455, 456], as well as the study of the electronic

properties of various correlated materials [452, 457, 458].

Despite this progress, a key aspect of the DMFT solution for the Hubbard model in $d = \infty$ remains to be understood. During the search for a self-consistent impurity model, the conduction bath is modified drastically in order to become correlated [459]. The numerical implementation of self-consistency, however, precludes a deeper understanding of the precise nature of the correlations present in the conduction bath of the impurity model, and its implications for the electron dynamics of the associated bulk lattice (Hubbard) model. Indeed, as we demonstrate below, our work paves the way for a more insightful exploration of the physics of the Mott Hubbard transition in strongly correlated electronic systems. Further, it has considerable consequences for the development of functional quantum materials and next-generation quantum technologies. Below, we lay out the specific questions addressed by us, and summarise our results at the end of this section.

- i. Is there a minimal but effective quantum impurity model Hamiltonian that describes the Mott MIT of the $1/2$ -filled Hubbard model on the Bethe lattice in $d = \infty$? Finding such an impurity model would also reduce the considerable computational effort that is presently required in self-consistent approaches.
- ii. What are the fluctuations that destroy the metal and lead to the insulating phase? Can we obtain a universal theory for these competing tendencies?
- iii. The coexistence of metallic and insulating phases at $T = 0$ within DMFT shows that the insulating solution is present within the many-body spectrum of the metallic phase. Can an associated impurity model Hamiltonian display the emergence of the insulating state prior to the transition? Analytic insight of this kind is particularly important because approaching a coexistence region numerically is often fairly tricky.
- iv. Is it possible to obtain a low-energy theory for the local gapless excitations precisely at the MIT, where the metal is on the brink of destruction? How do these excitations compare with those of the local Fermi liquid, e.g., in terms of self-energies and two-particle correlation functions?

Obtaining answers to these questions would go a long way towards understanding the dominant fluctuations at the heart of the Mott transition and the physical processes that become important as one approaches it. It would also help in understanding the mechanism of DMFT's search for the self-consistent impurity model within the space of Anderson impurity models.

The essence of our approach is to model phenomenologically the lattice self-energy obtained from DMFT in the form of additional bath correlations within an extended Anderson impurity model. In addition to the usual on-site repulsion (U) and single-particle hybridisation (V) between the impurity and the conduction bath of the Anderson impurity model (eq. (2.1)), we introduce (i) an additional on-site correlation (U_b) on the bath site with which the impurity couples, and (ii) an antiferromagnetic Kondo coupling (J) between the impurity and the conduction bath (eq. (2.2)). We note that a similar impurity model-based approach was taken towards understanding the physics of the heavy fermions several years ago by Si and Kotliar [460, 461]. We postpone a comparison of our work with theirs to the discussions section. The rest of the work is structured as follows. Sec. (2.2.1) describes the extended model that we will study, and the unitary renormalisation group

(URG) method that we employ to study it is presented in Sec. (2.2.2). In Sec. (2.3) and (2.4), we describe the phase diagram and various characteristics of the impurity phase transition. In Sec. (2.6), we use our extended model to explain various features of the coexistence region observed in DMFT. In Sec. (2.7), we describe the effect of the impurity on the low-lying excitations of the bath, near and at the transition. We conclude in Sec. (2.8) with some discussions and possible future directions. For the convenience of readers, we first present below a brief summary of our main results.

Summary of our main results

- *Presence of a local metal-insulator transition:* At a critical value of the parameter $r = -U_b/J$, the effective impurity model shows a transition from a Kondo screened phase into an un-screened local moment phase. The quantum critical point (QCP) involves a degeneration of the Kondo singlet and the local moment states.
- *The physics of Kondo screening and local pairing drives the transition:* The transition involves the frustration of the Kondo screening of the impurity by enhanced local pairing fluctuations in the bath and can be described by a universal theory written in terms of J and U_b .
- *Emergence of insulating solutions in the metallic phase:* Our analysis reveals that at a certain value of the parameter r prior to the transition, the single-particle hybridisation parameter (V) turns irrelevant (in the RG sense), and this leads to the emergence of the local moment solutions within the many-particle spectrum through an excited state quantum phase transition (ESQPT).
- *Critical fluctuations and the coexistence region:* We observe the appearance of long-ranged quantum fluctuations extending into the conduction bath in the vicinity of both the ESQPT at $r = r_{c1}$ and QPT at $r = r_{c2}$. We believe that these are the likely origin of the critical fluctuations observed above the finite temperature second-order critical point in DMFT [454, 462]. The two-step process at $T = 0$ also provides a natural explanation for the coexistence of metallic and insulating features in the phase diagram, in the regime $r_{c1} < r < r_{c2}$.
- *Emergence of non-Fermi liquid excitations at the QCP:* Precisely at the QCP, the local Fermi liquid is replaced by a quasi-local non-Fermi liquid (NFL) that spans the impurity, zeroth and first sites of the conduction bath. The NFL results from a degeneracy between the local moment and singlet states and leads to (i) “Andreev scattering” of incoming states into orthogonal outgoing states, (ii) anomalous power-law behaviour in the self-energies and two-particle correlations with universal exponents, and (iii) a fractional entanglement entropy of the impurity.
- *Correlated Fermi liquid excitations in the Hubbard sidebands:* A many-body perturbation theoretic treatment of the Hubbard sidebands reveals that they are comprised of the holon-doublon excitations created by the hybridisation of the impurity site with the conduction bath. These excitations consist of decoupled local Fermi liquids for the holons and doublons at the

lowest order, which, at higher orders, become coupled via correlated holon-doublon scattering between impurity and bath.

2.2 Model Hamiltonian and method

2.2.1 The extended Anderson impurity model

The single-impurity Anderson model (SIAM) [463, 464] consists of a single impurity site with local repulsive correlation U hybridising with a non-interacting fermionic conduction bath through a (momentum-independent) single-particle transfer whose coupling is V . For the case of a half-filled impurity site, the Hamiltonian of the SIAM is given by

$$\mathcal{H}_A = -\frac{U}{2} (\hat{n}_{d\uparrow} - \hat{n}_{d\downarrow})^2 + \sum_{\vec{k}, \sigma} \epsilon_{\vec{k}} \tau_{\vec{k}, \sigma} + V \sum_{\sigma} (c_{d\sigma}^\dagger c_{0\sigma} + \text{h.c.}), \quad (2.1)$$

where $\tau_{\vec{k}, \sigma} \equiv c_{\vec{k}, \sigma}^\dagger c_{\vec{k}, \sigma} - 1/2$ indicate the occupancy of the single-particle momentum state $|\vec{k}\rangle$. Also, $c_{d\sigma}$ and $c_{0\sigma} = \sum_k c_{k\sigma}$ are the fermionic annihilation operators of spin σ for the impurity and conduction bath site to which it couples (henceforth referred to as the *zeroth site*) respectively. The conduction bath is typically considered to possess a constant (i.e., energy-independent) density of states.

The SIAM (along with its $U \rightarrow \infty$ limit, the Kondo model) has been studied using several analytical and numerical techniques [36, 38, 39, 42, 444, 465–475]. The general conclusion for the positive U case at $T = 0$ is that on the particle-hole symmetric (that is, half-filled) line, the impurity local moment is always screened by the conduction electrons (referred to as the Kondo cloud [476–487]). Enhanced spin-flip scattering at low-energies leads to the formation of a macroscopic singlet ground state and local Fermi liquid gapless excitations [488, 489]. In order to enhance the SIAM, we introduce two extra two-particle interaction terms into the Hamiltonian:

- a spin-exchange term $J \vec{S}_d \cdot \vec{S}_0$ between the impurity spin $\vec{S}_d$ and the spin $\vec{S}_0$ of the zeroth site, and
- a local particle-hole symmetric correlation term $-U_b (\hat{n}_{0\uparrow} - \hat{n}_{0\downarrow})^2$ on the bath zeroth site.

With these additional terms, the Hamiltonian of the *extended single-impurity Anderson model* (henceforth referred to as the e-SIAM) is, at particle-hole symmetry, given by

$$\mathcal{H}_{\text{E-A}} = \mathcal{H}_A + J \vec{S}_d \cdot \vec{S}_0 - \frac{1}{2} U_b (\hat{n}_{0\uparrow} - \hat{n}_{0\downarrow})^2. \quad (2.2)$$

All the terms in the Hamiltonian have been depicted schematically in the left panel of Fig. (2.1). The additional interaction terms J and U_b enjoy particle-hole, SU(2)-spin and U(1)-charge symmetries. The e-SIAM Hamiltonian (eq. (2.2)) therefore preserves all the local symmetries of the half-filled Hubbard model on a lattice, and can potentially serve as an effective auxiliary quantum impurity model (with a correlated bath) describing the local physics of the latter.

2.2.2 The unitary renormalisation group method

In order to obtain the various low-energy phases of the e-SIAM, we perform a scaling analysis of the associated Hamiltonian (eq. (2.2)) using the recently developed unitary renormalisation group (URG) method [346, 347]. The method has been applied successfully on a wide variety of problems of correlated fermions [42, 74, 174, 183, 346, 347, 490–492]. The method proceeds by resolving quantum fluctuations in high-energy degrees of freedom, leading to a low-energy Hamiltonian with renormalised couplings and new emergent degrees of freedom. Typically, for a system with Fermi energy ϵ_F and bandwidth D_0 , the sequence of isoenergetic shells $\{D_{(j)}\}$, $D_{(j)} \in [\epsilon_F, D_0]$ define the states whose quantum fluctuations we sequentially resolve. The momentum states lying on shells $D_{(j)}$ that are far away from the Fermi surface comprise the UV states, while those on shells near the Fermi surface comprise the IR states. This scheme is shown in the right panel of Fig. (2.1).

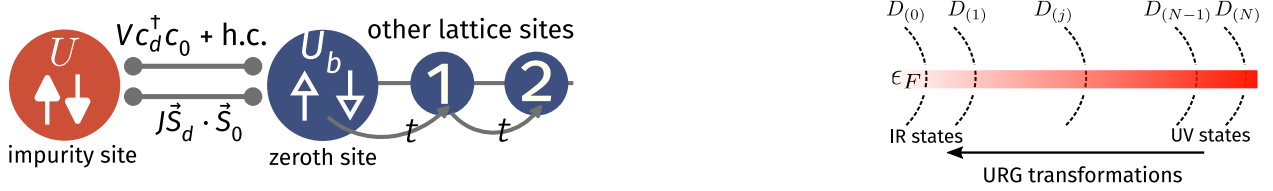


Figure 2.1: Left: Schematic 1D representation of the extended SIAM Hamiltonian. The red sphere is the impurity site with on-site Hubbard term U . It is connected via two couplings V and J to the bath zeroth site (large blue sphere) that has its own on-site Hubbard term U_b . The small blue spheres make up the rest of the bath, connected through the single-particle hopping t . Right: High energy - low energy scheme defined and used in the URG method. The states away from the Fermi surface form the UV subspace and are decoupled first, leading to a Hamiltonian which is more block-diagonal and comprised of only the IR states near the Fermi surface.

As a result of the URG transformations, the Hamiltonian $H_{(j)}$ at a given RG step j involves scattering processes between the k -states that have energies lower than $D_{(j+1)}$. The unitary transformation $U_{(j)}$ is then defined so as to remove the number fluctuations of the currently most energetic set of states $D_{(j)}$ [346, 347]:

$$H_{(j-1)} = U_{(j)} H_{(j)} U_{(j)}^\dagger, \text{ such that } [H_{(j-1)}, \hat{n}_j] = 0. \quad (2.3)$$

The eigenvalue of $\hat{n}_j$ has, thus, been rendered an integral of motion (IOM) under the RG transformation.

The unitary transformations can be expressed in terms of a generator $\eta_{(j)}$ that has fermionic algebra [346, 347]:

$$U_{(j)} = \frac{1}{\sqrt{2}} \left(1 + \eta_{(j)} - \eta_{(j)}^\dagger \right), \quad \left\{ \eta_{(j)}, \eta_{(j)}^\dagger \right\} = 1, \quad (2.4)$$

where $\{\cdot\}$ is the anticommutator. The unitary operator $U_{(j)}$ that appears in Eq. (2.4) can be cast into the well-known general form $U = e^S$, $S = \frac{\pi}{4} \left(\eta_{(j)}^\dagger - \eta_{(j)} \right)$ that a unitary operator can take, defined

by an anti-Hermitian operator $\mathcal{S}$. The generator $\eta_{(j)}$ is given by the expression [346, 347]

$$\eta_{(j)}^\dagger = \frac{1}{\hat{\omega}_{(j)} - \text{Tr}(H_{(j)}\hat{n}_j)} c_j^\dagger \text{Tr}(H_{(j)}c_j) . \quad (2.5)$$

The operators $\eta_{(j)}, \eta_{(j)}^\dagger$ behave as the many-particle analogues of the single-particle field operators $c_j, c_j^\dagger$ - they change the occupation number of the single-particle Fock space $|n_j\rangle$. The important operator $\hat{\omega}_{(j)}$ originates from the quantum fluctuations that exist in the problem because of the non-commutation of the kinetic energy terms and the interaction terms in the Hamiltonian:

$$\hat{\omega}_{(j)} = H_{(j-1)} - H_{(j)}^i . \quad (2.6)$$

$H_{(j)}^i$ is the part of $H_{(j)}$ that commutes with $\hat{n}_j$ but does *not* commute with at least one $\hat{n}_l$ for $l < j$. The RG flow continues up to energy D^* , where a fixed point is reached from the vanishing of the RG function. Detailed comparisons of the URG with other methods (e.g., the functional RG, spectrum bifurcation RG etc.) can be found in Refs. [174, 346]. More information on the unitary RG method is provided in Section 1 of the Supplementary Materials [493].

2.3 *T = 0 scaling theory for the e-SIAM*

2.3.1 URG equations for the e-SIAM

The derivation of the RG equations for the extended Anderson impurity model (e-SIAM) Hamiltonian is shown in Section 2 of the Supplementary Materials [493]. The bath coupling U_b is marginal. We provide below the RG equations of the remaining couplings for a given quantum fluctuation scale ω :

$$\begin{aligned} \Delta U &= 4V^2 n_j \left(\frac{1}{d_1} - \frac{1}{d_0} \right) - n_j \frac{J^2}{d_2}, \\ \Delta V &= -\frac{3n_j V}{8} \left[J \left(\frac{1}{d_2} + \frac{1}{d_1} \right) + \frac{4U_b}{3} \sum_{i=1}^4 \frac{1}{d_i} \right], \\ \Delta J &= -\frac{n_j J (J + 4U_b)}{d_2}, \end{aligned} \quad (3.1)$$

where the denominators d_i are given by

$$d_0 = \omega - \frac{D}{2} + \frac{U_b}{2} - \frac{U}{2}, \quad d_1 = \omega - \frac{D}{2} + \frac{U_b}{2} + \frac{U}{2} + \frac{J}{4}, \quad (3.2)$$

$$d_2 = \omega - \frac{D}{2} + \frac{U_b}{2} + \frac{J}{4}, \quad d_3 = \omega - \frac{D}{2} + \frac{U_b}{2}. \quad (3.3)$$

The symbols used in the RG equations have the following meanings: ΔU represents the renormalisation of the coupling U in going from the j^{th} Hamiltonian to the $(j-1)^{\text{th}}$ Hamiltonian by

decoupling the isoenergetic shell at energy $D_{(j)}$ (see right panel of Fig.(2.1)). n_j is the number of electronic states on the shell $D_{(j)}$. We note that the labels U_0, J_0, V_0 that appear in various figures (and elsewhere in the text) represent the bare values of the associated couplings U, J and V . The RG equations reduce, in the perturbative regime of the couplings U, V and J , to the well-known “poor man’s” scaling forms obtained for the SIAM [468] and the single-channel Kondo model [467] respectively.

The RG fixed point Hamiltonian describes the low-energy phase of the system. In general, if the RG fixed point is reached at an energy scale D^* , the fixed point Hamiltonian $\mathcal{H}^*$ is obtained simply from the fixed point values of the couplings (and by noting that the states above D^* are now part of the IOMs):

$$\mathcal{H}^* = \sum_{\sigma, \vec{k}}^{|e_{\vec{k}}| < D^*} \epsilon_{\vec{k}} \tau_{\vec{k}, \sigma} + V^* \sum_{\sigma} \left(c_{d\sigma}^\dagger c_{0\sigma} + \text{h.c.} \right) + J^* \vec{S}_d \cdot \vec{S}_0 - \frac{1}{2} U^* (\hat{n}_{d\uparrow} - \hat{n}_{d\downarrow})^2 - \frac{1}{2} U_b (\hat{n}_{0\uparrow} - \hat{n}_{0\downarrow})^2 .$$

The fixed point values of the couplings are obtained by solving the RG equations numerically.

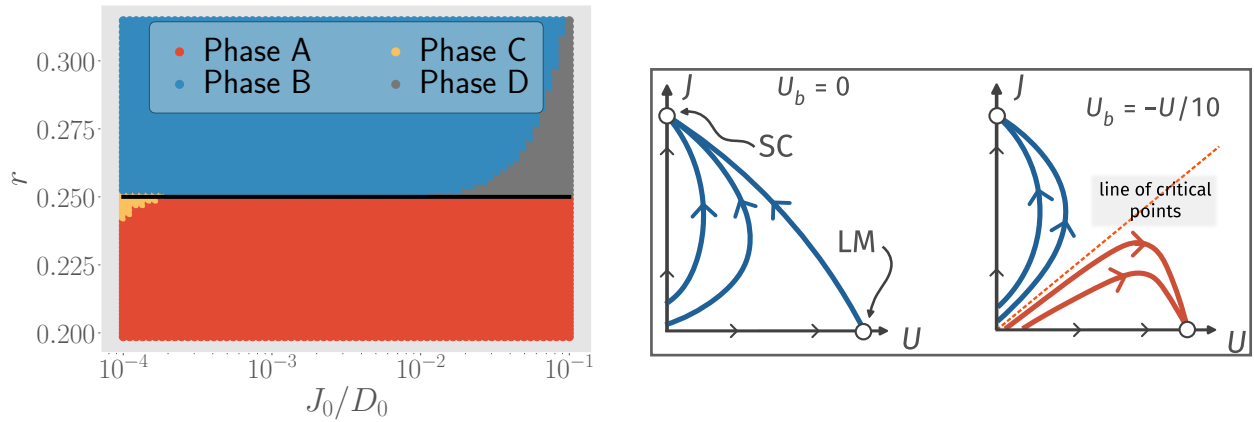


Figure 2.2: Left: Phase diagram of the e -SIAM, in the space of r and J_0/D_0 . U_b is set to $-U_0/10$. The various phases are described in the text. Right: Schematic nature of RG flows for $U_b = 0$ and $U_b = -U/10$. Blue curves represent RG flows towards the strong-coupling (SC) fixed point, while red curves represent RG flows towards the local moment (LM) fixed point.

2.3.2 Phase diagram

We will work in the low-energy regime where the quantum fluctuation scale ω is such that all the denominators are negative: $d_i < 0 \forall i$. Moreover, we constrain the impurity and local bath correlations (U and U_b respectively) through the relation $U_b = -U/10$. While the precise value of the factor of $1/10$ is unimportant, we have chosen a factor considerably smaller than 1 in order to demonstrate that an interesting body of results can be obtained with a value of $|U_b|$ that is much smaller than that of U . Further, the negative sign in the above relation is significant, as we shall see below that the MIT is obtained for the case of U_b being negative (i.e., attractive on-site correlations).

on the zeroth site of the bath). The relation between U_b and U is motivated on phenomenological grounds such that the MIT occurring in the bulk lattice model (obtained upon increasing the on-site Hubbard repulsion to large values) corresponds, in the auxiliary model mapping within DMFT, to the local MIT observed upon tuning the impurity correlation U (and the related bath correlation U_b) within the proposed effective impurity model Hamiltonian. The physical significance of attractive on-site correlations in the bath lies in providing a mechanism for the frustration of Kondo screening (in the RG equation for an antiferromagnetic Kondo coupling $J(> 0)$, eq. (9.36)) within an Anderson impurity coupled to a single channel of conduction electrons. We provide a detailed discussion of this point in the concluding section of our work.

Regime	Low-energy effective Hamiltonian	impurity ground-state
1. $0 < r < r_{c1}$	$\text{K.E}^* + V^* \sum_{\sigma} \left(c_{d\sigma}^{\dagger} c_{0\sigma} + \text{h.c.} \right) + J^* \vec{S}_d \cdot \vec{S}_0$	$\frac{1}{\sqrt{2}} (\uparrow_d\rangle \downarrow_0\rangle - \downarrow_d\rangle \uparrow_0\rangle + 2_d\rangle 0_0\rangle + 0_d\rangle 2_0\rangle)$
2. $r_{c1} < r < r_{c2}$	$\text{K.E}^* + J^* \vec{S}_d \cdot \vec{S}_0 - \frac{1}{2} U_b (\hat{n}_{0\uparrow} - \hat{n}_{0\downarrow})^2$	$\frac{1}{\sqrt{2}} (\uparrow_d\rangle \downarrow_0\rangle - \downarrow_d\rangle \uparrow_0\rangle)$
3. $r_{c2} < r$	$\text{K.E}^* - \frac{U^*}{2} (\hat{n}_{d\uparrow} - \hat{n}_{d\downarrow})^2 - \frac{U_b}{2} (\hat{n}_{0\uparrow} - \hat{n}_{0\downarrow})^2$	$\{ \uparrow_d\rangle, \downarrow_d\rangle\}$

Table 2.1: Effective Hamiltonians and ground-states of the three important parts of the phase diagram in Fig. (2.2). The ground-state is simplified to capture only the configuration of the impurity site and at most the bath zeroth site. K.E^* represents the kinetic energy of the conduction electron states residing within the fixed point window D^* .

The phase diagram is shown in the left panel of Fig. (2.2) in terms of the parameter $r = |U_b|/J$ (y-axis) and the ratio of the bare Kondo coupling (J_0) to the bare conduction bath bandwidth (D_0) (x-axis), we first define two important points in the space of couplings. These are values of the parameter r where there is a qualitative change in the nature of RG flows, and hence in the low-energy physics of the model

- $r = r_{c1} \left(= - \left(\frac{U_b}{J} \right)_{c1} = \frac{3}{20} = \left(\frac{U}{10J} \right)_{c1} > 0 \right)$: At this point, the coupling V becomes irrelevant,
- $r = r_{c2} \left(= - \left(\frac{U_b}{J} \right)_{c2} = \frac{1}{4} = \left(\frac{U}{10J} \right)_{c2} \right)$: At this point, the coupling J also turns irrelevant. Note that $r_{c2} > r_{c1}$.

We will use these two values of the parameter r as checkpoints around which we can describe the low-energy physics. There are three important parts in Fig. (2.2):

- red region, $0 < r < r_{c1}$: the $J - V$ model; V and J are both relevant, but U is irrelevant; spin and charge delocalisation on the impurity; spin-charge mixing in ground-state
- blue region, $r_{c1} < r < r_{c2}$: the $J - U_b$ model; J is relevant, but V and U are both irrelevant; charge localisation and spin delocalisation on the impurity; singlet ground-state
- violet region, $r_{c2} < r$: the $U - U_b$ model; U is relevant, but V and J are both irrelevant; spin and charge localisation on the impurity; local moment ground-state. This phase describes a

local Mott insulator on the impurity site, characterised by vanishing local double occupancy in ground state (black curve in left panel of Fig. 2.4).

Typical RG flows that lead to these phases are shown in Fig. (2.3). We also note that the grey region shown in the top right corner of Fig.(2.2) corresponds to a model in which all three couplings (J , U and V) are RG irrelevant. We find, however, that this phase is an artefact of solving the RG equations for an impurity coupled to a finite-sized conduction bath, and it gradually disappears upon increasing the bath size.

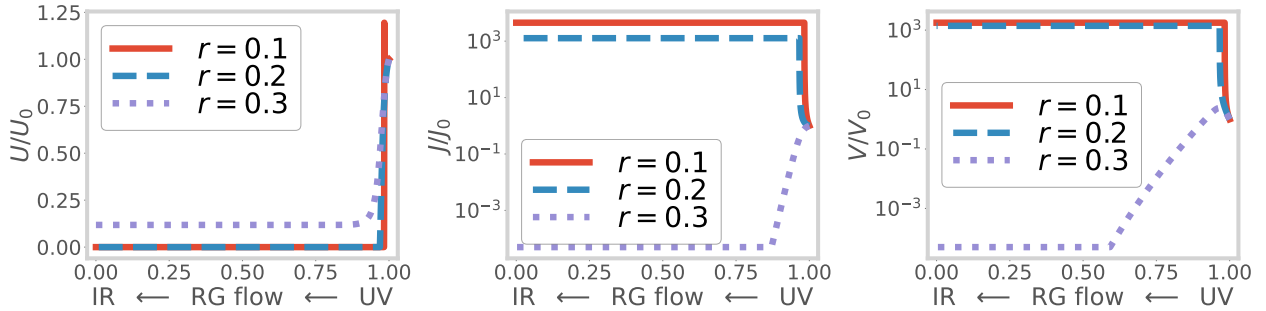


Figure 2.3: Variation of couplings U , V and J along the RG transformations, for three values of the transition-tuning ratio $r = -U_b/J_0$. The x-axis represents the distance of the running cutoff from the Fermi surface; the rightmost point is the first RG step (UV) and the leftmost point is the final RG step (IR). The red curves represent the flows for $r < r_{c1}$, where both V and J are relevant. The blue curves represent the RG flows for $r_{c1} < r < r_{c2}$, where J is relevant but V is irrelevant. The violet curves represent RG flows for $r > r_{c2}$, where both V and J are irrelevant but U flows to a finite value.

2.3.3 Phase transition at r_{c2}

The effective Hamiltonians and corresponding impurity ground-states for various phases have been listed in table (2.1). The variation in the ground state has been checked by numerically solving the fixed point Hamiltonian for various bare values of the couplings and is shown in the left panel of Fig. (2.4). A sharp change in the ground state from spin-singlet to local moment shows that the blue and violet phases are separated by an *impurity delocalisation-localisation transition* (black line in Fig. (2.2)). The transition occurs at finite values of the correlations: $r_{c2} = -\left(\frac{U_b}{J}\right)_{c2} = \frac{1}{4} = \left(\frac{U}{10J}\right)_{c2}$. This is a stark contrast from the standard SIAM, where the transition can happen only at on-site correlation $U \rightarrow \infty$. Therefore, the presence of the critical point at finite values of the various couplings transforms the landscape of RG phase diagram shown schematically in the right panel of Fig. (2.2) (right panel): the RG flows split into two classes - those that flow towards the strong-coupling Kondo screened fixed point and those that flow towards the local moment fixed point.

By a simple rewriting of the RG equation for J (eq. (9.36)), the impurity transition can be seen to arise from a competition between the Kondo screening physics of J and the local pairing physics

of U_b :

$$\Delta J = \frac{\overbrace{(J + 2U_b)^2 n_j}^{\text{usual Kondo physics}}}{|d_2|} - \frac{\overbrace{(2U_b)^2 n_j}^{\text{competing pairing physics}}}{|d_2|}, \quad (3.4)$$

where $d_2 = \omega - D/2 + U_b/2 + J/4$. The competition between the effective Kondo term $(J + 2U_b)^2$ and the competing pairing term $-4U_b^2$ leads to the presence of two stable phases - one that is Kondo screened and one that remains unscreened.

In the DMFT treatment of the 1/2-filled Hubbard model on the Bethe lattice in infinite dimensions, the vanishing of non-local contributions to the lattice self-energy means that the lattice Greens function can be computed self-consistently by solving a local quantum impurity problem [18]. In the rest of the work, we provide extensive evidence that the local transition observed in the e-SIAM is very similar to the Mott MIT observed in DMFT. We conclude thereby that the e-SIAM models faithfully the round-trip excursions of an electron on the Bethe lattice, and that the impurity phase transition observed in the e-SIAM offers a local description of the Mott MIT in the bulk Hubbard model. With this evidence in mind, we will henceforth refer to the transition at r_{c2} as a *local metal-insulator transition*.

2.3.4 The case of repulsive U_b

Henceforth, we will only consider the case of attractive U_b ($U_b > 0$), because that is the regime in which an impurity phase transition is realised at finite values of the couplings. However, before moving on, we will comment briefly on the effects of a repulsive U_b ($U_b > 0$). From the RG equation for J in eq. 9.36, we see that ΔJ is always positive for $U_b > 0$. Moreover, something similar happens in the RG equation for V , where the two terms in the square brackets now enforce each other (J and U_b now have the same sign). Together, they indicate that there will not be any Kondo destruction in the case of $U_b > 0$, and the only fixed points are the weak-coupling one ($J^* = 0$) and the strong coupling one ($J^* = \infty$); there is no local moment phase. This case is therefore adiabatically connected to the case of $U_b = 0$.

2.4 Descriptors of the local MIT

The present section gives more clarity on the nature of the impurity phase transition in the form of additional descriptors of the transition such as impurity spectral function and measures of entanglement, computed from the e-SIAM Hamiltonian. Throughout the rest of the work, we have set the value of the coupling V equal to the Kondo coupling J while generating the quantitative plots.

2.4.1 Evolution of the impurity spectral function

The impurity spectral function of the standard SIAM (eq. (2.1)) always displays a central peak at finite values of U (along with Hubbard sidebands at sufficiently large U), indicating the presence of

gapless local Fermi liquid excitations on the impurity site [40,44,444]. To demonstrate the impurity localisation transition, we compute the impurity local spectral function of the e-SIAM (eq. (2.2)). This involves numerically diagonalising the effective Hamiltonian at various energy scales along the RG flow and computing the spectral weight at a range of frequencies from UV to IR. We find that the spectral function (shown in the right panel of Fig. (2.4)) displays three notable features, in agreement with DMFT results [18]:

- The appearance of a *preformed gap* (flattening of spectral function between the central peak and sidebands) as r crosses r_{c1} ($= -\left(\frac{U_b}{J}\right)_{c1} = \frac{3}{20} = \left(\frac{U}{10J}\right)_{c1}$): The preformed optical gap simply indicates the separation of the spin (central peak) and charge (sidebands) degrees of freedom beyond r_{c1} . This separation is brought about by the irrelevance of V . We note that the consequences of an irrelevant hybridisation V can be accounted for in many-body perturbation theory, leading to “in-gap” processes within the optical gap of the impurity spectral function [153]. However, we expect that such contributions will be small, and cannot fill in the optical gap observed in Fig.2.4 completely.
- The sharpening of the central peak, and concomitant increase in the preformed gap, as r is increased in the range $r_{c1} < r < r_{c2}$.
- The vanishing of the central peak and appearance of a *hard gap* as r crosses r_{c2} : The irrelevance of J beyond r_{c2} destroys the Kondo screening and localises the impurity moment. This involves the destabilisation of the singlet ground state, and the stabilisation of the local moment states in its place.

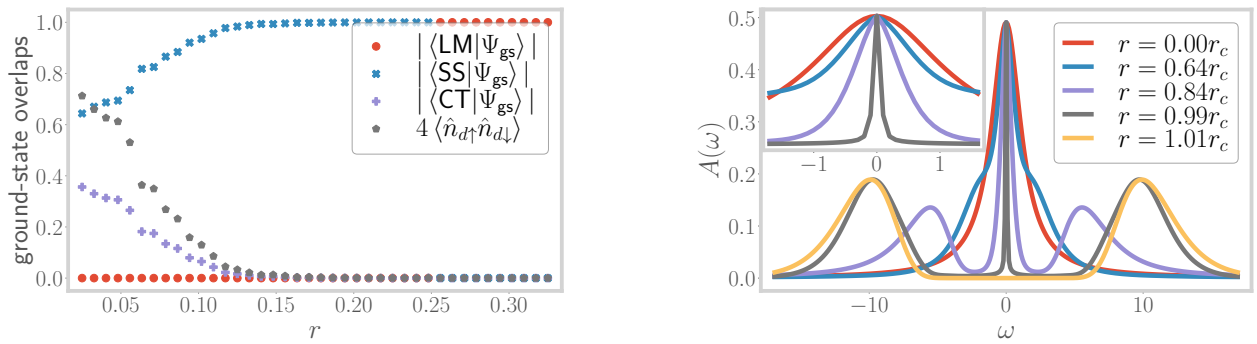


Figure 2.4: Left: Overlap of the RG fixed point ground state $|\Psi\rangle_{gs}$ with the spin-singlet state $|SS\rangle$, the zero charge member of the charge triplet states $|CT\rangle$ and the local moment states $|LM\rangle$, for all three regimes of the model. The black curve corresponds to the double occupancy on the impurity site and is observed to be strongly suppressed beyond r_{c1} . Beyond the critical point $r_{c2} = 0.25$, the overlaps with the entangled states vanish, indicating the transition to a decoupled local moment. Right: Variation of the impurity spectral function from $r = 0$ to $r > r_{c2}$. At small r , the central peak is broad, but at larger r , it sharpens, and the difference in spectral weight is used in creating the Hubbard sidebands. For $r > r_{c2}$, the central peak vanishes.

These features are also reflected in ground-state correlation measures like spin-flip and charge isospin-flip correlations (see left panel of Fig. (2.5)), which are defined as follows:

$$\frac{1}{2} \left(\langle S_i^+ S_j^- \rangle + \text{h.c.} \right) = \frac{1}{2} \left(\langle c_{i\uparrow}^\dagger c_{i\downarrow} c_{j\downarrow}^\dagger c_{j\uparrow} \rangle + \text{h.c.} \right), \quad \frac{1}{2} \left(\langle C_i^+ C_j^- \rangle + \text{h.c.} \right) = \frac{1}{2} \left(\langle c_{i\uparrow}^\dagger c_{i\downarrow}^\dagger c_{j\downarrow} c_{j\uparrow} \rangle + \text{h.c.} \right). \quad (4.1)$$

At small values of r , both the correlations are large as the ground-state has both spin and charge content (row 1 of table (2.1)). The subsequent decrease in the impurity-bath charge flip correlation (violet points of Fig. (2.5)), and the simultaneous increase in the impurity-bath spin-flip correlation (red points of Fig. (2.5)), can be attributed to the irrelevance of V at $r \simeq r_{c1}$ and the appearance of the preformed gap in the spectral function. Beyond r_{c2} , the spin-flip correlation sharply drops to zero due to the irrelevance of J . They are replaced by intra-bath correlations like the charge isospin-flip correlation between the zeroth site and the first site (blue points of Fig. (2.5)). Such intra-bath correlations are promoted by the bath on-site term U_b . The sudden change in the nature of the ground state, and resulting correlations, at r_{c2} indicates the presence of a quantum critical point whose nature will be analysed further in subsequent sections.

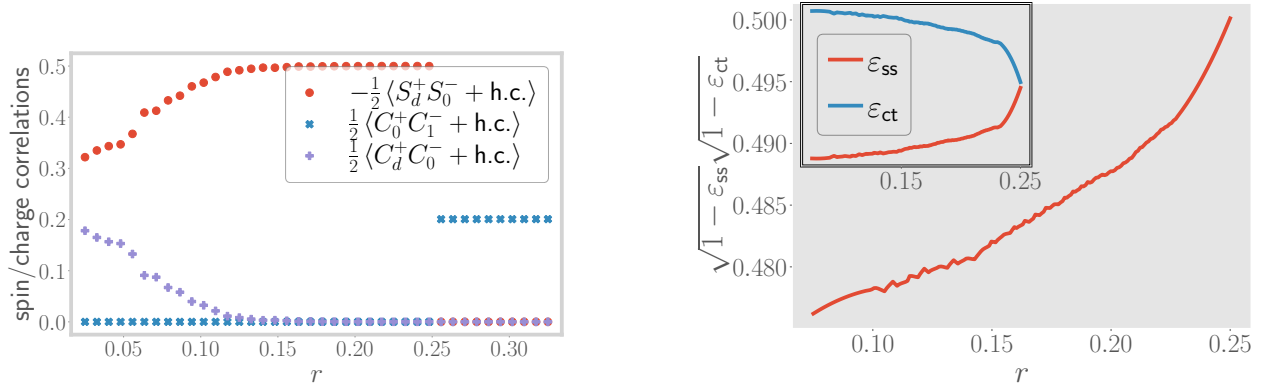


Figure 2.5: Left: Variation of impurity-bath spin-flip correlation (red) and charge isospin-flip correlation (violet), as well as intra-bath charge isospin-flip correlation (blue), from $r \sim 0$ to $r > r_{c2}$. The imp.-bath isospin correlation vanishes at r_{c1} , indicating the change in RG relevance of V . The imp.-bath spin correlation vanishes at r_{c2} , indicating the marginality of J at that point. The intra-bath correlation picks up after the transition, showing the decoupling of the impurity from the bath. Right: Variation of the geometric entanglement ϵ_{ss} (blue) w.r.t. the singlet state and that w.r.t the charge triplet zero state, ϵ_{ct} (red), with r . The former becomes maximum (unity) at r_{c1} , showing that the true ground state has no charge content. The latter discontinuously jumps to unity at r_{c2} , therefore acting as an order parameter for the transition.

2.4.2 Using entanglement to track correlations across the transition

We will now show that a certain measure of entanglement behaves as an order parameter for the transition. We can define a geometric measure of entanglement in terms of wavefunctions $|\psi_1\rangle$ and $|\psi_2\rangle$ [242, 494, 495]:

$$\epsilon(\psi_1, \psi_2) = 1 - |\langle \psi_1 | \psi_2 \rangle|^2. \quad (4.2)$$

From this definition, if $|\psi_1\rangle$ corresponds to a separable state, the entanglement content of the state $|\psi_2\rangle$ is small if it's overlap with $|\psi_1\rangle$ is large. For brevity, we will use the notation $\varepsilon_{ss} \equiv \varepsilon(\psi_{ss}, \psi_{gs}^{(2)})$, $\varepsilon_{ct} \equiv \varepsilon(\psi_{ct}, \psi_{gs}^{(2)})$ to represent the geometric entanglement between the e-SIAM ground-state $|\psi_{gs}\rangle$ and the singlet state $|\psi_{ss}\rangle$ or the charge triplet zero state $|\psi_{ct}\rangle$. The latter two states are shown in Table (2.1). The entanglement measures ε_{ss} and ε_{ct} can be related to the impurity Greens function through the following equation (details can be found in Sec. 3 of the Supplementary Materials [493]):

$$G_d(\omega) = \sum_n \left[(1 - \varepsilon_{ss}) G_{\Phi_{ss}, \Phi_{ss}}(\omega, n) + (1 - \varepsilon_{ct}) G_{\Phi_{ct}, \Phi_{ct}}(\omega, n) + 2\sqrt{(1 - \varepsilon_{ss})}\sqrt{(1 - \varepsilon_{ct})} G_{\Phi_{ss}, \Phi_{ct}}(\omega, n) \right], \quad (4.3)$$

where

$$G_{\psi_1, \psi_2}(\omega, n) = \frac{1}{2} \frac{\langle \psi_1 | c_{d\sigma} | \Psi_n \rangle \langle \Psi_n | c_{d\sigma}^\dagger | \psi_2 \rangle + \text{h.c.}}{\omega + E_{gs} - E_n} + \frac{1}{2} \frac{\langle \psi_1 | c_{d\sigma}^\dagger | \Psi_n \rangle \langle \Psi_n | c_{d\sigma} | \psi_2 \rangle + \text{h.c.}}{\omega - E_{gs} + E_n}. \quad (4.4)$$

This expression displays that the evolution of the impurity Greens function G_d with r (and related correlation functions shown in the left panel of Fig.(2.5)) is dependent on that of the entanglement measures ε_{ss} and ε_{ct} (shown in the right panel of Fig. (2.5)). Indeed, we find that the geometric entanglement ε_{ct} with the charge sector increases towards unity as the transition at r_{c2} is approached, and remains unity after the transition. A more sensitive measure is the singlet entanglement ε_{ss} : it initially decreases to zero at r_{c1} owing to the irrelevance of V , but rises discontinuously to unity at the transition and becomes equal to the charge entanglement in the local moment phase. Therefore, ε_{ss} acts as an order parameter for the local MIT at r_{c2} . We find that the cross-term $\sqrt{(1 - \varepsilon_{ss})}\sqrt{(1 - \varepsilon_{ct})}$ decreases monotonically to zero with increasing r (not shown), displaying a continual decrease in the mixing of the spin and charge sectors and the *immobilisation of the doublons and holons on the impurity site*.

In general, any one-particle or two-particle fluctuation $\langle O_1 O_2^\dagger \rangle$ that acts on the combined Hilbert space of the impurity and the zeroth site can be expressed in terms of these entanglement measures $\varepsilon_{ss}, \varepsilon_{ct}$. The detailed derivations and expressions are given in Section 3 of the Supplementary Materials [493]. As a demonstration, consider the spin-spin correlation $\langle S_d^+ S_0^- \rangle$ between the impurity and the zeroth site shown in the left panel of Fig. (2.5). The general expression given in Section 3 of the Supplementary Materials [493] is of the form

$$\begin{aligned} \langle S_d^+ S_0^- \rangle = & (1 - \varepsilon_{ss}) \langle \Phi_{ss} | S_d^+ S_0^- | \Phi_{ss} \rangle + (1 - \varepsilon_{ct}) \langle \Phi_{ct} | S_d^+ S_0^- | \Phi_{ct} \rangle \\ & + \sqrt{1 - \varepsilon_{ss}} \sqrt{1 - \varepsilon_{ct}} \left(\langle \Phi_{ss} | S_d^+ S_0^- | \Phi_{ct} \rangle + \langle \Phi_{ct} | S_d^+ S_0^- | \Phi_{ss} \rangle \right). \end{aligned} \quad (4.5)$$

From the expression given, we find that only the singlet overlap is non-zero, such that $\langle S_d^+ S_0^- \rangle$ is directly proportional to the quantity $1 - \varepsilon_{ss}$. As the entanglement measure ε_{ss} increases towards the transition (right panel of Fig. (2.5)), the quantity $1 - \varepsilon_{ss}$ decreases, in turn leading to the decrease in the magnitude of the correlation observed in the left panel of Fig. (2.5). For another explicit connection between correlations and entanglement measures, we present relations between the quantum

Fisher information (QFI) [496] and many-particle Greens functions in Section 3 of the Supplementary Materials [493]. There, we also plot the QFI for a number of two-particle operators as a function of r , notably the ones corresponding to the degree of compensation for the impurity ($\langle \vec{S}_d \cdot \vec{S}_0 \rangle$) and the impurity magnetisation ($\langle S_d^z \rangle$). These two quantities (and hence the corresponding QFI) are important because they track the local MIT and act as order parameters for the transition, and the QFI corresponding to these two operators quantify the quantum fluctuations present in the system corresponding to these order parameters. We show in Sec. 3 of the Supplementary Materials that the two phases on either side of the transition are characterised by distinct values of this pair of QFI: while the QFI corresponding to the degree of compensation is zero in the Kondo screened phase, it becomes non-zero in the local moment phase, and the opposite is true for the QFI arising from the impurity magnetisation. The phase precisely at the transition is distinct from those on either side because it displays a non-zero value for both of the QFI. While it is expected that a critical point would show enhanced fluctuations of multiple kinds (giving rise to universality), it is enlightening to find that this is also reflected in a measure of many-particle entanglement.

2.5 Universal theory for the local metal-insulator transition

In order to identify the competing tendencies near the transition at r_{c2} , we will now obtain the minimal effective Hamiltonian that displays the same transition. This involves integrating out the degrees of freedom that do not affect the low-energy physics near r_{c2} . We note that, due to the irrelevance of V , there is no scattering between the spin and charge states on the impurity at low energies. Further, the impurity charge states $|0\rangle$ and $|\uparrow_d \downarrow_d\rangle$ have been pushed to the high energy Hubbard sidebands because of the large value of U close to r_{c2} . As a result, the impurity charge states can be safely decoupled from the spin states through a Schrieffer-Wolff transformation. Up to second order in V^2/U , this transformation accounts for the effects of U and V by generating an additional (Kondo) spin-exchange term $\delta J \left(\sim \frac{V^2}{U+U_b} \right)$, as well as an additional on-site correlation $\delta U_b \left(\sim \frac{2V^2}{U+U_b+J/2} - \frac{8V^2}{U-U_b} \right)$ on the zeroth site of the conduction bath. In this way, we obtain the following renormalised effective Hamiltonian:

$$H_{\text{MIT}} = \mathcal{J} \vec{S}_d \cdot \vec{S}_0 - \frac{1}{2} \mathcal{U}_b (\hat{n}_{0\uparrow} - \hat{n}_{0\downarrow})^2 + H_{\text{K.E.}}, \quad (5.1)$$

where $\mathcal{J} = J + \delta J$ is the renormalised s-d interaction and $\mathcal{U}_b = U_b + \delta U_b$ is the renormalised local correlation on the bath zeroth site. A schematic 1D construction of the effective Hamiltonian is shown in the left panel of Fig. (2.6). We note that a similar approach towards extracting an effective theory for the Mott MIT has been employed in the past ([497, 498]). However, those works essentially led to renormalised Kondo models that do not possess any frustration of the Kondo screening. Thus, they cannot display an impurity transition in the absence of the requirement of self-consistency and can describe only the physics of the metallic regime. Importantly, the effective $\mathcal{J} - \mathcal{U}_b$ model (eq. (5.1)) is consistent with the IR fixed point Hamiltonian obtained in the appropriate regime $r_{c1} < r < r_{c2}$ (row 2 of table (2.1)).

The RG equations for $\mathcal{J}$ and $\mathcal{U}_b$ of the simplified effective impurity model eq. (5.1) can be

obtained by setting $U = V = 0$ in the RG equations (9.36):

$$\Delta\mathcal{J} = -\frac{n_j\mathcal{J}(\mathcal{J} + 4\mathcal{U}_b)}{d_2}, \quad \Delta\mathcal{U}_b = 0. \quad (5.2)$$

For $\mathcal{J} + 4\mathcal{U}_b > 0$, the Kondo coupling is relevant and the low-energy phase is a paramagnetic local Fermi liquid with gapless excitations. But for $\mathcal{J} + 4\mathcal{U}_b < 0$, the Kondo coupling becomes irrelevant and the ground state is a decoupled local moment that is isolated from the bath. Such an effective picture of the transition can be understood if one notes that a straightforward way to destroy the Kondo screening is to inhibit the coordinated spin fluctuations between the impurity and the bath. Such frustration of Kondo screening is precisely the effect of the $\mathcal{U}_b$ term: it promotes charge fluctuations on the bath site coupled directly to the impurity, reducing thereby the spectral weight for spin-flip scatterings between the impurity and bath.



Figure 2.6: Left: Schematic 1D construction of universal theory for the metal-insulator transition, obtained by integrating out the charge states of the impurity near the transition. Right: Zero bandwidth limit of the low-energy effective Hamiltonian for r between r_{c1} and r_{c2} (second row in table (2.1)). The bath is reduced to just a single degree of freedom - the zeroth site that is directly coupled to the impurity site.

Indeed, the local correlation $\mathcal{U}_b$ encourages entanglement between the sites of the bath and makes the formation of the impurity-bath singlet difficult. In other words, the $\mathcal{J} - \mathcal{U}_b$ model displays the destabilisation of the singlet by redistributing the entanglement from the impurity+bath system to purely within the bath (see left panel of Fig. (2.7)). Given the simplicity of these arguments, our analysis makes the case that local pairing fluctuations of the bath in eq. (5.1) offer a universal mechanism by which to frustrate the Kondo effect and lead thereby to an impurity transition that is the local counterpart of the Mott MIT obtained by auxiliary model approaches such as DMFT. We note that similar conclusions were reached in Refs. [460, 461, 472] for the emergence of non-Fermi liquid phases at critical points in the mixed valence regime of the periodic Anderson model.

Further insight into the destabilisation of the singlet can be obtained from a zero-bandwidth approximation of the bath. Under such an approximation, the IR effective Hamiltonian obtained from the RG flow (row 2 of Fig. (2.1) takes the form of a two-site model with modified couplings (shown in the right panel of Fig. (2.6)):

$$\tilde{\mathcal{J}} \vec{S}_d \cdot \vec{S}_0 - \frac{1}{2} \tilde{\mathcal{U}}_b (\hat{n}_{0\uparrow} - \hat{n}_{0\downarrow})^2. \quad (5.3)$$

The spectrum of this model is shown in the right panel of Fig. (2.7). For $|\tilde{\mathcal{U}}_b|/\tilde{\mathcal{J}} < 3/2$, the singlet ground-state (red) is separated from the excited local moment states (blue) by a gap of $\frac{-3\tilde{\mathcal{J}}}{4} - \frac{\tilde{\mathcal{U}}_b}{2}$. As we tune r towards r_{c2} , $|\tilde{\mathcal{U}}_b|$ increases and the fixed point value $\tilde{\mathcal{J}}$ decreases, leading to an overall reduction in the gap. At the critical point, the gap closes and the states become degenerate at zero

energy; this is the equivalent of the quantum critical point at r_{c2} of the full impurity model that was discussed earlier. We note that the vanishing of the energy of the metallic state was also observed by Brinkman and Rice from a Gutzwiller-type variational calculation of the Hubbard model at half-filling [19].

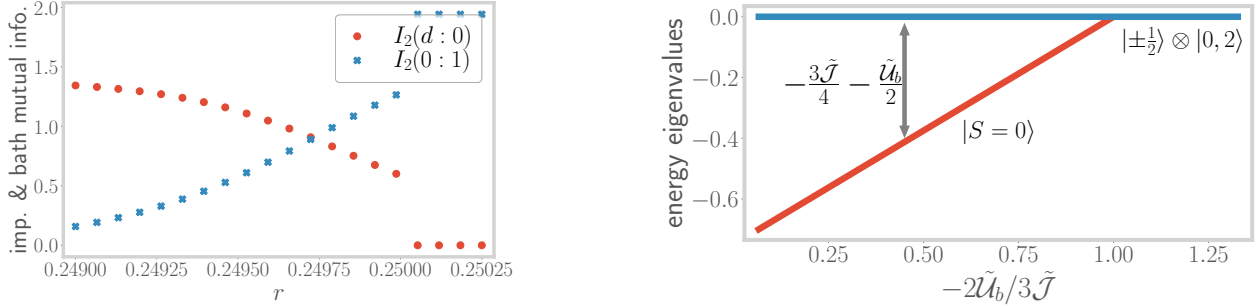


Figure 2.7: Left: Very close to the transition, the mutual information $I_2(d : 0)$ (red curve) between impurity and the zeroth site reduces, while that between the zeroth and the first site ($I_2(0 : 1)$, blue curve) increases, showing the redistribution of entanglement. Right: Spectrum of the zero bandwidth Hamiltonian of the right panel of Fig. (2.6). The blue line represents the local moment states at zero energy, while the red line represents the singlet state at an energy that continuously increases upon increasing r .

2.5.1 Local pairing and the destruction of Kondo screening

Very close to the transition at r_{c2} , we find signatures of the breakdown of the Kondo cloud in terms of decaying impurity-zeroth site correlations (blue curve in the left panel of Fig. (2.8)) and enhanced intra-bath correlations. Remarkably, we find an *increase in pairing correlations* between the bath zeroth site and first site (red curve in the left panel of Fig. (2.8)). As discussed above, these are observed to be responsible for the destruction of the Kondo screening: spin and charge degrees of freedom are mutually exclusive, and the increased charge fluctuations on the bath zeroth site suppress the spin-flip scattering processes between the impurity and the zeroth site. We will show later that these fluctuations lead to the destruction of the local Fermi liquid excitations in our model, and replace them with non-Fermi local excitations in the neighbourhood of r_{c2} . We believe that the observed growth in non-local pairing correlations near the transition at r_{c2} is likely tied to a putative low-energy divergence of the local pairing susceptibility. A similar observation was made in Ref. [460] from an auxiliary model-based analysis of a hole-doped extended Hubbard model in infinite dimensions that is pertinent to the physics of the heavy fermions. Similar to the conclusions of Ref. [460], we expect the pairing fluctuations of the bath to become dominant upon tuning the e-SIAM away from half-filling, signalling an instability towards a superconducting state.

2.5.2 Emergent self-consistency in the e-SIAM

As is well-established, the DMFT self-consistency equation is equivalent to requiring that the impurity Greens function become equal to the local Greens function in the bath [18]. Such a condition

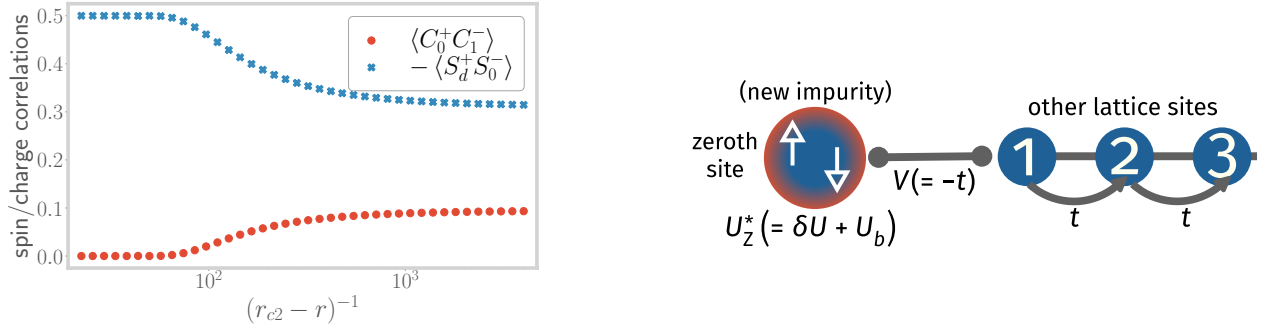


Figure 2.8: Left: Spin-flip correlation between the impurity and the bath (blue), and charge isospin-flip correlation between the bath zeroth and first sites (red), both as a function of the tuning parameter r , very close to the transition. The operators are defined as $S_i^+ = c_{i\uparrow}^\dagger c_{i\downarrow}$ and $C_i^+ = c_{i\uparrow}^\dagger c_{i\downarrow}^\dagger$. Right: New effective SIAM Hamiltonian generated upon integrating out the coupling between the impurity and the bath zeroth site. In the new SIAM, the zeroth site acts as the new impurity site with a renormalised on-site correlation $U_Z^* = U_b + \delta U_Z^*$.

is also used in the projective self-consistent technique of Moeller et al. [497] for the states within the central Kondo resonance. We will now show that our model displays a qualitatively similar emergent feature. To proceed, by employing a one-step URG transformation, we integrate out the impurity site from the rest of the fixed-point Hamiltonian of eq. (3.4). The details are shown in Sec. 4 of the Supplementary Materials [493]. The essential idea is similar to that of the Schrieffer-Wolff transformation: removing the impurity-bath couplings J and V generates an additional repulsive correlation δU_Z^* on the zeroth site. The new low-energy model H_Z is therefore an Anderson impurity model with a net local correlation $U_Z^* = \delta U_Z^* + U_b$ on the zeroth site (which is now the “new impurity site”), and a single-particle hybridisation $V_Z^* = -t$ coupling with a conduction bath formed by the remaining sites. This new impurity model is depicted schematically in the right panel of Fig. (2.8).

$$H_Z^* = \underbrace{-\frac{1}{2}U_Z^* (\hat{n}_{0\uparrow} - \hat{n}_{0\downarrow})^2}_{\text{new impurity} = 0^{\text{th}} \text{ site}} + \underbrace{V_Z^* \sum_{\langle j,0 \rangle} (c_{0\sigma}^\dagger c_{j\sigma} + \text{h.c.})}_{\text{hopping between } 0^{\text{th}} \text{ site \& new bath}} + \underbrace{(-t) \sum_{\langle i,j \rangle} (c_{i\sigma}^\dagger c_{j\sigma} + \text{h.c.})}_{\text{K.E. of new bath}}. \quad (5.4)$$

Since, for $r < r_{c2}$, the model always flows to strong coupling at low energies, the largest energy scales are J^* and V^* . Using this, the effective correlation U_Z^* on the zeroth site can be expressed to leading order as

$$U_Z^* \simeq \frac{J^*}{4} \frac{1}{1 + 2\gamma} - V^* \frac{\gamma}{\gamma^2 - \frac{1}{4}}, \quad \text{where } \gamma \equiv V^*/J^*. \quad (5.5)$$

From $r = 0$ to $r = r_{c1}$, the factor γ decreases due to the gradual removal of single-particle hopping from the impurity site (irrelevance of V), leading to an increase in the correlation U_Z^* . This is simply a restatement of the fact that the scattering processes that create the central impurity resonance induce a repulsive correlation on the bath zeroth site. The fact that we end up with a standard SIAM

on the zeroth site once the impurity site has been integrated out means that the spectral function of this new impurity will again go through a sharpening of the central peak (and the appearance of the Hubbard sidebands) upon increasing the parameter r of the original e-SIAM. This ensures that the impurity and zeroth site spectral functions look qualitatively similar up to r_{c1} . One can now repeat iteratively this process - decoupling the zeroth site generates a standard SIAM with a repulsive correlation on the first site, and so on. The fact that excursions starting from any point along the bath can be described by a positive U SIAM is, therefore, the emergent self-consistency in our model. Similar indications of a correlated spectral function on lattice sites far away from the impurity were also observed in Ref. [480] from finite- U slave boson calculations of the SIAM. Beyond r_{c1} , the irrelevance of V means that U_Z^* reduces to just $J^*/4$. As r is now increased towards r_{c2} , the correlation decreases because J is moving towards its critical point, indicating that the bath zeroth site is moving away from its local moment regime. This is another reflection of the increase in pairing fluctuations of the bath, as well as the lowering of spin-flip fluctuations between the impurity and the bath.

For $r > r_{c2}$, the impurity site decouples from the bath in the impurity model. This impurity model can be promoted to a bulk model as follows. Recall that in the auxiliary model mapping, any lattice site $\vec{r}_i$ of the bulk lattice can act as the impurity, and one can think of the impurity model with the impurity at $\vec{r}_i$ as a representation of the excursion of an electron from any such site $\vec{r}_i$ into the rest of the bath. When all such impurity models undergo the transition at $r = r_{c2}$ simultaneously, the result is the decoupling of all sites from their respective baths and a paramagnetic bulk insulator is obtained.

2.6 Coexistence of local metallic and insulating phases in the e-SIAM

The DMFT solution of the Hubbard model on the Bethe lattice exhibits a coexistence region of metallic and insulating solutions between two spinodal lines $U_{c1}(T)$ and $U_{c2}(T)$ [18], with the metallic solution having lower internal energy and the insulating solution being a metastable state at a higher energy [18, 32, 497, 499, 500]. The $T = 0$ transition is observed to be second order in nature and occurs through a merging of the metallic and insulating solutions at U_{c2} [497, 501]. On the other hand, at $T > 0$, the MIT happens along the first-order line $U_c(T)$ ($U_{c1} < U_c < U_{c2}$) where the free energies of the two solutions become equal, and the two spinodal lines merge into a second order critical point at a sufficiently high temperature T_c . We will now show that our Hamiltonian-based approach gives a clear zero temperature picture of various aspects that can lead to the emergence of the metal-insulator coexistence at $T > 0$.

2.6.1 Excited state quantum phase transition at r_{c1} : the Mott-Hubbard scenario

We have already discussed in detail the nature of the phase transition at r_{c2} : the zero-bandwidth picture provided (around eq. (5.3)) shows the merging of the metallic and insulating solutions at

that point, enabling identification of r_{c2} with the $T = 0$ continuous phase transition observed at U_{c2} in the DMFT phase diagram. The focus of this subsection is the physics of the other important point r_{c1} . As has been mentioned before, this point marks the RG irrelevance of the single-particle hybridisation amplitude V , and leads to the exclusion of the charge states from the ground-state (see left panel of Fig. (2.4)). This exclusion means that there is now one fewer scattering channel by which the impurity electron can hybridise with the bath. In turn, this leads to a *partial localisation of the impurity*, and can be thought of as the first step towards the more complete localisation that occurs at r_{c2} .

Apart from the change in the ground state, the physics at r_{c1} also involves a phase transition in certain excited states of the spectrum. To expose this, we consider the following states in the zero-bandwidth spectrum of the impurity model given in eq. (2.2):

$$\begin{aligned}
 |1, \sigma, \pm\rangle &= \alpha_{\pm} |\sigma_d\rangle |0_0\rangle \mp \sqrt{1 - \alpha_{\pm}^2} |0_d\rangle |\sigma_0\rangle, \quad |3, \sigma, \pm\rangle = \alpha_{\pm} |\sigma_d\rangle |2_0\rangle \mp \sqrt{1 - \alpha_{\pm}^2} |2_d\rangle |\sigma_0\rangle, \\
 E_+ &= -\frac{U_0}{4} + \sqrt{V_0^2 + \frac{U_0^2}{16}}, \quad E_- = -\frac{U_{\text{im}}}{4} - \sqrt{V_{\text{im}}^2 + \frac{U_{\text{im}}^2}{16}}, \\
 \alpha_+(U_0, V_0) &= \frac{V_0}{\sqrt{V_0^2 + \left(E_+ + \frac{U_0}{2}\right)^2}}, \quad \alpha_- = \frac{E_- + \frac{U_{\text{im}}}{2}}{\sqrt{V_{\text{im}}^2 + \left(E_- + \frac{U_{\text{im}}}{2}\right)^2}}.
 \end{aligned} \tag{6.1}$$

where $(\sigma = \uparrow, \downarrow)$ and $E_{\pm}$ is the energy of the states $|1(3), \sigma, \pm\rangle$ and the subscripts d and 0 refer to the impurity and bath zeroth site respectively. The states $|1(3), \sigma, +\rangle$ are high-energy states, so their energy E_+ and coefficient α_+ involve the bare single-particle hybridisation V_0 and impurity on-site repulsion U_0 . On the other hand, the other states $|1(3), \sigma, -\rangle$ are closer to the IR energy scale and involve renormalised intermediate-scale couplings V_{im} and U_{im} . Both E_+ and E_- are four-fold degenerate because of the SU(2) spin ($\uparrow \leftrightarrow \downarrow$) and particle-hole ($|0\rangle \leftrightarrow |2\rangle$) symmetries. For example, the degenerate subspace corresponding to E_+ is the set of states $\{|1, \uparrow, +\rangle, |1, \downarrow, +\rangle, |3, \uparrow, +\rangle, |3, \downarrow, +\rangle\}$.

All these states are in general delocalised - they involve the impurity hybridising with the bath via V . At $r = r_{c1}$, however, the coupling V becomes irrelevant and $V_{\text{im}} \rightarrow 0$, so that the coefficient α_- of the low-energy state becomes unity beyond that point. The low-energy states $|1(3), \sigma, -\rangle$ thus become localised at r_{c1} and give rise to a set of excited and *degenerate local moment states*, i.e., $\alpha_- \rightarrow 0$ as $r \rightarrow r_{c1}$ leads to

$$|1(3), \sigma, -\rangle \rightarrow \begin{cases} |\uparrow_d\rangle |0_0\rangle, |\downarrow_d\rangle |0_0\rangle \\ |\uparrow_d\rangle |2_0\rangle, |\downarrow_d\rangle |2_0\rangle \end{cases}. \tag{6.2}$$

The other coefficient α_+ remains non-zero because, as mentioned earlier, V is non-zero in the UV scales of the RG flow. As a result, the high-energy states $|1(3), \sigma, +\rangle$ remain delocalised, and are separated from the localised states by an energy-scale

$$\lim_{r \rightarrow r_{c1}^-} (E_+ - E_-) \simeq \sqrt{V_0^2 + \left(\frac{U_0}{4}\right)^2} + \sqrt{V_{\text{im}}^2 + \left(\frac{U_{\text{im}}}{4}\right)^2} + \frac{U_{\text{im}} - U_0}{4}. \tag{6.3}$$

This is shown schematically in the left panel of Fig. (2.9), and corresponds to the preformed gap in the zero mode spectrum. The point r_{c1} therefore represents a *delocalisation-localisation excited state quantum phase transition* (ESQPT) where degenerate local moment states are emergent as excited states in the many-body spectrum. This localisation of the impurity at r_{c1} is shown in the form of excited state mutual information in the right panel of Fig. (2.9). This ESQPT acts as a precursor to the QPT at r_{c2} , where the local moment states become degenerate with the spin-singlet ground state. The local moment states are stabilised as ground states in the insulating phase for $r > r_{c2}$.

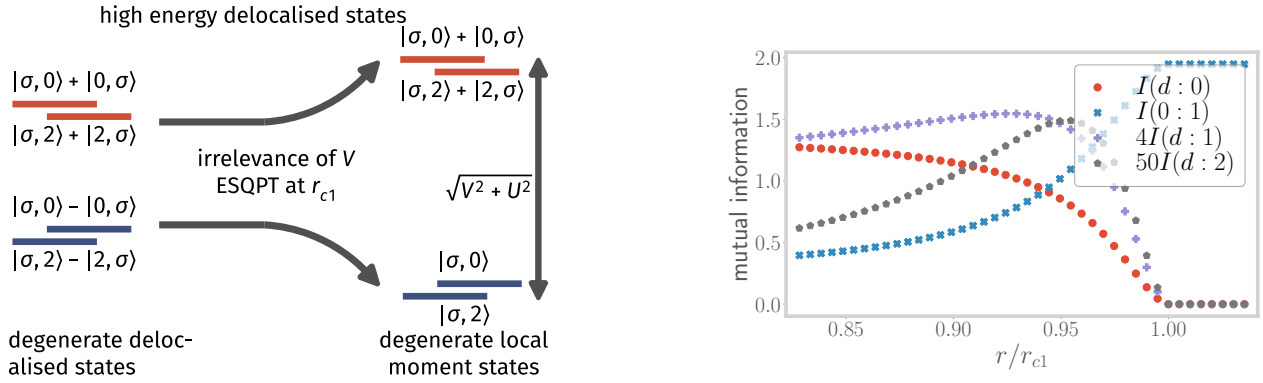


Figure 2.9: Left: Evolution of the excited states mentioned in Eqs. (6.1) as r is tuned through r_{c1} . The high-energy states (red) retain their impurity-bath coupling, while their low-energy counterparts (blue) become disentangled and form a degenerate set of local moment states. The high-energy states allow delocalisation into the bath and lead to the broad Hubbard sidebands. Right: Variation of mutual information (MI) between various parties, across the ESQPT at r_{c1} . The MI (red) between the impurity and the zeroth site vanishes at r_{c1} , showing the localisation of the impurity. The same between the bath zeroth site and the next site becomes maximum at r_{c1} . The MI between the impurity and the other sites (purple and gray) of the bath show an initial rise near r_{c1} , indicating that entanglement is becoming long-ranged near the ESQPT.

2.6.2 Theory for the charge excitations in the Hubbard sidebands

Beyond r_{c1} , excitations into the local moment states $|1, \sigma, -\rangle$ and $|3, \sigma, -\rangle$ reside at the edge of the central peak in the impurity spectral function (purple curve in the right panel of Fig.(2.4)) but provide no spectral weight due to the lack of electron mobility. This explains the development of a preformed gap in the impurity spectral function. That these local moment states have to reside at the edge of the central peak becomes clear when we note that as the width of the central peak shrinks continuously, the local moment states must also recede towards zero frequency and finally replace the zero frequency peak at $r = r_{c2}$ in order to give rise to the insulating local moment phase for $r > r_{c2}$. On the other hand, the still-delocalised high-energy states $|1, \sigma, +\rangle$ and $|3, \sigma, +\rangle$ are pushed into the Hubbard sidebands, and their hybridisation with the bath through V and t is responsible for the broadening of the sidebands.

This isolation of the delocalised states into the sidebands means that charge delocalisation processes are now excluded from the physics at low energies. As accessing the sidebands involves excitations at exorbitantly high energy scales, such processes can only happen virtually and involve very short time scales. The central Kondo resonance observed at low energies, therefore, does not support any charge delocalisation, and metallic excitations propagate only through spin-flip scattering processes of the impurity. This reveals that the Mott insulator comes about through a local binding of doublons and holons [11, 502, 503]. Closely related to this is the Mott-Hubbard scenario of the MIT, which is equivalent to our observation of the appearance of an optical gap in the spectrum after r_{c1} .

The fact that the sidebands are broad indicates that there are gapless excitations propagating from the impurity site into the conduction bath, but whose energy lies outside the Mott gap. These can also be viewed as metallic excitations of the bath that are induced by the coupled impurity. In order to expose the nature of these excitations, we perform a calculation similar to the local Fermi liquid calculation of Nozières [40]. This involves considering the states at $\omega \sim \pm U/2$ as the ground-states of the Hubbard sidebands, and then computing how excitations into the bath renormalise the ground-state subspace. Up to the second order in the hopping strength t , the effective Hamiltonian for the excitations of the bath can be expressed as

$$H_{\text{eff}}^{(2)} = \frac{4t^2(1 - \alpha_+^2)}{E_+ - E_{\text{gs}}} C_{\text{tot}}^z C_1^z - t \sum_{i>0, \sigma} \left(c_{i\sigma}^\dagger c_{i+1\sigma} + \text{h.c.} \right), \quad (6.4)$$

where E_+ and α_+ have been defined in eq. (6.1), $C_i^z |2(0)\rangle = +(-)\frac{1}{2} |2(0)\rangle$ is the z-component of the charge isospin operator at a particular site i , and $\vec{C}_{\text{tot}} = \vec{C}_d + \vec{C}_0$ is the total isospin for the impurity and zeroth sites. $E_{\text{gs}} = U/2$ is the two-site energy of the ground-state subspace within the sideband. These excitations are of the local Fermi liquid kind - the absence of any isospin-flip scattering term promotes the independent delocalisation of the doublon and holon states into the bath. The second-order effective Hamiltonian can therefore be written as the sum of two decoupled parts corresponding to holon and doublon propagation respectively, and suggesting adiabatic continuity with the $U = 0$ Hubbard model [504].

However, fourth-order corrections lead to the appearance of scattering processes that convert holon and doublons into one another, resulting in the local Fermi liquid becoming correlated

$$H_{\text{eff}}^{(4)} = \frac{\gamma^4 \alpha_+^2 \beta^2}{(1 - \alpha_+^2)(E'_+ - E_{\text{gs}})} \left[-C_{\text{tot}}^z C_{\text{tot}}^2 C_1^z + \sqrt{2} \mathcal{P}_{\text{tot}}^4 (C_0^+ - C_d^+) C_1^- + \text{h.c.} \right], \quad (6.5)$$

where $E'_+ = \frac{U}{4} - \frac{3J}{8} + \sqrt{4V^2 + \left(\frac{U}{4} + \frac{3J}{8}\right)^2}$ is the energy of the state containing the charge isospin triplet zero, and $\beta = \frac{2V}{\sqrt{4V^2 + (U+3J/4)^2}}$. $\mathcal{P}_{\text{tot}}^4$ projects on to the $n_d + n_0 = 4$ subspace. The scattering processes in eq. (6.5) are clearly non-Fermi liquid in nature as they allow the inter-conversion of the charge isospin eigenstates, and therefore reduce the lifetime of the quasiparticle excitations of the local Fermi liquid obtained in eq. (6.5). Additional details pertaining to the calculation of this effective Hamiltonian are present in Sec. 5 of the Supplementary Materials [493].

2.6.3 Hysteresis and the first-order line

As discussed above, the localisation of charge in the emergent local moment states $|1, \sigma, -\rangle$ and $|3, \sigma, -\rangle$ for $r > r_{c1}$ means they do not contribute any spectral weight to the impurity spectral function. Instead, they lead to an emergent preformed/optical gap between the central Kondo resonance and the Hubbard sidebands on each side. This *optical gap* finally becomes the true Mott gap at $r = r_{c2}$ when the central peak disappears. On the other hand, upon starting from $r > r_{c2}$ and reducing r , the system is initially in the (doubly degenerate) local moment ground states, and the impurity DOS is of course gapped. In the rest of the subsection, we will show that the presence of these two transitions leads to finite temperature behaviour for the e-SIAM that is qualitatively similar to that of the Hubbard model as seen from DMFT (that was described at the beginning of this section, and can also be seen in Fig. 2.10).

At non-zero temperatures, the system is described by not only the ground-state but also the excitations. The metallic and insulating solutions become temperature dependent, with energies $E_M(r, T)$ and $E_I(r, T)$ respectively. The curve $r_{c1}(T)$ at a given temperature again represents the values of the parameter r at which the local moment states enter the spectrum as excited eigenstates, while $r_{c2}(T)$ marks the point where the metallic ground-state leaves the spectrum: $E_M(r_{c2}(T), T) = E_I(r_{c2}(T), T)$. As a result, within the parameter range $r_{c1}(T) < r(T) < r_{c2}(T)$ at a finite temperature T , the low-energy part of the spectrum (states within the central Kondo resonance) involves both the metallic and insulating solutions. These two low-lying solutions then govern the partition function and hence the free energy:

$$Z(r, T) \simeq Z_M(r, T) + Z_I(r, T), \quad F(r, T) \simeq -k_B T \ln (Z_M(r, T) + Z_I(r, T)), \quad (6.6)$$

where $Z_M(r, T) = \exp(-\beta \mathcal{V} (E_M(r, T) - TS_M(r, T)))$ is the partition function corresponding to the metallic state with energy $E_M(r, T)$ and entropy $S_M(r, T)$ per unit volume $\mathcal{V}$, and

$$Z_I = \exp(-\beta \mathcal{V} (E_I(r, T) - TS_I(r, T))), \quad (6.7)$$

is similarly the partition function corresponding to the local moment states.

In the thermodynamic limit, only the larger of Z_M and Z_I contributes to the partition function $Z(r, T)$. Given a temperature T , there exists a value $r_c(T)$ lying in the range $r_{c1}(T) < r_c(T) < r_{c2}(T)$ at which the partition functions $Z_M(T)$ and $Z_I(T)$ become equal: $Z_M(r_c(T), T) = Z_I(r_c(T), T)$. This condition defines a curve $\{(T, r_c(T))\}$ at which the system undergoes a first-order transition between the metallic and insulating states:

$$\lim_{\mathcal{V} \rightarrow \infty} F(r, T) = \begin{cases} -k_B T \ln Z_M(r, T) & \text{if } r < r_c \text{ } (Z_M > Z_I) \\ -k_B T \ln Z_I(r, T) & \text{if } r > r_c \text{ } (Z_M < Z_I) \end{cases}. \quad (6.8)$$

The first-order curve $r_c(T)$ is determined from the partition function equality (or equivalently, the free energy equality) condition between the metal and the insulator:

$$T = \frac{E_M(r_c, T) - E_I(r_c, T)}{S_M(r_c, T) - S_I(r_c, T)}. \quad (6.9)$$

At $T = 0$, the first-order line $r_c(T)$ becomes identical to the critical point r_{c2} , as $E_M(r_c) = E_I(r_c)$ at $T = 0$. This simply means that the thermal entropy plays no role at zero temperature, and the transition is purely quantum-mechanical in nature. As temperature is now increased, the large entropy S_I of the doubly degenerate local moment states comes into play and the insulating state is able to take over at a smaller value of r . This suggests that a first-order transition can take place in the thermodynamic limit at $r_c(T) < r_{c2}(T)$. This has been shown schematically in Fig. (2.10).

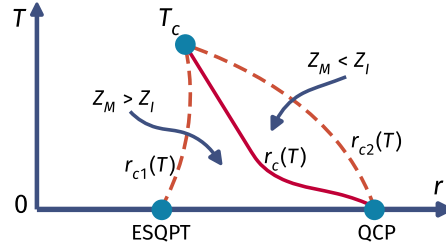


Figure 2.10: Qualitative structure of the finite temperature coexistence region of the $J - U_b$ model. The dotted lines on the left and the right represent the spinodals r_{c1} and r_{c2} where the insulating and metallic solutions become unstable, respectively. The solid red line represents the first-order line where the free energies and the partition functions of the two solutions become equal, $Z_M = Z_I$.

Further, away from the thermodynamic limit, the free energy in eq. (6.6) admits contributions from the stable metallic state as well as the metastable insulating state, and results in the coexistence of the two phases. This is because, depending on whether we are approaching from r_{c2}^+ or r_{c1}^- , the system will remain mostly in the minimum of either Z_I or Z_M respectively. There exist finite activation barriers in the free energy for passage into the other minimum, leading to hysteresis. The transition is observed only at $r_{c1}(T)$ or $r_{c2}(T)$ where the insulating and metallic solutions, respectively, become unstable. These two lines, therefore, mark the spinodals of the coexistence region. At a sufficiently high temperature T_c , the metallic and insulating solutions become indistinguishable at the first spinodal r_{c1} : $E_M(r_{c1}, T_c) = E_I(r_{c1}, T_c)$, $S_M(r_{c1}, T_c) = S_I(r_{c1}, T_c)$. We recall that of these two equalities, the first ($E_M = E_I$) also corresponds to the locus of the second spinodal (corresponding to r_{c2}). Moreover, combining the two equalities (energy and entropy) leads to the free energy equality condition, marking the locus of the first-order line $r_c(T)$. This has the consequence that the two spinodals and the first-order line merge into a second order critical point $r_c(T_c)$ at temperature T_c . This concludes our discussion of the finite temperature behaviour of the extended SIAM model and its qualitative similarities with the DMFT phase diagram for the half-filled Hubbard model on the Bethe lattice.

2.6.4 Zero temperature origin of critical fluctuations above the second order point

Above the second order point $r_c(T_c)$, the DMFT phase diagram shows a rapid crossover from the paramagnetic metallic solution to the paramagnetic insulating solution [18, 505]. Remarkably, signatures of quantum critical scaling in this crossover region have been recently predicted from theoretical analyses [454, 462], as well as detected experimentally in transport measurements of several

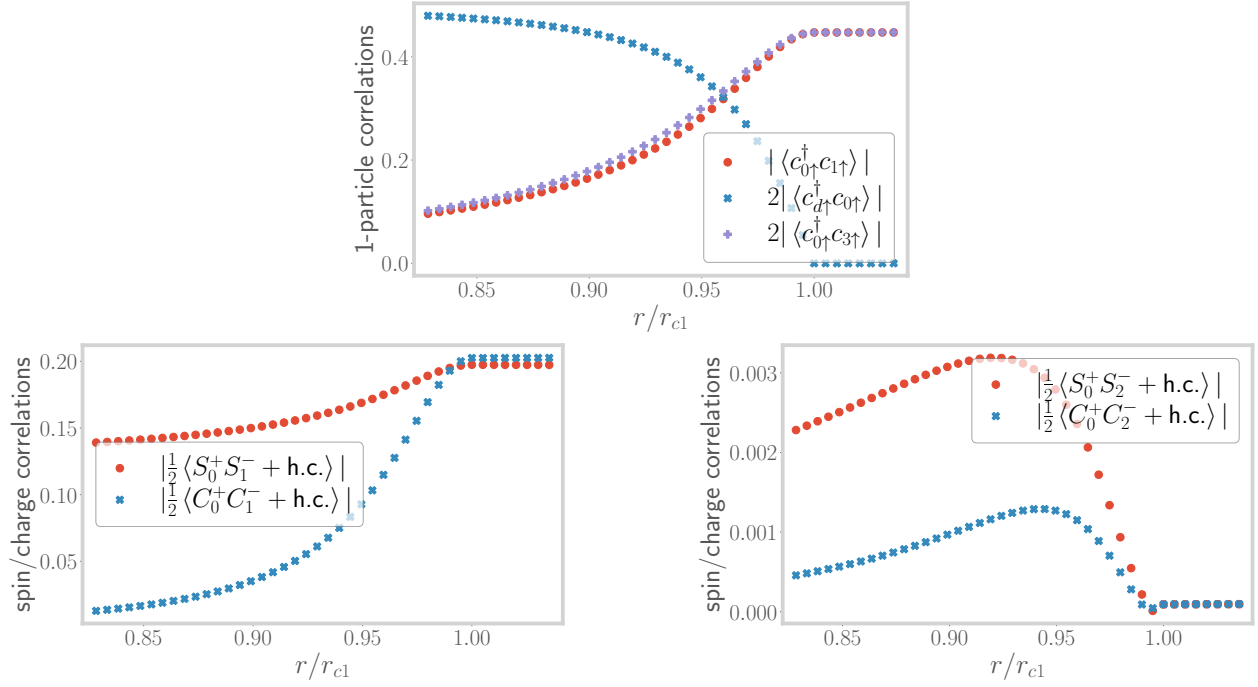


Figure 2.11: Top: Variation of the one particle correlations in the state $|1, \sigma, -\rangle$ between impurity and bath zeroth site (blue), as well as within the bath (red and violet). The former vanishes, depicting the excited state localisation transition at r_{c1} . Left: Variation of the spin-flip (red) and charge isospin-flip (blue) correlations between impurity and bath, close to r_{c1} . They both increase, indicating that the impurity is now less strongly coupled with the bath. Right: Variation of the spin-flip (red) and charge isospin-flip (blue) correlations between the zeroth and second sites of the bath, close to r_{c1} . Both show an initial increase, leading to the propagation of relatively long-range correlations into the bath.

organic compounds [506, 507]. In Refs. [454, 462], this has been ascribed to the existence of a *hidden quantum criticality* in the maximally frustrated 1/2-filled Hubbard model on the Bethe lattice with infinite coordination number. In order to locate the zero temperature origin of these signatures of quantum criticality, we inspect carefully the quantum-mechanical fluctuations near the two important points r_{c1} and r_{c2} in our model.

Close to r_{c1} , we compute correlations in the state $|1, \sigma, -\rangle$. As shown in the top panel of Fig. (2.11), we find that while the one-particle correlation between the impurity and the bath vanishes, the same between the zeroth and first sites of the bath picks up. Importantly, these correlations extend beyond the immediate neighbourhood of the impurity. This is observed in, for example, the one-particle correlation between the zeroth site and the second site (purple curve in the top panel of Fig. (2.11)). Some other signatures of correlations in the bath are shown in Fig. (2.11): while the left panel displays increased spin-flip and charge isospin-flip correlations between the zeroth and first sites of the bath, a similar phenomenon is observed between the zeroth and second sites of the bath in the right panel. Our findings are consistent with the presence of scale-invariant solutions obtained from recent NRG-DMFT calculations [160] that correspond to the metastable insulating solutions in the coexistence region.

The growth of similar long-ranged correlations within the bath (and leading away from the immediate neighbourhood of the impurity) near r_{c2} as well. We recall that correlations between the impurity and the zeroth site are observed to decrease (see left panel of Fig.(2.5)). Instead, as shown in Fig. (2.12), non-trivial two-particle correlations arise between the impurity and bath zeroth sites with bath sites that are farther away. The spreading of the spin-spin correlations shown in the left panel of Fig. (2.12) indicates a “stretching” of the Kondo singlet state prior to its destruction. We will show later that these enhanced correlations also result in a diverging quasiparticle mass at r_{c2} , indicating a breakdown of the local Fermi liquid metal. Further, the right panel of Fig. (2.12) indicates that longer-ranged pairing correlations develop between the bath zeroth site and bath sites farther away upon approaching the transition.

In general, within the metallic phase and away from r_{c1} or r_{c2} , the impurity is strongly coupled to the bath zeroth site and the impurity-bath entanglement follows an area law characteristic of the local nature of the impurity problem. However, near the excited state and ground state transitions at r_{c1} or r_{c2} respectively, these results indicate a spreading of entanglement into the bath: more and more bath sites beyond the zeroth site get correlated with the impurity site as the ESQPT or the QCP is approached. This is corroborated in Fig. 3 of the Supplementary Materials [493]. This suggests a significant enhancement of the entanglement beyond the area law and is consistent with the critical quantum fluctuations observed in the various two-particle correlations.

The results of this subsection indicate that it might be possible to experimentally detect the signatures of such critical quantum fluctuations around the spinodals of the DMFT phase diagram even at $T > 0$. We note that signatures of second-order criticality have been indeed observed recently in vanadium sesquioxide, in the form of critical slowing down and critical opalescence [508]. It is, therefore, tempting to speculate that these signatures arise from the presence of critical quantum fluctuations.

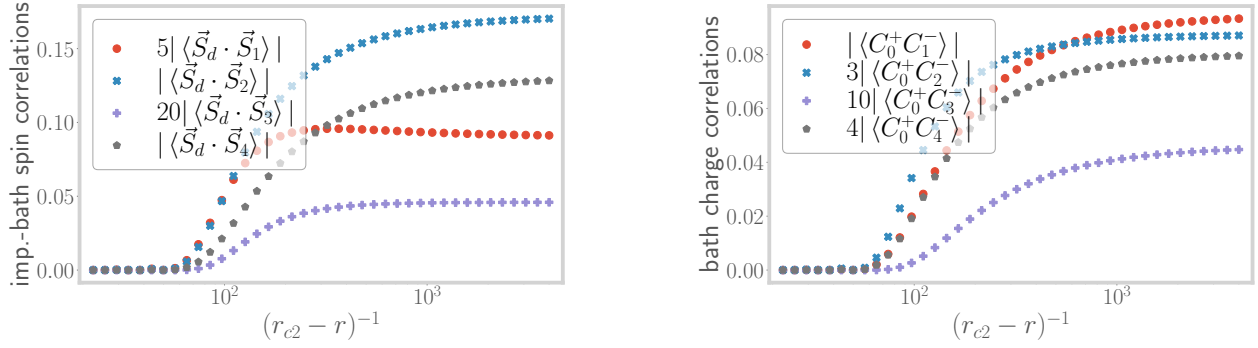


Figure 2.12: Left: Variation of the spin-spin correlations between the impurity and bath sites 1 through 4, close to r_{c2} . They show an increase, revealing the distribution of entanglement into the bath. Right: Variation of pairing correlations within the bath. These correlations show an increase, because of the weakening of the Kondo singlet.

2.7 Non-Fermi liquid signatures at the MIT

2.7.1 Death of the local Fermi liquid: the Brinkman-Rice scenario

The low-lying metallic excitations of the bath can be obtained by considering the singlet ground-state of the model (of energy $\sim -3\tilde{\mathcal{J}}/4$) in the metallic phase, and then studying the effect of an electron hopping term (with coupling t) between the singlet and the rest of the bath as a perturbation (see left panel of Fig. (2.13)). Such a strong-coupling expansion in powers of $t^2/\tilde{\mathcal{J}}$ leads to the usual local Fermi liquid effective Hamiltonian in the Kondo model [40, 42, 52, 444]:

$$H_{\text{LFL}} = \mathcal{F} \hat{n}_{1\uparrow} \hat{n}_{1\downarrow} + H_{\text{KE}}, \quad \mathcal{F} \sim t^4 / \tilde{\mathcal{J}}^3, \quad (7.1)$$

where $\hat{n}_{1\sigma}$ are the number operators for the first site (site adjacent to the zeroth site of the conduction bath), H_{KE} is the kinetic energy arising from the nearest-neighbour hopping among all sites in the bath apart from the zeroth site, and $\mathcal{F}$ is the local Fermi liquid correlation strength. The Kondo singlet (formed between the impurity spin and the bath zeroth site) has decoupled from the rest of the lattice, and eq. (7.1) describes the effective Hamiltonian for the rest of the bath beyond the zeroth site.

Following an identical approach, we find that the effective Hamiltonian for the low-lying excitations in the metallic phase is given by

$$H_{\text{LFL}} = \mathcal{F} [\hat{n}_{1\uparrow} \hat{n}_{1\downarrow} + (1 - \hat{n}_{1\uparrow})(1 - \hat{n}_{1\downarrow})] + H_{\text{KE}}, \quad \text{where } \mathcal{F} = \frac{2t^4}{\tilde{\mathcal{J}} (3\tilde{\mathcal{J}}/4 + \tilde{\mathcal{U}}_b)^2}. \quad (7.2)$$

We note that for our extended SIAM, the local Fermi liquid correlation $\mathcal{F}$ diverges as the transition is approached at $3\tilde{\mathcal{J}}/4 + \tilde{\mathcal{U}}_b \rightarrow 0$. The divergence of $\mathcal{F}$ leads to the divergence of the renormalised mass of the quasiparticles, and can also be obtained from a Gutzwiller variational calculation as shown by Brinkman and Rice [19]. This shows the breakdown of perturbation theory and is indicative of the fact that the ground state is about to change at the metal-insulator transition.

The death of the local Fermi liquid is also seen from the vanishing of the dynamically generated low-energy Kondo screening scale T_K . Towards obtaining T_K close to the transition, we follow the approach used by Moeller et al. 1995 [497] and Held et al. 2013 [498]. Near the transition, they obtained a Kondo model from the SIAM by applying a Schrieffer-Wolff transition that removes the charge fluctuations of the impurity site and retains the physics of only the Kondo coupling J . This amounts to removing the side peaks from the impurity spectral function and focusing on the low-energy central peak (right panel of Fig. (2.13)). Held et al. then integrated the RG equation for this Kondo model by using a Lorentzian DOS $\rho(D)$ in the bath: $\rho(D) = \frac{\rho_0 \Gamma^2}{D^2 + \Gamma^2}$. This is motivated by



Figure 2.13: Left: Setup for obtaining the low-energy excitations above the strong-coupling ground state. The effects of the rest of the bath is included by treating the hopping between the zeroth site (blue sphere on the left) and the first site (sphere labelled 1) as a perturbation on top of the singlet ground state formed by the impurity (red sphere) and the zeroth site. Right: At r_{c1} , the Hubbard sidebands have become isolated from the central peak in the local spectral function because of the finite optical gap. Removing the UV excitations that couple the low-energy and high-energy bands then leads to a theory of the Kondo model with a Lorentzian DOS.

the fact that the central peak of the impurity spectral function is a Lorentzian, and the bath becomes equivalent to the impurity site under self-consistency.

We implement the same approach on the e-SIAM, such that the Schrieffer-Wolff transformation leads to a $J - U_b$ model with a Lorentzian electronic DOS in the bath. The expression of the Kondo temperature for such a system has been derived in Sec. 6 of the Supplementary Materials [493]. Very close to the transition, we have $r \rightarrow 0.25^-$, and the Kondo temperature is given by

$$T_K = \frac{D_0}{k_B} \exp \left[-\frac{\ln(r_{c2} - r)}{4U_b \rho_0} \right], \quad (7.3)$$

where $r_{c2} = 0.25$ and D_0 is the bare bandwidth. Note that the pre-factor of the logarithm is positive as U_b is negative: $-4U_b \rho_0 = |4U_b \rho_0|$. As we approach the transition, the parameter r takes the limit $r \rightarrow r_{c2}^-$, and the Kondo temperature scale vanishes:

$$\lim_{r \rightarrow r_{c2}^-} T_K = \frac{D_0}{k_B} \lim_{r \rightarrow r_{c2}^-} (r_{c2} - r)^{|4U_b \rho_0|} \rightarrow 0. \quad (7.4)$$

Following the renormalised perturbation theory approach of Hewson [43, 509], the imaginary part of the self-energy of the local Fermi liquid quasiparticles is given by

$$\text{Im} [\Sigma(\omega)] \sim \frac{\mathcal{F}^2 \omega^2}{D_0 T_K^2}, \quad (7.5)$$

while the quasiparticle residue is given by $Z \sim T_K$. The divergence of $\text{Im} [\Sigma(\omega)]$ at the transition and the vanishing of T_K and the quasiparticle residue are important indicators of the loss of the local Fermi liquid excitations and the breakdown of Kondo screening.

The vanishing of the Kondo temperature scale also leads to the divergence of thermodynamic quantities such as the impurity contribution to local spin susceptibility χ_{imp} and the specific heat coefficient γ_{imp} . As the low-energy theory is a local Fermi liquid, χ_{imp} and γ_{imp} retain their Fermi liquid forms but involve the highly renormalised Kondo temperature scale [444, 510]:

$$\begin{aligned} \chi_{\text{imp}} &= \frac{w}{4k_B T_K}, \quad \lim_{r \rightarrow r_{c2}^-} \chi_{\text{imp}} = \frac{w}{4k_B} (r_{c2} - r)^{-4|U_b \rho_0|}, \\ \gamma_{\text{imp}} &= \frac{\pi^2 k_B^2 w}{6T_K}, \quad \lim_{r \rightarrow r_{c2}^-} \gamma_{\text{imp}} = \frac{\pi^2 k_B^2 w}{6} (r_{c2} - r)^{-4|U_b \rho_0|}, \end{aligned} \quad (7.6)$$

where $w \sim 0.4128$ is the Wilson number for the Kondo model [42, 444]. These results also show that the ratio of χ_{imp} and γ_{imp} , referred to as the Wilson ratio $R \equiv \frac{4\pi^2 k_B^2 \chi_{\text{imp}}}{3\gamma_{\text{imp}}}$, remains pinned at the single-channel Kondo value of $R_{\text{LFL}} = 2$ for $r \rightarrow r_{c2}^-$. This shows that the low-frequency Landau quasiparticles are able to survive until very close to the QCP. As we will show in the next subsection, the metallic excitations precisely at the QCP acquire non-Fermi liquid character and involve additional correlations between holons and doublons. These additional correlations are expected to lead to an enhancement of the Wilson ratio. Evidence for this can be found in the DMFT calculation of a local Wilson ratio very close to the Brinkman-Rice transition, $R_{\text{loc}} = 2.9 \pm 0.2$ [18].

2.7.2 Emergence of non-Fermi liquid excitations at the transition

In order to obtain an effective Hamiltonian for the excitations precisely at the critical point $r = 1/4$, we first note that the ground-state subspace of the zero bandwidth $J - U_b$ effective model (eq. (5.3)) becomes degenerate at this point:

$$\frac{1}{\sqrt{2}} (|\uparrow\rangle_d |\downarrow\rangle_0 - |\downarrow\rangle_d |\uparrow\rangle_0), \quad |\sigma\rangle_d |0\rangle_0, \quad |\sigma\rangle_d |2\rangle_0, \quad (\sigma = \uparrow, \downarrow), \quad (7.7)$$

with all the states lying at zero energy. Here, the first ket (with subscript d) represents the impurity spin configuration while the second ket (with subscript 0) represents the configuration of the bath zeroth site). We now diagonalise these states in the presence of the electron hopping t into the rest of the conduction bath, and obtain the effective Hamiltonian for the low-lying excitations mediated by t . This is described in detail in Sec. 7 of the Supplementary Materials [493]. We note that the ground state is four-fold degenerate at energy $-t$ in the presence of the hopping, and comprises the following states:

$$|N_{\text{tot}} = 2, S_{\text{tot}}^z = 0\rangle, \quad |N_{\text{tot}} = 4, S_{\text{tot}}^z = 0\rangle, \quad |N_{\text{tot}} = 3, S_{\text{tot}}^z = \frac{1}{2}\rangle, \quad |N_{\text{tot}} = 3, S_{\text{tot}}^z = -\frac{1}{2}\rangle, \quad (7.8)$$

where $N_{\text{tot}} = \hat{n}_d + \hat{n}_0 + \hat{n}_1$ and $S_{\text{tot}}^z = S_d^z + S_0^z + S_1^z$ are the total number operator and total magnetisation operator respectively for the impurity, bath zeroth and first sites taken together. It is worth observing that

- the presence of the unscreened states $|N_{\text{tot}} = 3, S_{\text{tot}}^z = \pm \frac{1}{2}\rangle$ in the ground-state subspace is a result of the inexact screening at the critical point, and
- the presence of a degenerate ground-state manifold indicates that these excitations of the bath will likely be of the non-Fermi liquid (NFL) kind [460, 511, 512].

The precise form of the effective Hamiltonian for this non-Fermi liquid, as well as the behaviour of various correlation functions, is obtained by considering the ground-state manifold in conjunction with the excited states at energy $-t$.

Since the total Hamiltonian conserves the total spin S_{tot}^z , the effective Hamiltonian separates into two sectors, $S_{\text{tot}}^z = 0$ and $|S_{\text{tot}}^z| = \frac{1}{2}$. Detailed calculations on obtaining this effective Hamiltonian from the RG fixed point theory are shown in Sec. 7 of the Supplementary Materials [493]. In order to highlight certain distinct features, we present simplified effective Hamiltonians only for the $S_{\text{tot}}^z = 0$ and $S_{\text{tot}}^z = \frac{1}{2}$ sectors. We first take a look at the $S_{\text{tot}}^z = 0$ sector:

$$H_{\text{eff}}^{S_{\text{tot}}^z=0} = t \vec{S}_d \cdot (\vec{S}_{1,-} + \vec{S}_{3,-}) . \quad (7.9)$$

The effective spin-1/2 ladder operators $\vec{S}_{1,\pm}$ act on the positive and negative parity sectors (labelled by $\pm$) of the $\hat{n}_0 + \hat{n}_1 = 1$ subspace spanned by the states

$$|\uparrow\uparrow\rangle_{1,\pm} = |\uparrow\rangle_0 |0\rangle_1 \pm |0\rangle_0 |\uparrow\rangle_1, \quad |\downarrow\downarrow\rangle_{1,\pm} = |\downarrow\rangle_0 |0\rangle_1 \pm |0\rangle_0 |\downarrow\rangle_1, \quad (7.10)$$

leading to a Pauli matrix representation of $\vec{S}_{1,\pm}$ in that basis. The Pauli matrix operators $\vec{S}_{3,\pm}$ similarly act on the $\hat{n}_0 + \hat{n}_1 = 3$ subspace spanned by the states $|\uparrow\uparrow\rangle_{3,\pm} = |\uparrow\rangle_0 |2\rangle_1 \pm |2\rangle_0 |\uparrow\rangle_1$ and $|\downarrow\downarrow\rangle_{3,\pm} = |\downarrow\rangle_0 |2\rangle_1 \pm |2\rangle_0 |\downarrow\rangle_1$, obtained by applying the transformation $|0\rangle \rightarrow 2$ on the $\hat{n}_0 + \hat{n}_1 = 1$ counterparts. We now point out two interesting features of this simplified effective Hamiltonian in eq. (7.9).

- The fact that the impurity is now interacting with emergent spins $\vec{S}_1$ and $\vec{S}_3$ that span over two lattice sites (0 and 1) instead of just the zeroth site indicates that the entanglement between the impurity and bath is now extended beyond the bath zeroth site, and that *the singlet is being stretched*.
- The presence of two spins that are trying to simultaneously screen the impurity spin introduces frustration in the dynamics of the impurity. This is reminiscent of the 2-channel Kondo effect, and the connection is made more precise in the concluding section of our work.

Both these features are precursors to the ultimate and complete destruction of screening in the local moment regime, and reflect the fact that the impurity-bath system *is now in an over-screened state*.

We now focus on the effective Hamiltonian of the $S_{\text{tot}}^z \neq 0$ subspace:

$$H_{\text{eff}}^{S_{\text{tot}}^z \neq 0} = -\frac{1}{2}t \left(S_d^- \mathcal{B}_\uparrow^+ + \text{h.c.} \right) . \quad (7.11)$$

The Pauli matrix operators $\mathcal{B}_\uparrow^+$ and $\mathcal{B}_\uparrow^-$ flip between the doublet of states $|\uparrow\rangle_0 |\uparrow\rangle_1$ and $\frac{1}{\sqrt{2}} (|2\rangle_0 |0\rangle_1 + |0\rangle_0 |2\rangle_1)$:

$$\begin{aligned}\mathcal{B}_\uparrow^- |\uparrow\rangle_0 |\uparrow\rangle_1 &= \frac{1}{\sqrt{2}} (|2\rangle_0 |0\rangle_1 + |0\rangle_0 |2\rangle_1), \\ \mathcal{B}_\uparrow^+ \frac{1}{\sqrt{2}} (|2\rangle_0 |0\rangle_1 + |0\rangle_0 |2\rangle_1) &= |\uparrow\rangle_0 |\uparrow\rangle_1.\end{aligned}\quad (7.12)$$

The structure of the effective Hamiltonian in eq. (7.11) leads to another drastic difference from the behaviour of the local Fermi liquid: it allows for the electrons to be “Andreev scattered” into an orthogonal state by hopping from the first site into the zeroth site. This is easily seen by considering the scattering processes in eq. (7.12): a state $|\uparrow\rangle_0 |\uparrow\rangle_1$ (i.e., with site 1 in $\uparrow$ configuration) has a finite probability of being flipped into the $|\downarrow\rangle_0 |2\rangle_1$ state (i.e., with site 1 in the doublon configuration). *An incident electron $c_{1\uparrow}^\dagger$ can therefore emerge as a doublon $c_{1\uparrow}^\dagger c_{1\downarrow}^\dagger$ or a hole $c_{1\uparrow}$ upon scattering from the zeroth site.* This is a direct consequence of the entanglement between the singlet and the first site at the critical point that was absent in the metallic regime of the e-SIAM. We note that similar orthogonal scattering processes also occur in a two-channel Kondo problem between effective pseudo-particles and pseudo-holes [513].

At this point, it is worth noting that the polarised ground-state $|S_{\text{tot}}^z = \frac{1}{2}\rangle$ can be written as an equal superposition of the singlet state $|\text{SS}\rangle_{d0} \otimes |\uparrow\rangle_1$ and the local moment states $\frac{1}{\sqrt{2}} (|\uparrow, 0, 2\rangle - |\uparrow, 2, 0\rangle)$. This symmetry-broken ground state, along with its counterpart $|S_{\text{tot}}^z = -\frac{1}{2}\rangle$, act as a bridge between the ground states of the metallic and insulating phases. They are thus important in displaying the breakdown of the Kondo cloud at the QCP, and the consequences arising from it. We will hence consider these states in the next subsection and demonstrate some exotic non-Fermi liquid properties of these states, e.g., inexact screening of the impurity, fractional impurity magnetisation and fractional impurity entanglement entropy.

Finally, by mapping the quantum impurity problem of the e-SIAM onto that of a classical Coulomb gas [460, 465], it can be shown that the local self-energies $\Sigma_{dd}(\omega)$ and $\Sigma_{00}(\omega)$ of the impurity and zeroth sites respectively and certain two-particle correlation functions have power-law behaviours in the frequency domain:

$$\text{Re} [\Sigma_{dd}] (\omega) \sim |\omega|^{\gamma_{dd}}, \quad \text{Re} [\Sigma_{00}] (\omega) \sim |\omega|^{\gamma_{00}}, \quad (7.13)$$

$$\langle S_d^+ \rangle \sim |\omega|^{(\alpha_1-1)/2}, \quad \langle c_{0\uparrow}^\dagger c_{0\downarrow}^\dagger \rangle \sim |\omega|^{(\alpha_3-1)/2}. \quad (7.14)$$

The precise forms of the exponents in terms of the conduction electron scattering phase shifts, as well as several other technical details of the calculation, are provided in Sec. 8 of the Supplementary Materials [493]. While the various exponents are found to be non-universal functions of the fixed point value of J and the coupling U_b for $r \rightarrow r_{c2}-$, we argue in subsection (2.7.4) that they assume universal values precisely at the QCP ($r = r_{c2}$). The algebraic behaviour of these local correlations is reminiscent of critical behaviour ascribed to the class of local quantum criticality [82, 514].

2.7.3 Impurity magnetisation and entanglement entropy

More indications of non-Fermi liquid behaviour at the QCP is obtained from a calculation of the impurity magnetisation and the impurity entanglement entropy ($S_{\text{EE}}(d)$) (shown in Sec. 9 of the

Supplementary Materials [493]) for the symmetry-broken states $|S_{\text{tot}}^z = \sigma/2\rangle$. This leads to a fractional entanglement entropy (in units of $\log 2$) for each of the two states:

$$\begin{aligned} \rho_{\text{imp}} &= \begin{pmatrix} \frac{1}{2} + m_{\text{imp}}^z & 0 \\ 0 & \frac{1}{2} - m_{\text{imp}}^z \end{pmatrix} = \begin{pmatrix} \frac{3}{4} & 0 \\ 0 & \frac{1}{4} \end{pmatrix}, \\ S_{\text{EE}}(d) &= -\text{Tr}(\rho_{\text{imp}} \ln \rho_{\text{imp}}) \approx 0.81 \log 2, \end{aligned} \quad (7.15)$$

where the impurity magnetisation m_{imp}^z takes the value $m_{\text{imp}}^z = 1/4$ at the QCP (i.e., half the value for a local moment). Further, $S_{\text{EE}}(d)$ can be written in terms of an effective impurity degeneracy g_{imp} , which we define using the impurity magnetisation: $g_{\text{imp}} \equiv 1 + 2|m_{\text{imp}}^z|$, such that it takes the expected values of 1 and 2 in the local Fermi liquid and local moment phases (with m_{imp}^z having values 0 and $\frac{1}{2}$ respectively). The effective degeneracy in the polarised subspace $S_{\text{tot}}^z = \pm 1/2$ at the QCP then turns out to be $3/2$, which is half-way between a unique state and a doublet. This corresponds to partial screening of the impurity degrees of freedom at the QCP, in contrast to complete screening in the metallic phase ($g_{\text{imp}} = 1$) and the absence of screening in the insulating phase ($g_{\text{imp}} = 2$).

In Sec. 9 of the Supplementary Materials [493], we show that the incomplete magnetisation (and hence the fractional value of S_{EE}) arises from the mixing of the local Fermi liquid ground-state $|\phi\rangle = \otimes_{k < k_F} |k \uparrow\rangle |k \downarrow\rangle$ and the gapless excitations above it, $|e\rangle_\sigma = e_\sigma^\dagger |\phi\rangle$, $e \equiv \sum_{k \in FS} c_{k\sigma}^\dagger$, leading to the decay of the local Fermi liquid quasiparticles [511]. This is captured by the modified ground-state $|\pm\rangle$ and lowest-lying excited states $|+\rangle$ at the QCP:

$$|\pm\rangle = \frac{1}{\sqrt{2}} (|SS\rangle \otimes |e\rangle_\sigma \pm |LM\rangle_\sigma \otimes |\phi\rangle). \quad (7.16)$$

The spin-singlet state $|SS\rangle \sim \sum_\sigma \sigma |\sigma\rangle_d |\bar{\sigma}\rangle_0$ and the local moment state $|LM\rangle_\sigma = |\sigma\rangle_d |2\rangle_0$ represent the configurations of the impurity and the bath zeroth sites. Such a ground-state $|\pm\rangle$ should be contrasted with the local Fermi liquid ground-state $|SS\rangle \otimes |\phi\rangle$ that is stable in the metallic phase for $r_{c1} < r < r_{c2}$. This change in the ground-state manifests in the vanishing of the quasiparticle residue of the local Fermi liquid excitations: $Z = |\langle + | e_\sigma^\dagger | - \rangle|^2 = |\langle + | (|LM\rangle_\sigma \otimes |e\rangle_\sigma) \rangle|^2 = 0$, consistent with the orthogonality catastrophe [515] described below eq. (7.5). As the local Fermi liquid quasiparticles are rendered unstable, they are replaced by stable composite three-site excitations of the form, $S_d^- c_{0\bar{\sigma}} e_\sigma^\dagger$. This is further reflected in the fact that the corresponding residue for such three-site composite excitation is non-zero at the QCP: $|\langle + | S_d^- c_{0\bar{\sigma}} e_\sigma^\dagger | - \rangle|^2 > 0$.

2.7.4 Friedel scattering phase shift and fractional excess charge

The orthogonality catastrophe faced by the infrared excitations of the conduction bath at the QCP points towards the presence of an “unrenormalised” phase shift [516] of the scattering k -states. Indeed, using the Friedel sum rule [48, 517, 518], we obtain a $\pi/2$ phase shift for the extended states at the Fermi surface for the QCP, which is only half the unitary limit. This can be seen by recalling that the ground state in eq. (7.16) is an equal admixture of a decoupled local moment state and a singlet state. Through the singlet state, the impurity contributes an excess charge of unity to the Fermi surface of the conduction bath (because of the presence of gapless spin-flip fluctuations). The

local moment state does not contribute any excess charge, because the impurity is decoupled from the bath. The net excess charge then comes out to be $n_{\text{exc}} = \frac{1}{2} (1 + 0) = \frac{1}{2}$. From the Friedel sum rule, the phase shift at the fixed point is $\delta^* = \pi n_{\text{exc}} = \pi/2$, as mentioned above. Expectedly, n_{exc} is zero in the local moment phase. Also, as the exponents of the algebraic form for the local quantities shown in eq. (7.13) are functions of the scattering phase shift, we find that these exponents take universal values precisely at the QCP.

The values of the excess charge at the QCP and in the local moment phase lead to important corollaries. Firstly, the jump in the excess charge by unity in going from $r < r_{c2}$ to $r > r_{c2}$ is in fact a reduction in the Luttinger volume V_L of the conduction bath [24, 519]. V_L counts the number of extended states present at and below the Fermi volume, and corresponds to a topological quantum number [50, 51, 346, 520]. Since the excess charge n_{exc} calculated above is simply the impurity contribution to V_L at the Fermi surface, the difference in the excess charge across the transition implies that the two phases acquire different values for the invariant V_L . Thus, n_{exc} tracks the change (ΔV_L) in the Luttinger volume, and the transition is topological in nature. Secondly, in a recent work, Sen et al. [521] have shown that the Mott MIT of the infinite-dimensional Hubbard model proceeds through the dissociation of domain walls in a fictitious Su-Schrieffer-Heeger chain connected to the physical lattice sites. If one couples their observation with the fact that the two dissociated domain walls at the ends of the SSH chain are together known to host a single charge [522, 523], it becomes evident that the fractional excess charge we obtain at the QCP corresponds to the state that is localised in the SSH chain near the physical lattice site as obtained by Sen et al. The well-known bulk-boundary correspondence of the SSH model (see, e.g., [524]) suggests that the excess charge n_{exc} and the change ΔV_L in the Luttinger volume are the respective topological invariants of the boundary and bulk in the e-SIAM that are tied to one another. Further, the appearance of a half-quantised n_{exc} at the MIT of the e-SIAM corresponds, in the SSH model, to the dissociation of a domain wall-anti domain wall pair with each carrying a half charge [522, 523].

We show in Sec. 9 of the Supplementary Materials [493] that it is in fact possible to connect the scattering phase shift (δ^*), impurity magnetisation (m_{imp}^z), effective degeneracy ($\tilde{g}$) and excess charge (n_{exc}) in a single relation:

$$\delta^* = \pi n_{\text{exc}} = \left[1 + \left(\frac{2 - \tilde{g}}{\tilde{g} - 1} \right)^2 \right]^{-1} = \frac{4m_{\text{imp}}^z{}^2}{1 - 4|m_{\text{imp}}^z| + 8m_{\text{imp}}^z{}^2}. \quad (7.17)$$

As we now discuss, this provides a unified picture of the effect of frustration on the impurity spin. As we have seen earlier, the presence of U_b in the Hamiltonian of the e-SIAM introduces local moment states into the spectrum and leads to states with non-vanishing impurity magnetisation at the QCP. This non-zero magnetisation can be interpreted as a partial screening of the impurity spin, in turn resulting in an effective impurity degeneracy ($1 < \tilde{g} < 2$ between that of a unique screened state and an unscreened local moment. The partial screening also manifests in only half the excess charge being contributed to the conduction bath. This reduction in the excess charge acts as a change in the boundary conditions felt by the conduction electrons coupled to the impurity and manifests as a phase shift that is less than the unitarity limit of π .

2.8 Discussions and Outlook

In summary, we have shown that an attractive correlation on the bath zeroth site is enough to frustrate the Kondo effect and stabilise the local moment phase. The destruction of the Kondo cloud, and the associated local Fermi liquid, occurs through *pairing fluctuations in the bath and proximate to the impurity* (left panel of Fig. (2.8)). This is reminiscent of a subdominant superconducting tendency that was observed in the half-filled Hubbard model at $T = 0$ from a unitary RG treatment in Refs. [174,435]. We also note the finding of non-local attractive effective interactions in a recent theoretical study of the 2D Hubbard model [525]. We find that the critical point displays non-Fermi liquid behaviour with a vanishing quasiparticle residue Z . The strong agreement of our results with several aspects of DMFT suggests that the local self-energy obtained self-consistently in that method can be represented quite faithfully through the e-SIAM. It will, thus, be important to test the predictions offered by our approach directly within the DMFT method. We note that the non-Fermi liquid at the critical point displays a *partial correlation of doublons and holons* that is distinct from not only the correlated Fermi liquid metal (unbound holons and doublons), but also the paramagnetic insulator (comprised of bound holons and doublons). The excitations that propagate through the two unscreened states in the ground-state subspace (Eq. (7.8)) involve the simultaneous creation of a doublon and a holon: in these channels, the doublons and holons cannot propagate in the absence of each other. Such an incomplete correlation is an intermediate step towards complete confinement in the local moment phase. It appears interesting to experimentally test some of these ideas on a mesoscopic quantum dot [526,527] which is, apart from the usual electron tunnel coupling to an electronic reservoir, additionally coupled through the proximity effect to a superconducting lead.

We stress here that the e-SIAM analysed in this work represents an impurity model that (i) has a single correlated channel of conduction electrons, (ii) is consistent with the symmetries of a half-filled Hubbard model (SU(2)-spin, U(1)-charge and particle-hole), and (iii) shows a local metal-insulator transition. The model, therefore, provides a minimal route towards obtaining a Hamiltonian-based understanding of DMFT. Further, at the level of the renormalisation group flows that involve the frustration of Kondo screening, the effect of the attractive on-site interaction U_b introduced by us is equivalent to that of additional bath correlations that have been introduced in related impurity models. These include a single impurity Anderson model with multiple anisotropic conduction channels studied by Giamarchi et al. [528], and a periodic Anderson model enhanced with explicit s-d coupling (as well as a density-density interaction) between the conduction and impurity bands, studied by Si and Kotliar [460,472,529] as well as by Ruckenstein et al., [530]. In the former K -channel model of Ref. [528], the z -component (V_l , $l \in [1, K - 1]$) of the Kondo interaction term in the $K - 1$ additional conduction channels acts as a source of frustration for the spin-flip term γ_0 of the Kondo interaction in the original $l = 0$ channel, leading to a breakdown of Kondo screening. The U_b coupling in the e-SIAM acts as a similar source of frustration: the equivalence lies at the level of the renormalisation group equations, such that both V_l and U_b oppose the RG relevance of γ_0 and J respectively. This can be made more precise by comparing the RG equations of our work (eqs.(9.36)) and that of Giamarchi et al. [528]: the mapping between the two frustration terms U_b and V_l is found to be of the form $U_b \sim \sum_{l=1}^{K-1} V_l^2 \rho$, where ρ is the conduction bath DOS. A similar relationship can be found between the RG equations obtained by us and those for the extended periodic Anderson model studied by Si and Kotliar [460]. In this sense, our analysis applies

to a wide variety of models where the Kondo effect is stable in a certain regime of parameters but is destroyed in other parameter regimes by some form of quantum-mechanical frustration of the impurity spin degree of freedom.

Extensions of the present work involve analysing the mixed valence regime of the impurity site (i.e., away from half-filling) within the e-SIAM auxiliary model. This allows the possibility that the finite-temperature critical end-point of the DMFT first-order transition could turn into a quantum critical point from the merging of the $T = 0$ ESQPT and QPT observed in the present work. Recent DMFT calculations [158] show a shrinking of the coexistence region with hole doping but fall short of revealing a QCP. If such a QCP does exist, can it harbour pair fluctuations between the impurity and zeroth bath sites that become dominant upon doping, signalling thereby a putative superconducting instability of a related bulk model? We recall that a superconducting state of matter was indeed observed to be emergent from a QCP in the hole-doped Hubbard model at $T = 0$ via a unitary RG treatment in Refs. [183, 435]. The phenomenon of electronic differentiation in k -space can perhaps be captured by considering cluster variants of the e-SIAM, i.e., multiple impurities connected to one another through single-particle hopping and/or RKKY-like interactions [531, 532]. In general, the presence of multiple and varied classes of correlations in the auxiliary model - localisation from Mott physics, delocalisation from spin and charge fluctuations, and pairing from local attractive correlations - makes this a strong candidate for an auxiliary model that carries the potential for describing the emergence of a variety of novel phases of correlated quantum matter.

2.9 Appendix: Derivation of renormalisation group equations for the extended-SIAM

The Hamiltonian we are working with is the extended Anderson impurity model (described in Section III of the main manuscript):

$$\mathcal{H} = -\frac{1}{2}U (\hat{n}_{d\uparrow} - \hat{n}_{d\downarrow})^2 + \sum_{\vec{k}, \sigma} \epsilon_{\vec{k}} \tau_{\vec{k}, \sigma} + J \vec{S}_d \cdot \vec{S}_0 + V \sum_{\sigma} \left(c_{d\sigma}^{\dagger} c_{0\sigma} + \text{h.c.} \right) - \frac{1}{2}U_b (\hat{n}_{0\uparrow} - \hat{n}_{0\downarrow})^2. \quad (9.1)$$

The renormalisation in the Hamiltonian $H_{(j)}$ at the j^{th} RG step, upon decoupling an electronic state $q\beta$, is given by the expression:

$$(\Delta H_{(j)})_{\vec{q}, \beta} = H_{(j-1)} - H_{(j)} = c_{\vec{q}, \beta}^{\dagger} T_{\vec{q}, \beta} \frac{1}{\omega_e - H_D} T_{\vec{q}, \beta}^{\dagger} c_{\vec{q}, \beta} + T_{\vec{q}, \beta}^{\dagger} c_{\vec{q}, \beta} \frac{1}{\omega_h - H_D} c_{\vec{q}, \beta}^{\dagger} T_{\vec{q}, \beta}, \quad (9.2)$$

where $\omega_{e,h}$ are the quantum fluctuation scales for the electron and hole scattering channels, H_D is the part of the Hamiltonian that is diagonal in k -space, and $T_{\vec{q}, \beta} + T_{\vec{q}, \beta}^{\dagger}$ is the part of the Hamiltonian that does not commute with $\hat{n}_{\vec{q}, \beta}$ (in short, it is the part that is off-diagonal with respect to a particular state $\vec{q}, \beta$). This off-diagonal part is made up of contributions from V, J and U_b : $T_{\vec{q}\uparrow} = \left[V c_{d\uparrow}^{\dagger} + J \sum_{\vec{k}} S_d^+ c_{\vec{k}\downarrow}^{\dagger} + U_b \sum_{\vec{k}_1, \vec{k}_2, \vec{k}_3} c_{\vec{k}_1\uparrow}^{\dagger} c_{\vec{k}_3\downarrow}^{\dagger} c_{\vec{k}_4\downarrow} \right] c_{\vec{q}\uparrow}$, while the diagonal part is given by $H_D = \epsilon_{\vec{q}} \tau_{q\uparrow} + \frac{1}{4} J S_d^z (\hat{n}_{q\uparrow} - \hat{n}_{q\downarrow}) - \frac{1}{2} U_b (\tau_{q\uparrow} - \tau_{q\downarrow})^2$. The contribution from U_b is obtained by taking the

Hartree contributions corresponding to the states $|q\sigma\rangle$ from the full interacting term in the Hamiltonian. Note that we have ignored potential scattering terms in the off-diagonal part $T_{q\beta}$. In order to allow both spin-flip and charge transfer scattering processes, we start from initial states with $\tau_{q\uparrow} = -\tau_{q\downarrow}$. As a result, the contribution to H_D from the kinetic energy is $\epsilon_q \tau_{q\uparrow} = (\pm D) \times \left(\pm \frac{1}{2}\right) = D/2$, while the contribution from U_b is $-\frac{1}{2}U_b \left(\pm \frac{1}{2} - \left(\mp \frac{1}{2}\right)\right)^2 = -U_b/2$.

2.9.1 Renormalisation of U_b

U_b can renormalise only via itself. The relevant renormalisation term in the particle sector is

$$U_b^2 \sum_{q\beta} \sum_{\substack{k_1, k_2, k_3, \\ k'_1, k'_2, k'_3}} c_{q\beta}^\dagger c_{k_1\beta} c_{k_3\beta}^\dagger c_{k'_1\bar{\beta}} \frac{1}{\omega - H_D} c_{k'_2\bar{\beta}}^\dagger c_{k'_3\bar{\beta}} c_{k_2\beta}^\dagger c_{q\beta} . \quad (9.3)$$

In order to renormalise U_b , we need to contract one more pair of momenta. There are two choices. The first is by setting $k_3 = k'_3 = q$. The two internal states, then, are $q\beta$ and $q\bar{\beta}$. As discussed above, the intermediate state energy is $-U_b/4$. We therefore have

$$\frac{U_b^2 n_j}{\omega - D/2 + \frac{U_b}{2}} \sum_{\beta} \sum_{k_1, k_2, k'_1, k'_2} c_{k_1\beta} c_{k'_1\bar{\beta}} c_{k'_2\bar{\beta}}^\dagger c_{k_2\beta}^\dagger = \frac{U_b^2 n_j}{\omega - D/2 + \frac{U_b}{2}} \sum_{\beta} \sum_{k_1, k_2, k'_1, k'_2} c_{k'_2\bar{\beta}}^\dagger c_{k'_1\bar{\beta}} c_{k_2\beta}^\dagger c_{k_1\beta} \quad (9.4)$$

Another way to contract the momenta is by setting $k'_1 = k'_2 = q$, which gives a renormalisation of

$$\frac{U_b^2 n_j}{\omega - D/2 + \frac{U_b}{2}} \sum_{\beta} \sum_{k_1, k_2, k_3, k'_3} c_{k_1\beta} c_{k_3\bar{\beta}}^\dagger c_{k'_3\bar{\beta}} c_{k_2\beta}^\dagger = -\frac{U_b^2 n_j}{\omega - D/2} \sum_{\beta} \sum_{k_1, k_2, k_3, k'_3} c_{k_3\bar{\beta}}^\dagger c_{k'_3\bar{\beta}} c_{k_2\beta}^\dagger c_{k_1\beta} . \quad (9.5)$$

The two contributions cancel each other. The same cancellation happens in the hole sector as well.

2.9.2 Renormalisation of the impurity correlation U

The coupling U is renormalised by two kinds of vertices: V^2 and J^2 . We will consider these processes one at a time. For convenience, we define $\epsilon_d = -U/2$.

The renormalisation arising from the first kind of terms, in the particle sector, is

$$\begin{aligned} \sum_{q\beta} c_{q\beta}^\dagger c_{d\beta} \frac{V^2}{\omega - H_D} c_{d\beta}^\dagger c_{q\beta} &= \sum_{q\beta} V^2 \hat{n}_{q\beta} (1 - \hat{n}_{d\beta}) \left(\frac{1 - \hat{n}_{d\bar{\beta}}}{\omega_0 - E_0} + \frac{\hat{n}_{d\bar{\beta}}}{\omega_1 - E_1} \right) \\ &= V^2 n_j \sum_{\beta} (1 - \hat{n}_{d\beta}) \left(\frac{1 - \hat{n}_{d\bar{\beta}}}{\omega_0 - E_0} + \frac{\hat{n}_{d\bar{\beta}}}{\omega_1 - E_1} \right) . \end{aligned} \quad (9.6)$$

q runs over the momentum states that are being decoupled at this RG step: $|q| = \Lambda_j$. $E_{1,0}$ are the diagonal parts of the Hamiltonian at $\hat{n}_{d\bar{\beta}} = 1, 0$ respectively: $E_1 = \frac{D}{2}$ and $E_0 = \frac{D}{2} + \epsilon_d - \frac{J}{4}$. In order to relate ω_0 and ω_1 with the common fluctuation scale ω for the conduction electrons, we

will replace these quantum fluctuation scales with the current renormalised values of the single-particle self-energy for the initial state from which we started scattering. For $\hat{n}_{d\bar{\beta}} = 0$, there is no additional self-energy because the impurity does not have any spin: $\omega_0 = \omega$. For $\hat{n}_{d\bar{\beta}} = 1$, we have an additional self-energy of ϵ_d arising from the correlation on the impurity: $\omega_1 = \omega + \epsilon_d$. Substituting the values of $E_{0,1}$ and $\omega_{0,1}$, we get

$$V^2 n_j \sum_{\beta} (1 - \hat{n}_{d\beta}) \left(\frac{1 - \hat{n}_{d\bar{\beta}}}{\omega - \frac{D}{2} - \epsilon_d + \frac{J}{4}} + \frac{\hat{n}_{d\bar{\beta}}}{\omega - \frac{D}{2} + \epsilon_d} \right). \quad (9.7)$$

Performing a similar calculation for the hole sector gives the contribution:

$$V^2 n_j \sum_{\beta} \hat{n}_{d\beta} \left(\frac{1 - \hat{n}_{d\bar{\beta}}}{\omega - \frac{D}{2} + \epsilon_d} + \frac{\hat{n}_{d\bar{\beta}}}{\omega - \frac{D}{2} - \epsilon_d + \frac{J}{4}} \right). \quad (9.8)$$

We now come to the second class of terms: spin-spin. We first look at the particle sector:

$$\frac{J^2}{4} \sum_{q\beta} c_{d\bar{\beta}}^{\dagger} c_{d\beta} c_{q\bar{\beta}}^{\dagger} c_{-q\bar{\beta}} \frac{1}{\omega - H_D} c_{d\beta}^{\dagger} c_{d\bar{\beta}} c_{q\bar{\beta}}^{\dagger} c_{q\beta} = \frac{J^2}{4} n_j \frac{1}{\omega - \frac{D}{2} + \frac{J}{4}} \sum_{\beta} \hat{n}_{d\bar{\beta}} (1 - \hat{n}_{d\beta}). \quad (9.9)$$

The diagonal part in the denominator was simple to deduce in this case because the nature of the scattering requires the spins S_d^z and $\frac{\beta}{2} (\hat{n}_{q\beta} - \hat{n}_{q\bar{\beta}})$ to be anti-parallel. This ensures that the intermediate state has an energy of $E = \frac{D}{2} + \epsilon_d - \frac{J}{4}$, and the quantum fluctuation scale is $\omega' = \omega + \epsilon_d$, such that $\omega' - E = \omega - \frac{D}{2} + \frac{J}{4}$. In the hole sector, we have

$$\frac{J^2}{4} n_j \frac{1}{\omega - \frac{D}{2} + \frac{J}{4}} \sum_{\beta} \hat{n}_{d\beta} (1 - \hat{n}_{d\bar{\beta}}). \quad (9.10)$$

From eqs. (9.7), (9.8), (9.9) and (9.10), we write

$$\begin{aligned} \Delta U &= \Delta\epsilon_2 + \Delta\epsilon_0 - 2\Delta\epsilon_1 \\ &= -\frac{4V^2 n_j}{\omega + \frac{U_b}{2} - \frac{D}{2} - U/2} + \frac{4V^2 n_j}{\omega + \frac{U_b}{2} - \frac{D}{2} + U/2 + \frac{J}{4}} - \frac{J^2 n_j}{\omega + \frac{U_b}{2} - \frac{D}{2} + \frac{J}{4}}, \end{aligned} \quad (9.11)$$

where we have restored the contribution from U_b in the denominator.

2.9.3 Renormalisation of the hybridisation V

Renormalisation of V happens through two kinds of processes: VJ and VU_b . Within the first kind, the scattering can be either via S_d^z or through $S_d^{\pm}$. For the first kind, we have the following contribution in the particle sector:

$$\begin{aligned} &\sum_{q\beta} V c_{q\beta}^{\dagger} c_{d\beta} \frac{1}{\omega - H_D} \frac{1}{4} J \sum_k (\hat{n}_{d\beta} - \hat{n}_{d\bar{\beta}}) c_{k\beta}^{\dagger} c_{q\beta} \\ &= \frac{1}{4} V J n_j \frac{1}{2} \left(\frac{1}{\omega'_1 - E} + \frac{1}{\omega'_2 - E} \right) \sum_{k\beta} (1 - \hat{n}_{d\bar{\beta}}) c_{d\beta} c_{k\beta}^{\dagger}. \end{aligned} \quad (9.12)$$

The transformation from $\frac{1}{\omega - H_D}$ to $\frac{1}{2} \left(\frac{1}{\omega'_1 - E} + \frac{1}{\omega'_2 - E} \right)$ is made so that we can account for both the initial state and the final state energies through the two fluctuation scales ω'_1 and ω'_2 respectively; we calculate the denominators for both the initial and final states and then take the mean of the two (hence the factor of half in front). This was not required previously because, in the earlier scattering processes, the impurity returned to its initial state at the end, at least in terms of $\epsilon_d (\hat{n}_{d\uparrow} - \hat{n}_{d\downarrow})^2$, and so we had $\omega'_1 = \omega'_2 = \omega'$. Substituting the energies and the ω -scales, we get

$$- \left(\frac{\frac{n_j}{4} V J \frac{1}{2}}{\omega - \frac{D}{2} + \frac{J}{4}} + \frac{\frac{n_j}{4} V J \frac{1}{2}}{\omega - \frac{D}{2} - \epsilon_d + \frac{J}{4}} \right) \sum_{k\beta} (1 - \hat{n}_{d\bar{\beta}}) c_{k\beta}^\dagger c_{d\beta}. \quad (9.13)$$

One can generate another such process by exchanging the single-particle process and the spin-exchange process:

$$\sum_{q\beta} \frac{1}{4} J \sum_k (\hat{n}_{d\beta} - \hat{n}_{d\bar{\beta}}) c_{q\beta}^\dagger c_{k\beta} \frac{1}{\omega - H_D} V c_{d\beta}^\dagger c_{q\beta}. \quad (9.14)$$

This is simply the Hermitian conjugate of the previous contribution. Combining this with the previous then gives

$$-\frac{n_j}{8} V J \left(\frac{1}{\omega - \frac{D}{2} + \frac{J}{4}} + \frac{1}{\omega - \frac{D}{2} - \epsilon_d + \frac{J}{4}} \right) \sum_{k\beta} (1 - \hat{n}_{d\bar{\beta}}) \times (c_{d\beta}^\dagger c_{k\beta} + \text{h.c.}). \quad (9.15)$$

If we similarly calculate the contributions from the spin-exchange processes involving $S_d^\pm$, we get

$$-\frac{1}{4} V J n_j \left(\frac{1}{\omega - \frac{D}{2} + \frac{J}{4}} + \frac{1}{\omega - \frac{D}{2} - \epsilon_d + \frac{J}{4}} \right) \sum_{k\beta} (1 - \hat{n}_{d\beta}) (c_{k\bar{\beta}}^\dagger c_{d\bar{\beta}} + \text{h.c.}). \quad (9.16)$$

The contributions from the hole sector are obtained by making the transformation $\hat{n}_{d\bar{\beta}} \rightarrow 1 - \hat{n}_{d\bar{\beta}}$ on the particle sector contributions. The total renormalisation to V from VJ processes are

$$-\frac{3n_j}{8} V J \left(\frac{1}{\omega + U_b/4 - \frac{D}{2} + \frac{J}{4}} + \frac{1}{\omega + U_b/4 - \frac{D}{2} + U/2 + \frac{J}{4}} \right) \sum_{k\beta} (c_{d\beta}^\dagger c_{k\beta} + \text{h.c.}). \quad (9.17)$$

The renormalisation in V from U_b arises through terms of VU_b and U_bV kind. One particular scattering process in this category involves adding an electron on the impurity site:

$$\sum_{k,\beta,q} V c_{d\beta}^\dagger c_{q\beta} \frac{1}{\omega - H_D} U_b c_{q\beta}^\dagger c_{k\beta} n_{\bar{q},\bar{\beta}} = \sum_{k,\beta,q,i} V c_{d\beta}^\dagger c_{q\beta} \frac{1}{2} \left(\frac{n_{d\bar{\beta}}}{\omega_i^1 - E_1} + \frac{1 - n_{d\bar{\beta}}}{\omega_i^0 - E_0} \right) U_b c_{q\beta}^\dagger c_{k\beta} n_{\bar{q},\bar{\beta}}, \quad (9.18)$$

where E_1 and E_0 are excited state energies associated with the occupied and unoccupied configurations of $d\bar{\beta}$, while $\omega_0^{1/0}$ and $\omega_1^{1/0}$ represent the energy associated with the initial and final states in the scattering process, for the particle and hole projectors of $d\bar{\beta}$ respectively. When projected

by $n_{d\bar{\beta}}$, the initial and excited states have $n_d = 1$, leading to $E_1 = D/2 - U_b/2 - U/2 - J/4$ and $\omega_0^1 = \omega - U/2$ while the final state has $n_d = 2$ which gives $\omega_1^1 = \omega$. When projected by $1 - n_{d\bar{\beta}}$, we have $n_d = 0$ in the initial and excited state resulting in $E_0 = D/2 - U_b/2$ and $\omega_0^0 = \omega$ while the final state has $n_d = 1$ leading to $\omega_1^0 = \omega - U/2$. Combining them gives

$$\begin{aligned} & \sum_{k,\beta,q} V U_b c_{d\beta}^\dagger c_{k\beta} (1 - n_{q\beta}) n_{\bar{q},\bar{\beta}} \frac{1}{2} \left[\frac{n_{d\bar{\beta}}}{\omega - D/2 + U_b/2 + J/4} + \frac{n_{d\bar{\beta}}}{\omega - D/2 + U_b/2 + J/4 + U/2} \right] \\ & + \sum_{k,\beta,q} V U_b c_{d\beta}^\dagger c_{k\beta} (1 - n_{q\beta}) n_{\bar{q},\bar{\beta}} \frac{1}{2} \left[\frac{1 - n_{d\bar{\beta}}}{\omega - D/2 + U_b/2} + \frac{1 - n_{d\bar{\beta}}}{\omega - D/2 + U_b/2 + U/2} \right] \end{aligned} \quad (9.19)$$

Another process is constructed by removing an electron from the impurity site:

$$\sum_{k,\beta,\bar{q}} V c_{\bar{q}\beta}^\dagger c_{d\beta} \frac{1}{\omega - H_D} U_b c_{k\beta}^\dagger c_{\bar{q}\beta} n_{\bar{q},\bar{\beta}} = \sum_{k,\beta,q,i} V c_{\bar{q}\beta}^\dagger c_{d\beta} \frac{1}{2} \left(\frac{n_{d\bar{\beta}}}{\omega_i^1 - E_1} + \frac{1 - n_{d\bar{\beta}}}{\omega_i^0 - E_0} \right) U_b c_{k\beta}^\dagger c_{\bar{q}\beta} n_{\bar{q},\bar{\beta}}. \quad (9.20)$$

By identifying the various configurations, we again obtain the various energies: $E_1 = D/2 - U_b/2$, $\omega_0^1 = \omega$, $\omega_1^1 = \omega - U/2$, and $E_0 = D/2 - U_b/2 - U/2 - J/4$, $\omega_0^0 = \omega - U/2$, $\omega_1^0 = \omega$. Substituting these, we get

$$\begin{aligned} & - \sum_{k,\beta,q,i} V U_b c_{k\beta}^\dagger c_{d\beta} n_{\bar{q}\beta} n_{\bar{q},\bar{\beta}} \frac{1}{2} \left[\frac{n_{d\bar{\beta}}}{\omega - D/2 + U_b/2} + \frac{n_{d\bar{\beta}}}{\omega - D/2 + U_b/2 - U/2} \right] \\ & - \sum_{k,\beta,q,i} V U_b c_{k\beta}^\dagger c_{d\beta} n_{\bar{q}\beta} n_{\bar{q},\bar{\beta}} \frac{1}{2} \left[\frac{1 - n_{d\bar{\beta}}}{\omega - D/2 + U_b/2 + J/4} + \frac{1 - n_{d\bar{\beta}}}{\omega - D/2 + U_b/2 + U/2 + J/4} \right] \end{aligned} \quad (9.21)$$

Another set of processes is generated by reversing the two operations. Corresponding to the first one, we have

$$\sum_{k,\beta,q} U_b c_{\bar{q}\beta}^\dagger c_{k\beta} n_{\bar{q},\bar{\beta}} \frac{1}{\omega - H_D} V c_{d\beta}^\dagger c_{\bar{q}\beta}, \quad (9.22)$$

but this is just the hermitian conjugate of the process analysed in eq. 9.20, and we can write down the final result directly:

$$\begin{aligned} & - \sum_{k,\beta,q,i} V U_b c_{d\beta}^\dagger c_{k\beta} n_{\bar{q}\beta} n_{\bar{q},\bar{\beta}} \frac{1}{2} \left[\frac{n_{d\bar{\beta}}}{\omega - D/2 + U_b/2} + \frac{n_{d\bar{\beta}}}{\omega - D/2 + U_b/2 - U/2} \right] \\ & - \sum_{k,\beta,q,i} V U_b c_{d\beta}^\dagger c_{k\beta} n_{\bar{q}\beta} n_{\bar{q},\bar{\beta}} \frac{1}{2} \left[\frac{1 - n_{d\bar{\beta}}}{\omega - D/2 + U_b/2 + J/4} + \frac{1 - n_{d\bar{\beta}}}{\omega - D/2 + U_b/2 + U/2 + J/4} \right]. \end{aligned} \quad (9.23)$$

The final processes we have to consider is the flipped version of eq. 9.20:

$$\sum_{k,\beta,q} U_b c_{k\beta}^\dagger c_{q\beta} n_{\bar{q},\bar{\beta}} \frac{1}{\omega - H_D} V c_{q\beta}^\dagger c_{d\beta}, \quad (9.24)$$

but this is again the hermitian conjugate of eq. 9.18, so we get

$$\begin{aligned} & \sum_{k,\beta,q} V U_b c_{k\beta}^\dagger c_{d\beta} (1 - n_{q\beta}) n_{\bar{q},\bar{\beta}} \frac{1}{2} \left[\frac{n_{d\bar{\beta}}}{\omega - D/2 + U_b/2 + J/4 + U/2} + \frac{n_{d\bar{\beta}}}{\omega - D/2 + U_b/2 + J/4} \right] \\ & + \sum_{k,\beta,q} V U_b c_{k\beta}^\dagger c_{d\beta} (1 - n_{q\beta}) n_{\bar{q},\bar{\beta}} \frac{1}{2} \left[\frac{1 - n_{d\bar{\beta}}}{\omega - D/2 + U_b/2 - U/2} + \frac{1 - n_{d\bar{\beta}}}{\omega - D/2 + U_b/2} \right] \end{aligned} \quad (9.25)$$

The first term gives

$$\begin{aligned} & - \sum_{k\beta} c_{d\beta}^\dagger c_{k\beta} \left[\frac{\hat{n}_{d\bar{\beta}}}{2} \left(\frac{n_j U_b V}{\omega - \frac{D}{2} - \frac{U}{2} + \frac{U_b}{4}} + \frac{n_j U_b V}{\omega - \frac{D}{2} + \frac{U_b}{4}} \right) \right. \\ & \left. + \frac{1 - \hat{n}_{d\bar{\beta}}}{2} \left(\frac{n_j U_b V}{\omega - \frac{D}{2} + \frac{U_b}{4} + \frac{U}{2} + \frac{J}{4}} + \frac{n_j U_b V}{\omega - \frac{D}{2} + \frac{U_b}{4} + \frac{J}{4}} \right) \right]. \end{aligned} \quad (9.26)$$

The second term is of the form $\sum_{q\beta} \sum_k U_b V c_{q\beta}^\dagger c_{d\beta} \frac{1}{\omega - H_D} \hat{n}_{q\bar{\beta}} c_{k\beta}^\dagger c_{q\beta}$, and this is just the Hermitian conjugate of the previous term, so these two terms together lead to

$$\begin{aligned} & -n_j U_b V \sum_{k\beta} \left(c_{d\beta}^\dagger c_{k\beta} + \text{h.c.} \right) \left[\frac{\hat{n}_{d\bar{\beta}}}{2} \left(\frac{1}{\omega - \frac{D}{2} - \frac{U}{2} + \frac{U_b}{4}} + \frac{1}{\omega - \frac{D}{2} + \frac{U_b}{4}} \right) \right. \\ & \left. + \frac{1 - \hat{n}_{d\bar{\beta}}}{2} \left(\frac{1}{\omega - \frac{D}{2} + \frac{U_b}{4} + \frac{U}{2} + \frac{J}{4}} + \frac{1}{\omega - \frac{D}{2} + \frac{U_b}{4} + \frac{J}{4}} \right) \right]. \end{aligned} \quad (9.27)$$

In the hole sector, we have

$$-n_j U_b V \sum_{k\beta} c_{k\beta}^\dagger \left[\frac{\hat{n}_{d\bar{\beta}}}{2} \left(\frac{1}{\omega_1 - E_1} + \frac{1}{\omega'_1 - E_1} \right) + \frac{1 - \hat{n}_{d\bar{\beta}}}{2} \left(\frac{1}{\omega_0 - E_0} + \frac{1}{\omega'_0 - E_0} \right) \right] c_{d\beta}. \quad (9.28)$$

$E_1 = D/2 - U_b/4 - U/2 - J/4$, $E_0 = D/2 - U_b/4 - K/4$. The fluctuation scales are $\omega_1 = \omega = \omega'_0$, $\omega'_1 = \omega - U/2 = \omega_0$. Substituting these gives

$$\begin{aligned} & - \sum_{k\beta} c_{d\beta}^\dagger c_{k\beta} \left[\frac{1 - \hat{n}_{d\bar{\beta}}}{2} \left(\frac{n_j U_b V}{\omega - \frac{D}{2} - \frac{U}{2} + \frac{U_b}{4}} + \frac{n_j U_b V}{\omega - \frac{D}{2} + \frac{U_b}{4}} \right) + \frac{\hat{n}_{d\bar{\beta}}}{2} \left(\frac{n_j U_b V}{\omega - \frac{D}{2} + \frac{U_b}{4} + \frac{U}{2} + \frac{J}{4}} \right. \right. \\ & \left. \left. + \frac{n_j U_b V}{\omega - \frac{D}{2} + \frac{U_b}{4} + \frac{J}{4}} \right) \right]. \end{aligned} \quad (9.29)$$

The other term, obtained by exchanging V and U_b , gives the Hermitian conjugate, so the overall contribution from the hole sector is the same as the total contribution from the particle sector, but with $\hat{n}_{d\bar{\beta}} \rightarrow 1 - \hat{n}_{d\bar{\beta}}$. Combining both the sectors, we get

$$\begin{aligned} & - \sum_{k\beta} \left(c_{d\beta}^\dagger c_{k\beta} + \text{h.c.} \right) \left[\left(\frac{n_j U_b V/2}{\omega - \frac{D}{2} - \frac{U}{2} + \frac{U_b}{4}} + \frac{n_j U_b V/2}{\omega - \frac{D}{2} + \frac{U_b}{4}} \right) + \left(\frac{n_j U_b V/2}{\omega - \frac{D}{2} + \frac{U_b}{4} + \frac{U}{2} + \frac{J}{4}} \right. \right. \\ & \left. \left. + \frac{n_j U_b V/2}{\omega - \frac{D}{2} + \frac{U_b}{4} + \frac{J}{4}} \right) \right]. \end{aligned} \quad (9.30)$$

2.9.4 Renormalisation of the spin-exchange coupling J

The term $J\vec{S}_d \cdot \vec{S}_0$ can be split into three parts: $J^z S_d^z$, $\frac{1}{2}J^+ S_d^+ S_0^-$ and $\frac{1}{2}J^- S_d^- S_0^+$. We will only calculate the renormalisation in J^+ , which will be equal to that of J^- , J^z due to spin-rotation symmetry. The terms that renormalise J^+ are of the form $S_d^+ S_d^z$ and $S_d^z S_d^+$. In the particle sector, we have

$$\begin{aligned} - \sum_q \sum_{kk'} \frac{1}{4} J^2 S_d^+ c_{q\downarrow}^\dagger c_{k'\uparrow} \frac{1}{\omega - H_D} S_d^z c_{k\downarrow}^\dagger c_{q\downarrow} &= n_j \frac{1}{4} J^2 \left(-\frac{1}{2} S_d^+ \right) \sum_{kk'} c_{k\downarrow}^\dagger c_{k'\uparrow} \frac{1}{\omega + \frac{U_b}{2} - \frac{D}{2} + \frac{J}{4}}, \\ \sum_q \sum_{kk'} \frac{1}{4} J^2 S_d^z c_{q\uparrow}^\dagger c_{k'\uparrow} \frac{1}{\omega - H_D} S_d^+ c_{k\downarrow}^\dagger c_{q\uparrow} &= -n_j \frac{1}{4} J^2 \left(\frac{1}{2} S_d^+ \right) \sum_{kk'} c_{k\downarrow}^\dagger c_{k'\uparrow} \frac{1}{\omega + \frac{U_b}{2} - \frac{D}{2} + \frac{J}{4}}. \end{aligned} \quad (9.31)$$

The denominator is determined using $E = \frac{D}{2} + \epsilon_d - \frac{J}{4}$ and $\omega' = \omega + \epsilon_d$. In the hole sector, we similarly have

$$\begin{aligned} \sum_q \sum_{kk'} \frac{1}{4} J^2 S_d^+ c_{k\downarrow}^\dagger c_{q\uparrow} \frac{1}{\omega - H_D} S_d^z c_{q\uparrow}^\dagger c_{k'\uparrow} &= n_j \frac{1}{4} J^2 \left(-\frac{1}{2} S_d^+ \right) \sum_{kk'} c_{k\downarrow}^\dagger c_{k'\uparrow} \frac{1}{\omega + \frac{U_b}{2} - \frac{D}{2} + \frac{J}{4}}, \\ - \sum_q \sum_{kk'} \frac{1}{4} J^2 S_d^z c_{k\downarrow}^\dagger c_{q\downarrow} \frac{1}{\omega - H_D} S_d^+ c_{q\downarrow}^\dagger c_{k'\uparrow} &= -n_j \frac{1}{4} J^2 \left(\frac{1}{2} S_d^+ \right) \sum_{kk'} c_{k\downarrow}^\dagger c_{k'\uparrow} \frac{1}{\omega + \frac{U_b}{2} - \frac{D}{2} + \frac{J}{4}}. \end{aligned} \quad (9.32)$$

The renormalisation due to U_b also happens through multiple terms. One of the terms is

$$\frac{1}{2} J U_b \sum_q \sum_{k,k'} S_d^+ c_{q\downarrow}^\dagger c_{k\uparrow} \frac{1}{\omega - H_D} \hat{n}_{q\uparrow} c_{k'\downarrow}^\dagger c_{q\downarrow} = -\frac{1}{2} \frac{J U_b n_j}{\omega - \frac{D}{2} + \frac{U_b}{2} + \frac{J}{4}} \sum_{k,k'} S_d^+ c_{k'\downarrow}^\dagger c_{k\uparrow}. \quad (9.33)$$

The factor of half in front is the same half factor that appears in front of the $S_1^+ S_2^-$, $S_1^- S_2^+$ terms when we rewrite $\vec{S}_1 \cdot \vec{S}_2$ in terms of $S^z, S^\pm$. Another term is obtained by switching J and U_b :

$$\frac{1}{2} J U_b \sum_q \sum_{k,k'} \hat{n}_{q\downarrow} c_{q\uparrow}^\dagger c_{k\uparrow} \frac{1}{\omega - H_D} S_d^+ c_{k'\downarrow}^\dagger c_{q\uparrow} = -\frac{1}{2} \frac{J U_b n_j}{\omega - \frac{D}{2} + \frac{U_b}{2} + \frac{J}{4}} \sum_{k,k'} S_d^+ c_{k'\downarrow}^\dagger c_{k\uparrow}. \quad (9.34)$$

The corresponding terms in the hole sector are

$$-\frac{1}{2} \frac{J U_b n_j}{\omega - \frac{D}{2} + \frac{U_b}{2} + \frac{J}{4}} \sum_{k,k'} S_d^+ c_{k'\downarrow}^\dagger c_{k\uparrow}, -\frac{1}{2} \frac{J U_b n_j}{\omega - \frac{D}{2} + \frac{U_b}{2} + \frac{J}{4}} \sum_{k,k'} S_d^+ c_{k'\downarrow}^\dagger c_{k\uparrow}. \quad (9.35)$$

2.9.5 Final URG equations

In summary, the renormalisation in the couplings takes the form:

$$\begin{aligned} \Delta U &= 4V^2 n_j \left(\frac{1}{d_1} - \frac{1}{d_0} \right) - n_j \frac{J^2}{d_2}, & \Delta V &= -\frac{3n_j V}{8} \left[J \left(\frac{1}{d_2} + \frac{1}{d_1} \right) + \frac{4U_b}{3} \sum_{i=1}^4 \frac{1}{d_i} \right], \\ \Delta J &= -\frac{n_j J (J + 4U_b)}{d_2}, & \Delta U_b &= 0, \end{aligned} \quad (9.36)$$

where the denominators d_i are given by

$$d_0 = \omega - \frac{D}{2} + \frac{U_b}{2} - \frac{U}{2}, \quad d_1 = \omega - \frac{D}{2} + \frac{U_b}{2} + \frac{U}{2} + \frac{J}{4}, \quad (9.37)$$

$$d_2 = \omega - \frac{D}{2} + \frac{U_b}{2} + \frac{J}{4}, \quad d_3 = \omega - \frac{D}{2} + \frac{U_b}{2}. \quad (9.38)$$

For the sake of completeness, we present the RG equation for a charge isospin Kondo coupling K between the impurity and the bath:

$$\Delta K = -\frac{n_j K (K + 4U_b)}{\omega - \frac{D}{2} + \frac{U_b}{2} + \frac{K}{4}}. \quad (9.39)$$

Indeed, the RG equation for K is observed to be identical in form to that for the spin Kondo coupling J , obtained through the transformation $J \rightarrow K$. This indicates that charge fluctuations between the bath zeroth and first sites (that are incited by an attractive interaction U_b) lead to the RG irrelevance of $K(> 0)$, and thereby a destabilisation of the charge Kondo effect similar to that presented in the main manuscript for J .

2.10 Appendix: Expressing correlation functions in terms of entanglement

We will first relate the impurity Greens function to the geometric entanglement. Given a ground state $|\Psi\rangle_{\text{gs}}$ and a spectrum of energies $\{E_n\}$, the retarded impurity Greens function (in time domain) is defined as $G_{d\sigma}(t) = -i\theta(t) \langle \{c_{d\sigma}(t), c_{d\sigma}^\dagger\} \rangle$. It can be given a spectral representation in terms of the eigenstates $\{|\Psi\rangle_n\}$ of the e-SIAM:

$$G_{d\sigma}(\omega) = \frac{1}{Z} \sum_n \left[\frac{|\langle \Psi_{\text{gs}} | c_{d\sigma} | \Psi_n \rangle|^2}{\omega + E_{\text{gs}} - E_n} + \frac{|\langle \Psi_n | c_{d\sigma} | \Psi_{\text{gs}} \rangle|^2}{\omega - E_{\text{gs}} + E_n} \right], \quad (10.1)$$

where $Z \equiv \sum_n e^{-\beta E_n}$ is the partition function. We will now insert a complete basis into the expression. The basis will be the set of eigenstates of the Hamiltonian $H = H_1 + H_2$, where H_1 is the two-site $J - U_b$ Hamiltonian formed by the impurity and zeroth sites, and H_2 is a tight-binding Hamiltonian formed by the remaining sites. Since the Hamiltonians are decoupled, the eigenstates $|\Psi\rangle_{m,n}$ will be direct product states formed by combining the eigenstates $|\phi\rangle_m, |\psi\rangle_n$ of the individual Hamiltonians H_1 and H_2 respectively: $|\Phi\rangle_{m,n} = |\phi\rangle_m \otimes |\psi\rangle_n$. Inserting this basis leads to the expression:

$$|\Psi\rangle_{\text{gs}} = \sum_{m,n} |\Phi\rangle_{m,n} (\langle \phi_m | \otimes \langle \psi_n |) |\Psi_{\text{gs}}\rangle. \quad (10.2)$$

The ground state of H_1 is the spin-singlet: $|\phi\rangle_0 = |ss\rangle$. We denote the ground state of the tight-binding Hamiltonian H_2 by $|\psi\rangle_0$. Because of the highly renormalised couplings V and J , the impurity site is almost maximally entangled with the zeroth site, such that the ground state $|\Psi\rangle_{\text{gs}}$ in the $r <$

0.25 regime can be thought of as a direct product of the two-site ground state, $c|ss\rangle + \sqrt{1-c^2}|ct\rangle$, in direct product with the tight-binding ground state:

$$|\Psi\rangle_{\text{gs}} \simeq \left(c|ss\rangle + \sqrt{1-c^2}|ct\rangle \right) \otimes |\psi\rangle_0 = |\Phi\rangle_{\text{ss}} + |\Phi\rangle_{\text{ct}} , \quad (10.3)$$

where $|\Phi\rangle_{\text{ss(ct)}} = |ss(ct)\rangle \otimes |\psi\rangle_0$. This suggests that not all terms in the summation of eq. (10.2) contribute; out of all $\{|\psi\rangle_n\}$, only the ground state $n = 0$ contributes, while only $|ss\rangle$ and $|ct\rangle$ contribute from the set $\{|\phi\rangle_m\}$. The summation then simplifies to

$$|\Psi\rangle_{\text{gs}} = |\Phi\rangle_{\text{ss}} \langle ss|\Psi_{\text{gs}}^{(2)}\rangle + |\Phi\rangle_{\text{ct}} \langle ct|\Psi_{\text{gs}}^{(2)}\rangle , \quad (10.4)$$

where $|\Psi_{\text{gs}}^{(2)}\rangle$ is the two-site part of the ground state and can be obtained by starting with the full ground state $|\Psi\rangle_{\text{gs}}$ and tracing over the other lattice sites of the system.

We assume that the global phases of the wavefunctions are real, and since the internal weights of the wavefunctions are real as well, the overlaps $\langle ss|\Psi_{\text{gs}}^{(2)}\rangle$, $\langle ct|\Psi_{\text{gs}}^{(2)}\rangle$ are also real. We can use these overlaps to define a geometric measure of entanglement [242, 494, 495]:

$$\varepsilon(\psi_1, \psi_2) = 1 - |\langle \psi_1 | \psi_2 \rangle|^2 . \quad (10.5)$$

If $|\psi_1\rangle$ is thought of as a separable state, then $|\psi_2\rangle$ should be less entangled if its overlap with $|\psi_1\rangle$ is large, which is indeed borne out by the definition. The overlaps then become $\langle ss|\Psi_{\text{gs}}^{(2)}\rangle = \sqrt{1 - \varepsilon(ss, \Psi_{\text{gs}}^{(2)})}$, and similarly for the state $|ct\rangle$. For brevity, we will use the notation $\varepsilon_{\text{ss}} \equiv \varepsilon(ss, \Psi_{\text{gs}}^{(2)})$, $\varepsilon_{\text{ct}} \equiv \varepsilon(ct, \Psi_{\text{gs}}^{(2)})$. The retarded impurity Greens function for spin σ can now be written in terms of these measures:

$$\begin{aligned} G_{d\sigma}(\omega) = \frac{1}{Z} \sum_n & \left[(1 - \varepsilon_{\text{ss}}) \left(\frac{|\langle \Phi_{\text{ss}} | c_{d\sigma} | \Psi_n \rangle|^2}{\omega + E_{\text{gs}} - E_n} + \frac{|\langle \Psi_n | c_{d\sigma} | \Phi_{\text{ss}} \rangle|^2}{\omega - E_{\text{gs}} + E_n} \right) + (1 - \varepsilon_{\text{ct}}) \frac{|\langle \Phi_{\text{ct}} | c_{d\sigma} | \Psi_n \rangle|^2}{\omega + E_{\text{gs}} - E_n} \right. \\ & + (1 - \varepsilon_{\text{ct}}) \frac{|\langle \Psi_n | c_{d\sigma} | \Phi_{\text{ct}} \rangle|^2}{\omega - E_{\text{gs}} + E_n} + \sqrt{(1 - \varepsilon_{\text{ss}})} \sqrt{(1 - \varepsilon_{\text{ct}})} \frac{\langle \Phi_{\text{ss}} | c_{d\sigma} | \Psi_n \rangle \langle \Psi_n | c_{d\sigma}^\dagger | \Phi_{\text{ct}} \rangle + \text{h.c.}}{\omega + E_{\text{gs}} - E_n} \\ & \left. + \sqrt{(1 - \varepsilon_{\text{ss}})} \sqrt{(1 - \varepsilon_{\text{ct}})} \frac{\langle \Phi_{\text{ct}} | c_{d\sigma} | \Psi_n \rangle \langle \Psi_n | c_{d\sigma}^\dagger | \Phi_{\text{ss}} \rangle + \text{h.c.}}{\omega - E_{\text{gs}} + E_n} \right] . \quad (10.6) \end{aligned}$$

We now generalise this to any real-space two-particle correlation involving operators O_1, O_2 that are at most two-particle operators and act on the combined Hilbert space of the impurity site and the zeroth site. The so-called lesser Greens function corresponding to these operators is defined as

$$G_{O_1^\dagger, O_2}^<(t) \equiv i \langle O_1^\dagger O_2(t) \rangle , \quad G_{O_1^\dagger, O_2}^<(t \rightarrow \infty) = i \langle \Psi_{\text{gs}} | O_1^\dagger O_2 | \Psi_{\text{gs}} \rangle . \quad (10.7)$$

We focus on the static case ($t \rightarrow \infty$) here. Following eq. (10.4), we can rewrite the ground states in terms of the entanglement measures mentioned above.

$$|\Psi\rangle_{\text{gs}} = |\Phi\rangle_{\text{ss}} \sqrt{1 - \varepsilon_{\text{ss}}} + |\Phi\rangle_{\text{ct}} \sqrt{1 - \varepsilon_{\text{ss}}} , \quad (10.8)$$

where $|\Phi\rangle_{\text{ss(ct)}}$ are the zero spin and zero charge isospin eigenstates of the Hamiltonian $H = H_1 + H_2$ defined below eq. (10.1). Substituting this in the static lesser Greens function gives

$$\begin{aligned} \frac{1}{i} G_{O_1^\dagger, O_2}^<(t \rightarrow \infty) = & (1 - \varepsilon_{\text{ss}}) \langle \Phi_{\text{ss}} | O_2 O_1^\dagger | \Phi_{\text{ss}} \rangle + (1 - \varepsilon_{\text{ct}}) \langle \Phi_{\text{ct}} | O_2 O_1^\dagger | \Phi_{\text{ct}} \rangle \\ & + \sqrt{1 - \varepsilon_{\text{ss}}} \sqrt{1 - \varepsilon_{\text{ct}}} \left(\langle \Phi_{\text{ss}} | O_2 O_1^\dagger | \Phi_{\text{ct}} \rangle + \langle \Phi_{\text{ct}} | O_2 O_1^\dagger | \Phi_{\text{ss}} \rangle \right). \end{aligned} \quad (10.9)$$

This can be extended to generalised retarded Greens functions $G_{O_1^\dagger, O_2}^R(t) \equiv -i\theta(t) \langle \{O_1^\dagger(t), O_2\} \rangle$, whose spectral representation is of the form

$$G_{O_1^\dagger, O_2}^R(\omega) = \frac{1}{Z} \sum_n \left[\frac{\langle \Psi_{\text{gs}} | O_2 | \Psi_n \rangle \langle \Psi_n | O_1^\dagger | \Psi_{\text{gs}} \rangle}{\omega + E_{\text{gs}} - E_n} + \frac{\langle \Psi_n | O_2 | \Psi_{\text{gs}} \rangle \langle \Psi_{\text{gs}} | O_1^\dagger | \Psi_n \rangle}{\omega - E_{\text{gs}} + E_n} \right]. \quad (10.10)$$

Following an approach very similar to the one that led to eq. (10.6), we can cast the generalised Greens function in terms of the geometric entanglement measures:

$$\begin{aligned} G_{O_1^\dagger, O_2}^R(\omega) = \frac{1}{Z} \sum_n \left[(1 - \varepsilon_{\text{ss}}) \left(\frac{(O_1)_{\text{ss},n}^* (O_2)_{\text{ss},n}}{\omega + E_{\text{gs}} - E_n} + \frac{(O_1)_{n,\text{ss}}^* (O_2)_{n,\text{ss}}}{\omega - E_{\text{gs}} + E_n} \right) + (1 - \varepsilon_{\text{ct}}) \frac{(O_1)_{\text{ct},n}^* (O_2)_{\text{ct},n}}{\omega + E_{\text{gs}} - E_n} \right. \\ \left. + (1 - \varepsilon_{\text{ct}}) \frac{(O_1)_{n,\text{ct}}^* (O_2)_{n,\text{ct}}}{\omega - E_{\text{gs}} + E_n} + \sqrt{(1 - \varepsilon_{\text{ss}})} \sqrt{(1 - \varepsilon_{\text{ct}})} \frac{(O_2)_{\text{ss},n} (O_1)_{\text{ct},n}^* + \text{h.c.}}{\omega + E_{\text{gs}} - E_n} \right. \\ \left. + \sqrt{(1 - \varepsilon_{\text{ss}})} \sqrt{(1 - \varepsilon_{\text{ct}})} \frac{(O_2)_{\text{ct},n} (O_1)_{\text{ss},n}^* + \text{h.c.}}{\omega - E_{\text{gs}} + E_n} \right], \end{aligned} \quad (10.11)$$

where $(O_2)_{\text{ss},n}$ is the matrix element $\langle \Phi_{\text{ss}} | O_2 | \Psi_n \rangle$, and similar definitions exist for $(O_1)_{\text{ss},n}$ and its $|\Phi_{\text{ct}}\rangle$ counterparts.

Closely related to geometric entanglement is the quantum Fisher information (QFI) F_Q [496, 533], which quantifies how sensitive a state ρ is to unitary transformations generated by an observable $\hat{O}$. For a pure state $\rho = |\psi\rangle \langle \psi|$, $F_Q(\psi, \hat{O})$ is defined in terms of the variance of the operator $\hat{O}$ (i.e., the connected correlation function):

$$F_Q(\psi, \hat{O}) = 4\Delta(\hat{O})^2 = 4 \left(\langle \psi | \hat{O}^2 | \psi \rangle - \langle \psi | \hat{O} | \psi \rangle^2 \right), \quad (10.12)$$

and it provides an upper bound on the precision that can be achieved in measuring the parameter θ that is dual to the observable $\hat{O}$: $\mathcal{N}(\Delta\theta)^2 \geq F_Q^{-1}$, $\mathcal{N}$ being the number of independent measurements [496, 533]. $F_Q(\psi, \hat{O})$ is thus a measure of the entanglement arising from quantum fluctuation content in $|\psi\rangle$ (considered with respect to an eigenstate of $\hat{O}$). This can be made more manifest by considering a traceless operator $\hat{M}$ with eigenstates $\{|m\rangle\}$. Without loss of generality, we can rescale the operator such that $\sum m^2 = 1/2$. There are several examples of such operators in the context of the present work, such as the impurity magnetisation operator (S_d^z), the Kondo spin-flip operator ($S_d^+ S_0^- + \text{h.c.}$) and the Kondo isospin-flip operator ($C_d^+ C_0^- + \text{h.c.}$). If $\varepsilon_m(\psi) \equiv 1 - |\langle m | \psi \rangle|^2$ is the geometric entanglement between a given state $|\psi\rangle$ and the eigenstates of $\hat{M}$, the QFI corresponding to $\hat{M}$ in the state ψ can be written as

$$F_Q(\psi, \hat{M}) = 4 \left[\frac{1}{2} - \sum_m m^2 \varepsilon_m - \left(\sum_m m \varepsilon_m \right)^2 \right]. \quad (10.13)$$

The total geometric entanglement $\sum_m \varepsilon_m$ is constrained to be $d_M - 1$, where d_M is the dimension of the operator $\hat{M}$. Because each eigenvalue m has a magnitude of at most $1/\sqrt{2}$ (all m^2 must add up to $1/2$), the maximum magnitude of the last two terms in eq. (10.13) (and hence the minimum value of the QFI) is attained for the case when some of the ε_m are zero. However, because the eigenstates are orthogonal, only one of them can have perfect overlap with the state $|\psi\rangle$ and only one ε_m can be zero at a time. The case of minimum QFI therefore corresponds to $\varepsilon_{m^*} = 0, \varepsilon_{m \neq m^*} = 1$, leading to $F_Q = 0$. The opposite situation arises when all the geometric entanglement measures are non-zero and equal, $\varepsilon_m = 1/d_M$, leading to a maximum QFI value of $F_Q = 2/d_M$. A uniformly spread geometric entanglement distribution, therefore, leads to a larger QFI, while a more focused distribution leads to a smaller QFI. This sums up the relation between the quantum Fisher information and geometric entanglement.

If the operator $\hat{M}$ is such that its eigenstates are separable states from the perspective of a certain party, the vanishing QFI arises when $|\psi\rangle$ has zero geometric entanglement with respect to one of the separable states, and the QFI saturates when $|\psi\rangle$ has a finite geometric entanglement with all of the states. A large QFI can thus be associated with more entanglement. Moreover, if the eigenstates of $\hat{M}$ are symmetry-broken states, a vanishing QFI indicates that $|\psi\rangle$ is very close to such a symmetry-broken state, while a large QFI indicates that $|\psi\rangle$ is a uniform superposition of such symmetry-broken states and therefore preserves the symmetry as a whole. A larger QFI points towards the presence of more quantum fluctuations and the lack of susceptibility towards symmetry-breaking.

In order to demonstrate these ideas in the present problem, we computed the QFI in the ground-state of the e-SIAM for a number of Hermitian operators, as a function of the parameter $r = -U_b/J_0$. These are shown in the left panel of Fig. (2.14) (left panel). In order to explain the behaviour depicted in the figure, we point out again that the ground-states for $r \ll r_{c2}$, $r \lesssim r_{c2}$ and $r > r_{c2}$ are $|\text{SS}\rangle + |\text{CT}\rangle$, $|\text{SS}\rangle$ and $|\text{LM}\rangle$ respectively, where $|\text{SS}\rangle$, $|\text{CT}\rangle$ and $|\text{LM}\rangle$ are the spin-singlet state, charge triplet and local moment states respectively. We focus on the spin-flip QFI, corresponding to the operator $\hat{O} = S_d^+ S_0^- + \text{h.c.}$.

- For $r < r_{c1}$, the ground-state involves both $|\text{SS}\rangle$ and $|\text{CT}\rangle$, meaning that neither of the two entanglement measures ε_{SS} or ε_{CT} will be zero. As mentioned in the preceding paragraph, this leads to the maximal QFI, as can be seen in the figure as well.
- At $r = r_{c1}$, the charge triplet content vanishes and the ground-state is purely a singlet beyond that point. As a result, in the range $r_{c1} < r < r_{c2}$, the geometric entanglement corresponding to the singlet state is zero, leading to a minimal and vanishing QFI. This is again shown in the figure.
- Finally, for $r > r_{c2}$, the local moment states become the ground-states, where the bath zeroth site displays all four configurations as a superposition, because of the hopping into the rest of the bath. This state is again not an eigenstate of the spin-flip operator, and the geometric entanglement will be uniformly distributed across the states, none being zero. This explains the large QFI in the local moment phase.

Similar explanations hold for the single-particle hopping FQI between the impurity and the zeroth site, and the charge isospin FQI between the zeroth site and the first site. In Fig. (2.14) (right

panel), we show the QFI for the same three operators very close to the MIT (i.e., for $r \lesssim r_{c2}$). Two important quantities that track the local MIT and act as order parameters for the transition are the degree of compensation for the impurity ($\langle \vec{S}_d \cdot \vec{S}_0 \rangle$) and the impurity magnetisation ($\langle S_d^z \rangle$), and the first two QFI (red and blue curves) in Fig. 2.14 are therefore particularly important because that they quantify the quantum fluctuations present in the system corresponding to these order parameters. Our analysis shows that the two phases on either side of the transition are characterised by distinct values of this pair of QFI: while the QFI corresponding to the degree of compensation zero in the Kondo screened phase, it becomes non-zero in the local moment phase, and the opposite is true for the QFI arising from the impurity magnetisation. The phase precisely at the transition is distinct from those on either side because it displays a non-zero value for both the QFI. While it is expected that a critical point would show enhanced fluctuations of multiple kinds (giving rise to universality), it is enlightening to find that this is also reflected in a measure of many-particle entanglement.

All three QFIs are observed to saturate to finite, non-zero values at the MIT, indicating that the non-Fermi liquid state therein possesses quantum fluctuations of all three varieties.

We will now relate the QFI to some other measures of correlation. The QFI corresponding to an observable can be directly related to the static lesser Greens functions associated with that observable:

$$F_Q(\psi, \hat{O}_2) = 4 \left(\langle \hat{O}_2 \hat{O}_1^\dagger \rangle \Big|_{\hat{O}_1^\dagger = \hat{O}_2} - \langle \hat{O}_2 \hat{O}_1^\dagger \rangle^2 \Big|_{\hat{O}_1 = 1} \right) = 4 \left[G_{O_2, O_2}^<(t \rightarrow \infty) - \left(G_{1, O_2}^<(t \rightarrow \infty) \right)^2 \right] \quad (10.14)$$

In the context of the present work, eq. (10.9) allows relating the QFI to the geometric entanglement measured with respect to the spin-singlet and charge triplet states. By combining eqs. (10.9) and (10.14), we get

$$\begin{aligned} F_Q(\psi, \hat{O}_2) &= 4 \left[(1 - \varepsilon_{ss}) \langle \Phi_{ss} | \hat{O}_2^2 | \Phi_{ss} \rangle + (1 - \varepsilon_{ct}) \langle \Phi_{ct} | \hat{O}_2^2 | \Phi_{ct} \rangle \right. \\ &\quad + \sqrt{1 - \varepsilon_{ss}} \sqrt{1 - \varepsilon_{ct}} \left(\langle \Phi_{ss} | \hat{O}_2^2 | \Phi_{ct} \rangle + \langle \Phi_{ct} | \hat{O}_2^2 | \Phi_{ss} \rangle \right) - \left\{ (1 - \varepsilon_{ss}) \langle \Phi_{ss} | \hat{O}_2 | \Phi_{ss} \rangle \right. \\ &\quad \left. \left. + (1 - \varepsilon_{ct}) \langle \Phi_{ct} | \hat{O}_2 | \Phi_{ct} \rangle + \sqrt{1 - \varepsilon_{ss}} \sqrt{1 - \varepsilon_{ct}} \left(\langle \Phi_{ss} | \hat{O}_2 | \Phi_{ct} \rangle + \langle \Phi_{ct} | \hat{O}_2 | \Phi_{ss} \rangle \right) \right\}^2 \right] \quad (10.15) \end{aligned}$$

Further, the retarded Greens function defined above in eq. (10.10) can very generally be related to the QFI. By using the simplified forms $\frac{i}{2} G_{O_2, O_2}^R(t \rightarrow \infty) = \langle (O_2)^2 \rangle$ and $\frac{i}{2} G_{1, O_2}^R(t \rightarrow \infty) = \langle O_2 \rangle$, we get

$$F_Q(\psi, \hat{O}_2) = 2i G_{O_2, O_2}^R(t \rightarrow \infty) + \left(G_{1, O_2}^R(t \rightarrow \infty) \right)^2. \quad (10.16)$$

For the sake of completeness, we generalise the above relations to finite temperatures by adopting the approach of Hauke et al. [496]. For a mixed state at inverse temperature β characterised by the density matrix $\rho = \sum_n f_n |n\rangle \langle n|$ where $(E_n, |n\rangle)$ are the eigenvalue-eigenstate pairs and $f_n = e^{-\beta E_n} / \sum_n e^{-\beta E_n}$ are the Boltzmann weights, the FQI can be assigned a spectral representation of the form [496]

$$F_Q(\rho, \hat{O}) = 2 \sum_{m,n} \frac{(f_m - f_n)^2}{f_m + f_n} |\langle m | \hat{O} | n \rangle|^2. \quad (10.17)$$

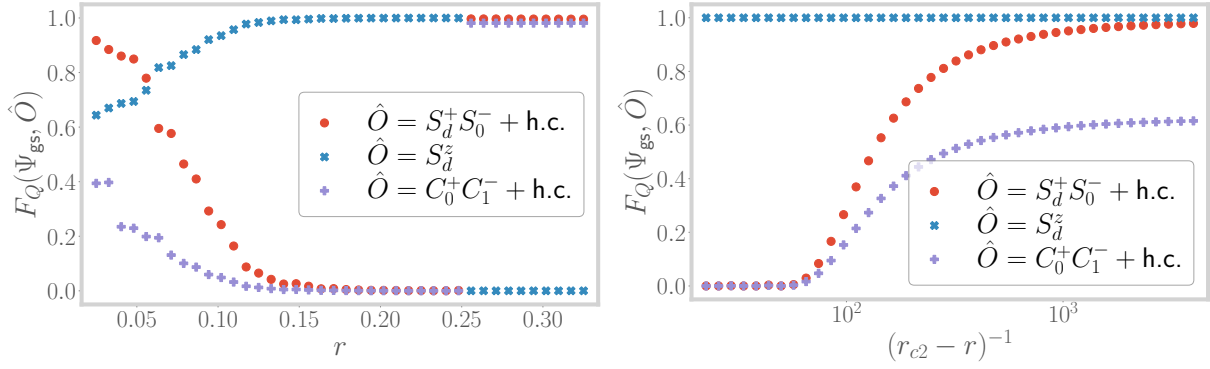


Figure 2.14: Left: Variation of the quantum Fisher information for three separate fluctuations, across the entire range of r . Right: The QFI corresponding to the same three operators, but very close to the QCP at r_{c2} .

By using the identity $\frac{f_m - f_n}{f_m + f_n} = \int_{-\infty}^{\infty} d\omega \delta(\omega + E_m - E_n) \tanh(\beta\omega/2)$, the finite temperature FQI can be written in the form

$$F_Q(\rho, \hat{O}) = 2 \int_{-\infty}^{\infty} d\omega \tanh(\beta\omega/2) \sum_{m,n} (f_m - f_n) \delta(\omega + E_m - E_n) |\langle m | \hat{O} | n \rangle|^2. \quad (10.18)$$

In order to relate this with a correlation function, we now point out that the lesser Greens function $G^<$ defined in eq. (10.7), as well as the greater Greens function $G^>$ that we will now define, also admit spectral representations in terms of the eigenstates $|n\rangle$ of the full problem:

$$G_{O^\dagger, O}^<(t) \equiv i \langle \hat{O}^\dagger \hat{O}(t) \rangle \implies G_{O^\dagger, O}^<(\omega) = 2\pi i \sum_{m,n} f_n \delta(\omega + E_m - E_n) |\langle m | \hat{O}^\dagger | n \rangle|^2 \quad (10.19)$$

$$G_{O^\dagger, O}^>(t) \equiv -i \langle \hat{O}(t) \hat{O}^\dagger \rangle \implies G_{O^\dagger, O}^>(\omega) = -2\pi i \sum_{m,n} f_m \delta(\omega + E_m - E_n) |\langle m | \hat{O} | n \rangle|^2 \quad (10.20)$$

If $\hat{O}$ is a Hermitian operator, eqs. (10.19) can be used to rewrite the QFI in terms of $G_{O^\dagger, O}^<(\omega)$. Indeed, by combining eqs. (10.19) and eq. (10.18) with the assumption $\hat{O} = \hat{O}^\dagger$, we get

$$F_Q(\rho, \hat{O}) = -\frac{1}{\pi i} \int_{-\infty}^{\infty} d\omega \tanh(\beta\omega/2) \left(G_{O, O}^>(\omega) + G_{O, O}^<(\omega) \right). \quad (10.21)$$

This constitutes the general relation between the finite temperature QFI for a general Hermitian many-particle operator $\hat{O}$ and the corresponding many-particle correlations at zero and finite frequencies.

The Kubo linear response function associated with the operator $\hat{O}(t)$ due to a perturbation from the same operator is associated with a correlation function/susceptibility $C_{\hat{O}}(t, t') = i\theta(t - t') \langle [\hat{O}(t), \hat{O}(t')] \rangle$. By going to the frequency domain, the imaginary part of such a function can

be related to the sum of the lesser and greater Greens functions:

$$\begin{aligned}
 C_{\hat{O}}(\omega) &= i \int_0^\infty d(t-t') e^{i\omega(t-t')} \left(\langle \hat{O}(t) \hat{O}(t') \rangle - \langle \hat{O}(t') \hat{O}(t) \rangle \right) \\
 \Rightarrow C_{\hat{O}}(\omega) - C_{\hat{O}}(\omega)^* &= i \int_0^\infty d(t-t') \left(e^{i\omega(t-t')} - e^{-i\omega(t-t')} \right) \left(\langle \hat{O}(t) \hat{O}(t') \rangle - \langle \hat{O}(t') \hat{O}(t) \rangle \right) \\
 &= i \int_{-\infty}^\infty d(t-t') e^{i\omega(t-t')} \left(\langle \hat{O}(t) \hat{O}(t') \rangle - \langle \hat{O}(t') \hat{O}(t) \rangle \right) \\
 &= - \left(G_{O,O}^>(\omega) + G_{O,O}^<(\omega) \right) \tag{10.22}
 \end{aligned}$$

This allows us to connect the QFI with the imaginary part of the Kubo correlation function/susceptibility obtained in Ref. [496]:

$$F_Q(\rho, \hat{O}) = \frac{1}{\pi i} \int_{-\infty}^\infty d\omega \tanh(\beta\omega/2) (C_{\hat{O}}(\omega) - C_{\hat{O}}(\omega)^*) . \tag{10.23}$$

2.11 Appendix: One-shot decoupling of impurity from zeroth site

We will decouple the impurity states from the fixed point Hamiltonian using a single URG transformation. This will, of course, generate correlations on the zeroth site. The resulting Hamiltonian will be an AIM with the hopping between the zeroth site and the rest of the chain acting as the effective hybridisation. Schematically, we will have

$$\begin{aligned}
 H^* = H_{\text{imp}} + V_{\text{imp-0}} + H_0 + H_{0-1} + H_{\text{rest}} &\xrightarrow{\text{decouple imp.}} E_{\text{imp}} + \tilde{H}_0 + H_{0-1} + H_{\text{rest}} \\
 &= H_{\text{new imp}} + V_{\text{new imp - rest}} + H_{\text{rest}} . \tag{11.1}
 \end{aligned}$$

Since the impurity is not coupled with any site beyond the zeroth site, the parts of the Hamiltonian that involve "rest" will not change in the process. This also means that the decoupling can be performed by looking at the smaller Hamiltonian

$$\begin{aligned}
 H_{\text{imp+0}} = H_{\text{imp}} + H_0 + V_{\text{imp-0}} = & -\frac{U^*}{2} (\hat{n}_{d\uparrow} - \hat{n}_{d\downarrow})^2 - U_b (\hat{n}_{0\uparrow} - \hat{n}_{0\downarrow})^2 + J^* S_d^z S_0^z \\
 & + V^* \sum_{\sigma} \left(c_{d\sigma}^\dagger c_{0\sigma} + \text{h.c.} \right) + \frac{J^*}{2} \left(c_{d\uparrow}^\dagger c_{d\downarrow} c_{0\downarrow}^\dagger c_{0\uparrow} + \text{h.c.} \right) . \tag{11.2}
 \end{aligned}$$

2.11.1 Renormalisation from V^*

From the off-diagonal term involving V^* , we generate the following term, in the particle sector for 0:

$$\sum_{\sigma} c_{0\sigma}^\dagger c_{d\sigma} \frac{V^{*2}}{\tilde{\omega} - H_d} c_{d\sigma}^\dagger c_{0\sigma} = \sum_{\sigma} c_{0\sigma}^\dagger c_{d\sigma} \frac{V^{*2}}{\tilde{\omega} + \frac{U^*}{2} (1 - \hat{n}_{d\bar{\sigma}})^2 + U_b \hat{n}_{0\bar{\sigma}} + \frac{J^*}{4} (1 - \hat{n}_{d\bar{\sigma}}) \hat{n}_{0\bar{\sigma}}} c_{d\sigma}^\dagger c_{0\sigma} \tag{11.3}$$

where $\tilde{\omega}$ is the quantum fluctuation operator for the impurity site and $H_d = -\frac{U^*}{2} (\hat{n}_{d\uparrow} - \hat{n}_{d\downarrow})^2 - U_b (\hat{n}_{0\uparrow} - \hat{n}_{0\downarrow})^2 + J^* S_d^z S_0^z$, is the diagonal part of the Hamiltonian for the imp+0 system. In order to

resolve the operators in the denominator, we expand the unity in the numerator using the identity $1 = \hat{n}_{d\bar{\sigma}}\hat{n}_{0\bar{\sigma}} + \hat{n}_{d\bar{\sigma}}\hat{h}_{0\bar{\sigma}} + \hat{h}_{d\bar{\sigma}}\hat{n}_{0\bar{\sigma}} + \hat{h}_{d\bar{\sigma}}\hat{h}_{0\bar{\sigma}}$, where $\hat{h} = 1 - \hat{n}$ is hole operator. On Substituting this, we get

$$\begin{aligned} \sum_{\sigma} c_{0\sigma}^{\dagger} c_{d\sigma} \frac{V^{*2}}{\bar{\omega} - H_d} c_{d\sigma}^{\dagger} c_{0\sigma} &= V^{*2} \sum_{\sigma} c_{0\sigma}^{\dagger} c_{d\sigma} \frac{\hat{h}_{d\bar{\sigma}}\hat{h}_{0\bar{\sigma}} + \hat{h}_{d\bar{\sigma}}\hat{n}_{0\bar{\sigma}} + \hat{n}_{d\bar{\sigma}}\hat{h}_{0\bar{\sigma}} + \hat{n}_{d\bar{\sigma}}\hat{n}_{0\bar{\sigma}}}{\bar{\omega} + \frac{U^*}{2}\hat{h}_{d\bar{\sigma}} + U_b\hat{n}_{0\bar{\sigma}} + \frac{J^*}{4}\hat{h}_{d\bar{\sigma}}\hat{n}_{0\bar{\sigma}}} c_{d\sigma}^{\dagger} c_{0\sigma} \\ &= V^{*2} \sum_{\sigma} \hat{h}_{d\sigma}\hat{n}_{0\sigma} \left[\frac{\hat{h}_{d\bar{\sigma}}\hat{h}_{0\bar{\sigma}}}{\bar{\omega}_{00} + \frac{U^*}{2}} + \frac{\hat{h}_{d\bar{\sigma}}\hat{n}_{0\bar{\sigma}}}{\bar{\omega}_{01} + \frac{U^*}{2} + U_b + \frac{J^*}{4}} + \frac{\hat{n}_{d\bar{\sigma}}\hat{h}_{0\bar{\sigma}}}{\bar{\omega}_{10}} + \frac{\hat{n}_{d\bar{\sigma}}\hat{n}_{0\bar{\sigma}}}{\bar{\omega}_{11} + U_b} \right]. \quad (11.4) \end{aligned}$$

$\bar{\omega}_{(0,1),(0,1)}$ represents the quantum fluctuation scale corresponding to the configuration in the numerator.

In order to "freeze" the impurity dynamics, we will substitute $\hat{n}_{d\sigma} = \hat{n}_{d\bar{\sigma}} = \frac{1}{2}$, because of the Z_2 symmetry and the particle-hole symmetry of the impurity levels. This gives

$$\frac{1}{4} V^{*2} \sum_{\sigma} \hat{n}_{0\sigma} \left[\frac{\hat{h}_{0\bar{\sigma}}}{\bar{\omega}_{00} + \frac{U^*}{2}} + \frac{\hat{n}_{0\bar{\sigma}}}{\bar{\omega}_{01} + \frac{U^*}{2} + U_b + \frac{J^*}{4}} + \frac{\hat{h}_{0\bar{\sigma}}}{\bar{\omega}_{10}} + \frac{\hat{n}_{0\bar{\sigma}}}{\bar{\omega}_{11} + U_b} \right]. \quad (11.5)$$

The state that most closely represents the metallic ground state is $\bar{\omega}_{10}$, we take that as the reference quantum fluctuation scale $\bar{\omega}$. It is of the order of $\bar{\omega} \sim -\frac{J^*}{4} - \frac{U^*}{2} - U_b$. We will now relate the other scales to $\bar{\omega}$ by expressing them in terms of the energy of the initial state:

$$\bar{\omega}_{00} \sim -U_b = \bar{\omega} + \frac{J^*}{4} + \frac{U^*}{2}, \quad \bar{\omega}_{01} \sim 0 = \bar{\omega} + \frac{J^*}{4} + \frac{U^*}{2} + U_b, \quad \bar{\omega}_{11} \sim -\frac{U^*}{2} = \bar{\omega} + \frac{J^*}{4} + U_b. \quad (11.6)$$

Substituting these gives

$$\frac{1}{4} V^{*2} \sum_{\sigma} \hat{n}_{0\sigma} [\hat{h}_{0\bar{\sigma}}\alpha_1 + \hat{n}_{0\bar{\sigma}}\alpha_2], \quad (11.7)$$

where $\alpha_1 = \left(\bar{\omega} + U^* + \frac{J^*}{4}\right)^{-1} + (\bar{\omega})^{-1}$ and $\alpha_2 = \left(\bar{\omega} + U^* + 2U_b + \frac{J^*}{2}\right)^{-1} + \left(\bar{\omega} + 2U_b + \frac{J^*}{4}\right)^{-1}$.

Because of the particle-hole symmetry on the impurity as well as in the bath, the renormalisation from the hole sector is obtained simply by transforming $\hat{n} \leftrightarrow \hat{h}$:

$$\frac{1}{4} V^{*2} \sum_{\sigma} \hat{h}_{0\sigma} [\hat{n}_{0\bar{\sigma}}\alpha_1 + \hat{h}_{0\bar{\sigma}}\alpha_2]. \quad (11.8)$$

The total renormalisation arising from V is therefore

$$\frac{1}{4} V^{*2} \sum_{\sigma} \left[\alpha_1 (\hat{n}_{0\sigma}\hat{h}_{0\bar{\sigma}} + \hat{h}_{0\sigma}\hat{n}_{0\bar{\sigma}}) + \alpha_2 (\hat{n}_{0\sigma}\hat{n}_{0\bar{\sigma}} + \hat{h}_{0\sigma}\hat{h}_{0\bar{\sigma}}) \right] = \frac{1}{2} V^{*2} (\alpha_1 - \alpha_2) (\hat{n}_{0\uparrow} - \hat{n}_{0\downarrow})^2 + \text{constant}. \quad (11.9)$$

2.11.2 Renormalisation from J^*

The renormalisation arising from decoupling the Kondo coupling has two terms. The first term arises when the spin of the zeroth site is initially up:

$$\begin{aligned} \frac{J^{*2}}{4} c_{d\downarrow}^{\dagger} c_{d\uparrow} c_{0\uparrow}^{\dagger} c_{0\downarrow} \frac{1}{\bar{\omega} - H_d} c_{0\downarrow}^{\dagger} c_{0\uparrow} c_{d\uparrow}^{\dagger} c_{d\downarrow} &= c_{d\downarrow}^{\dagger} c_{d\uparrow} c_{0\uparrow}^{\dagger} c_{0\downarrow} \frac{J^{*2}/4}{\bar{\omega} + \frac{U^*}{2} + U_b + \frac{J^*}{4}} c_{0\downarrow}^{\dagger} c_{0\uparrow} c_{d\uparrow}^{\dagger} c_{d\downarrow} \\ &= \frac{J^{*2}}{4} \frac{\hat{n}_{d\downarrow}\hat{h}_{d\uparrow}\hat{n}_{0\uparrow}\hat{h}_{0\downarrow}}{\bar{\omega} + \frac{U^*}{2} + U_b + \frac{J^*}{4}}. \quad (11.10) \end{aligned}$$

The quantum fluctuation scale $\tilde{\omega}$ for this process can be similarly related to $\bar{\omega}$: $\tilde{\omega} \sim -\frac{U^*}{2} - U_b - \frac{J^*}{4} = \bar{\omega}$. This gives

$$\frac{J^{*2}}{4} \frac{\hat{n}_{d\downarrow} \hat{h}_{d\uparrow} \hat{n}_{0\uparrow} \hat{h}_{0\downarrow}}{\bar{\omega} + \frac{U^*}{2} + U_b + \frac{3J^*}{4}} = \frac{J^{*2}}{16} \frac{\hat{n}_{0\uparrow} \hat{h}_{0\downarrow}}{\bar{\omega} + \frac{U^*}{2} + U_b + \frac{J^*}{4}}, \quad (11.11)$$

as the renormalisation for this configuration. At the final step, we substituted $\hat{n}_{d\sigma} = \hat{h}_{d\sigma} = \frac{1}{2}$. For the other configuration where the spin of the zeroth site is down, we get

$$\frac{J^{*2}}{16} \frac{\hat{n}_{0\downarrow} \hat{h}_{0\uparrow}}{\bar{\omega} + \frac{U^*}{2} + U_b + \frac{J^*}{4}}. \quad (11.12)$$

Adding both sectors, we get

$$\frac{J^{*2}}{16} \frac{1}{\bar{\omega} + \frac{U^*}{2} + U_b + \frac{J^*}{4}} (\hat{n}_{0\uparrow} - \hat{n}_{0\downarrow})^2. \quad (11.13)$$

2.11.3 Total renormalisation

Adding the contributions from both V^* and J^* , the net renormalisation is the generation of a local correlation term $-U' (\hat{n}_{0\uparrow} - \hat{n}_{0\downarrow})^2$ on the zeroth site, where U' is given by

$$U' = \frac{-J^{*2}/16}{\bar{\omega} + \frac{U^*}{2} + U_b + \frac{J^*}{4}} - \frac{1}{2} V^{*2} \left(\frac{1}{\bar{\omega} + U^* + \frac{J^*}{4}} + \frac{1}{\bar{\omega}} - \frac{1}{\bar{\omega} + U^* + 2U_b + \frac{J^*}{2}} - \frac{1}{\bar{\omega} + 2U_b + \frac{J^*}{4}} \right) \quad (11.14)$$

The effective Hamiltonian for the zeroth site is therefore

$$H_{0+\text{rest}} = \underbrace{-(U' + U_b) (\hat{n}_{0\uparrow} - \hat{n}_{0\downarrow})^2}_{\text{new correlated impurity}} - \underbrace{t \sum_{\substack{j \in \text{n.n. of } 0, \\ \sigma}} (c_{0\sigma}^\dagger c_{j\sigma} + \text{h.c.})}_{\text{hopping between new impurity \& new bath}} - \underbrace{t \sum_{\langle i,j \rangle} (c_{i\sigma}^\dagger c_{j\sigma} + \text{h.c.})}_{\text{K.E. of new bath}} \quad (11.15)$$

2.12 Appendix: Effective Hamiltonian for the excitations within the Hubbard sidebands

Beyond r_{c1} , the Hubbard sidebands can be seen to separate from the central Kondo resonance in the impurity spectral function. These sidebands are broad, indicating that the impurity site is hybridising with the bath from within the sidebands, at high energies $\omega \sim U/2$. These processes lead to the presence of gapless excitations within the rest of the bath, outside the Mott gap. To find the effective Hamiltonian for the rest of the bath (sites 1 and onwards), we will take the eigenstates of the two-site Hamiltonian H that reside within the sideband as our ground-states, and treat the hopping from the zeroth site to the rest of the bath as a perturbation. The two-site Hamiltonian is

$$H_{\text{two site}} = V \sum_{\sigma} (c_{d\sigma}^\dagger c_{0\sigma} + \text{h.c.}) + J \vec{S}_d \cdot \vec{S}_0 - \frac{U}{2} (\hat{n}_{d\uparrow} - \hat{n}_{d\downarrow})^2 + \frac{U}{2}. \quad (12.1)$$

In order to align the doubly-occupied impurity level at the more conventional value of $U/2$, we have shifted the Hamiltonian by the same constant value. The effect of U_b has been ignored because it is small compared to U at the frequencies pertinent to the physics of the sidebands. The two-site states that reside within the Hubbard sidebands are

$$\begin{aligned} |\Psi_{cc}\rangle &\equiv |c\rangle_d |c\rangle_0; \quad c \in \{0, 2\}; \quad |\Psi_{CS}\rangle \equiv \frac{1}{\sqrt{2}} (|2\rangle_d |0\rangle_0 - |0\rangle_d |2\rangle_0); \\ |\Psi_{\sigma,c}\rangle &= \alpha |\sigma\rangle_d |c\rangle_0 + \sqrt{1 - \alpha^2} |c\rangle_d |\sigma\rangle_0; \quad \sigma \in \{\uparrow, \downarrow\}; \\ |\Psi_{CT}\rangle &= \beta \frac{1}{\sqrt{2}} (|\uparrow\rangle_d |\downarrow\rangle_0 - |\downarrow\rangle_d |\uparrow\rangle_0) + \sqrt{1 - \beta^2} \frac{1}{\sqrt{2}} (|2\rangle_d |0\rangle_0 + |0\rangle_d |2\rangle_0), \end{aligned} \quad (12.2)$$

where the coefficients of the eigenstates are given by

$$\alpha \simeq \frac{V}{\sqrt{V^2 + (E_{X1} + \frac{U}{2})^2}}, \quad \beta = \frac{2V}{\sqrt{4V^2 + (E_{CS} + \frac{U}{2} + \frac{3J}{4})^2}}. \quad (12.3)$$

The energies of the states are

$$\begin{aligned} E_{gs} \equiv E_{00} = E_{22} = E_{CS} &= \frac{U}{2}, \quad E_{X1} \equiv E_{\sigma,c} = \frac{U}{4} + \sqrt{V^2 + \frac{U^2}{16}}, \\ E_{X2} \equiv E_{CS} &= \frac{U}{4} - \frac{3J}{8} + \sqrt{4V^2 + \left(\frac{U}{4} + \frac{3J}{8}\right)^2}. \end{aligned} \quad (12.4)$$

The ground state (comprising the states shown in eq. (12.2)) is triply degenerate. In order to study the dynamics of the bath, we now introduce the perturbation $H_I = -t \sum_{\sigma} (c_{0\sigma}^\dagger c_{1\sigma} + \text{h.c.})$. We also take an expanded set of eigenstates $\{|\Psi_n\rangle \otimes |a\rangle\}$, where $\{|\Psi_n\rangle\}$ is the set of eigenstates of $H_{\text{two site}}$, and $|a\rangle \in \{|0\rangle, |\uparrow\rangle, |\downarrow\rangle, |2\rangle\}$ represents the configuration of the first site. The eigenstates of $H_{\text{two site}}$ are also eigenstates the total number operator $\hat{n}_d + \hat{n}_0 + \hat{n}_1$, and since H_I does not conserve the total number, all odd order shifts are zero: $\langle \Psi_n | H_I^b | \Psi_n \rangle$, $b = 1, 3, \dots$, is zero for all the eigenstates.

For the second order shift, we need to calculate the overlaps $\langle \Psi_m | H_I | \Psi_n \rangle$, $m \neq n$ for n in the ground state. Since all three ground-states involve the impurity and zeroth sites in charge configuration (0 or 2), $H_I | \Psi_n \rangle$ will result in states of the form $|0(2)\rangle_d |\sigma\rangle_0 |a\rangle_1$. This resulting state has non-zero overlap only with the part $\sqrt{1 - \alpha^2} |c\rangle_d |\sigma\rangle_0 |a\rangle$ of the expanded state $|\Psi_{\sigma,c}\rangle |a\rangle$. The square of each such overlap is, therefore, $t^2 (1 - \alpha^2)$.

- Excitations from the ground-states $|\Psi_{00}\rangle |0\rangle$ and $|\Psi_{22}\rangle |2\rangle$ have zero overlap with any other state, such that the second order shifts in these states is zero.
- Excitations from the ground-states $|\Psi_{00}\rangle |\sigma\rangle$ and $|\Psi_{22}\rangle |\sigma\rangle$ have non zero overlap with one excited state each, such that the second order shift in these states are $\gamma^2 \equiv t^2 (1 - \alpha^2) / (E_{X1} - E_{gs})$.
- Excitations from the remaining ground states have non-zero overlaps with two excited states each, such that the second order shift in these states are $2\gamma^2$.

Upon dropping the zeroth order part (because it has no dynamics for the rest of the bath), the effective Hamiltonian at second order in t takes the form

$$H_{\text{eff}}^{(2)} = \gamma^2 \left[(|\Psi_{00}\rangle \langle \Psi_{00}| + |\Psi_{22}\rangle \langle \Psi_{22}| + 2|\Psi_{\text{CS}}\rangle \langle \Psi_{\text{CS}}|) (\sum_{\sigma} |\sigma\rangle_1 \langle \sigma|_1) \right. \\ \left. + 2(|\Psi_{22}\rangle \langle \Psi_{22}| + |\Psi_{\text{CS}}\rangle \langle \Psi_{\text{CS}}|) |0\rangle_1 \langle 0|_1 + 2(|\Psi_{00}\rangle \langle \Psi_{00}| + |\Psi_{\text{CS}}\rangle \langle \Psi_{\text{CS}}|) |2\rangle_1 \langle 2|_1 \right] \quad (12.5)$$

By using the completeness relation $1 = \sum_{\sigma} |\sigma\rangle_1 \langle \sigma|_1 + |0\rangle_1 \langle 0|_1 + |2\rangle_1 \langle 2|_1$ of the Hilbert space of the first site, this evaluates to

$$H_{\text{eff}}^{(2)} = \text{constant} + \gamma^2 (|\Psi_{22}\rangle \langle \Psi_{22}| - |\Psi_{00}\rangle \langle \Psi_{00}|) (|2\rangle_1 \langle 2|_1 - |0\rangle_1 \langle 0|_1) , \quad (12.6)$$

where the "constant" term refers to operators that do not depend on the dynamics of the rest of the bath. By defining the charge isospin operator $C^z |2(0)\rangle = +(-)\frac{1}{2} |2(0)\rangle$, the effective Hamiltonian can be written as (dropping the non-dynamical part)

$$H_{\text{eff}}^{(2)} = 4\gamma^2 C_{\text{tot}}^z C_1^z , \quad (12.7)$$

where C_i^z is the charge isospin for site i and $C_{\text{tot}}^z \equiv C_d^z + C_0^z$ is the z-component of the total charge isospin for the impurity and the zeroth sites.

The fourth order shift in the energy of the eigenstate $|\Psi_n\rangle$ will be given by

$$E_n^{(4)} = \sum_{j,k,l} \frac{(H_I)_{nl} (H_I)_{lk} (H_I)_{kj} (H_I)_{jn}}{(E_l - E_n) (E_k - E_n) (E_j - E_n)} - E_n^{(2)} \left[\frac{(H_I)_{kn}}{E_k - E_n} \right]^2 , \quad (12.8)$$

where the matrix elements $(H_I)_{in}$ are defined as $\langle i|H_I|n\rangle$. This fourth-order shift vanishes for almost all the ground-states. For the remaining states, it becomes important to treat their degeneracy. The "appropriate" combinations of states that provide non-zero fourth order contribution are

$$|+\rangle = \sqrt{\frac{2}{3}} |\Psi_{22}\rangle |0\rangle_1 + \sqrt{\frac{1}{3}} |\Psi_{\text{CS}}\rangle |2\rangle_1 , \quad |-\rangle = -\sqrt{\frac{2}{3}} |\Psi_{00}\rangle |2\rangle_1 + \sqrt{\frac{1}{3}} |\Psi_{\text{CS}}\rangle |0\rangle_1 . \quad (12.9)$$

Corresponding to these ground states, the fourth-order shift in the eigenvalue is

$$E^{(4)} = \frac{3\gamma^4 \alpha^2 \beta^2}{(1 - \alpha^2) (E_{\text{X2}} - E_{\text{gs}})} , \quad (12.10)$$

such that the fourth-order contribution to the effective Hamiltonian is

$$H_{\text{eff}}^{(4)} = E^{(4)} (|+\rangle \langle +| + |-\rangle \langle -|) \\ = \frac{\gamma^4 \alpha^2 \beta^2}{(1 - \alpha^2) (E_{\text{X2}} - E_{\text{gs}})} \left[-C_{\text{tot}}^z C_{\text{tot}}^2 C_1^z + \sqrt{2} \mathcal{P}_{\text{tot}}^4 (C_0^+ - C_d^+) C_1^- + \text{h.c.} \right] . \quad (12.11)$$

The operator $\mathcal{P}_{\text{tot}}^4$ acts on states belonging to the combined $d + 0$ Hilbert space and projects on to the $\hat{n}_d + \hat{n}_0 = 4$ subspace.

2.13 Appendix: Behaviour of the Kondo scale T_K close to the transition

Towards obtaining the emergent Kondo temperature scale T_K close to the transition, we follow the approach used by Moeller et al. 1995 [497] and Held et al. 2013 [498]. Near the transition, they obtained a Kondo model from the full AIM by applying a Schrieffer-Wolff transition that removes the charge fluctuations of the impurity site and retains the physics of only the s-d coupling J . This amounts to removing the side peaks from the impurity spectral function and focusing on the low-energy central peak. Held et al. then integrated the RG equation for this Kondo model by using a Lorentzian DOS in the bath. This is motivated by the fact that the central peak of the impurity spectral function is a Lorentzian, and the bath becomes equivalent to the impurity site under self-consistency. We implement the same approach but on our extended impurity model, such that the transformation then leads to a $J - U_b$ model. The relevant RG equations are mentioned in Section IV of the main manuscript:

$$\Delta J = -\frac{n_j J (J + 4U_b)}{\omega - D/2 + U_b/2 + J/4}, \quad \Delta U_b = 0, \quad (13.1)$$

where $n_j = \rho(D)|\Delta D|$ is proportional to the bath density of states. In Section VII of the main manuscript, we show that the spectral function of the zeroth site is tracked by that on the impurity site through the coherent spin-flip fluctuations, and this shows that the appropriate density of states for the bath in this regime is of the Lorentzian kind (same as the shape of the central peak of the impurity spectral function). We therefore set

$$\rho(D) = \frac{\rho_0 \Gamma^2}{D^2 + \Gamma^2}, \quad (13.2)$$

where Γ represents the half-width of the Lorentzian. A similar line of arguments was presented by Held et al. [498]. Close to the transition, the RG flow of the Kondo coupling J is stunted by the competing pairing term, and the RG equation can be simplified into the Poor man's scaling 1-loop form:

$$\frac{dJ}{dD} \simeq -2\rho_0 \Gamma^2 \frac{J (J + 4U_b)}{(\Gamma^2 + D^2) D}. \quad (13.3)$$

We now integrate this RG equation over the range $D \in [D_0, D^*]$, where D_0 is the bare bandwidth, and D^* is the fixed-point bandwidth. D^* represents the energy scale for the conduction electrons in the IR.

$$\begin{aligned} \int_{J_0}^{J^*} \frac{dJ}{J (J + 4U_b)} &= - \int_{D_0}^{D^*} \frac{2\rho_0 \Gamma^2 dD}{(\Gamma^2 + D^2) D} \\ \Rightarrow \frac{1}{4U_b \rho_0} \left(\ln \frac{J^*}{J_0} - \ln \frac{J^* + 4U_b}{J_0 + 4U_b} \right) &= \ln \frac{D^* + \Gamma^2}{D_0 + \Gamma^2} - \ln \frac{D^*}{D_0}. \end{aligned} \quad (13.4)$$

We will now take the limit of $J_0 + 4U_b \rightarrow 0^-$ and *equate the singular terms on both sides* of the equation:

$$\ln \frac{D^*}{D_0} = -\frac{1}{4U_b \rho_0} \ln \frac{J_0 + 4U_b}{J_0} = -\frac{1}{4U_b \rho_0} \ln (r_{c2} - r) + \text{non-singular part}. \quad (13.5)$$

We have replaced the couplings J and U_b with the dimensionless parameters $r = -U_b/J_0$ and $r_{c2} = 1/4$. Dropping the non-singular part and solving for D^* then gives the IR energy scale:

$$D^* = D_0 \exp \left[-\frac{\ln(r_{c2} - r)}{4U_b\rho_0} \right]. \quad (13.6)$$

The dynamically generated Kondo temperature scale is then obtained as

$$T_K = \frac{D^*}{k_B} = \frac{D_0}{k_B} \exp \left[-\frac{\ln(r_{c2} - r)}{4U_b\rho_0} \right]. \quad (13.7)$$

Note that the prefactor of the logarithm is positive because U_b is negative: $-4U_b\rho_0 = |4U_b\rho_0|$. As we approach the transition, the parameter r takes the limit $r \rightarrow r_{c2}^-$, and the Kondo temperature scale vanishes:

$$\lim_{r \rightarrow r_{c2}^-} T_K = \frac{D_0}{k_B} \lim_{r \rightarrow r_{c2}^-} (r_{c2} - r)^{4|U_b\rho_0|} \rightarrow 0. \quad (13.8)$$

2.14 Appendix: Effective Hamiltonian for the low-lying excitations at the critical point

In order to obtain an effective Hamiltonian for the low-lying excitations in the bath, we will introduce a hopping term V between the zeroth site and the first site into the two-site Hamiltonian H_0 , and treat that term perturbatively.

$$H = \underbrace{J\vec{S}_d \cdot \vec{S}_0 - U_b (\hat{n}_{0\uparrow} - \hat{n}_{0\downarrow})^2}_{H_0} - t \underbrace{\sum_{\sigma} (c_{0\sigma}^\dagger c_{1\sigma} + \text{h.c.})}_V. \quad (14.1)$$

As mentioned before, H_0 is our zeroth Hamiltonian, while V is the perturbation. At the critical point, the zeroth Hamiltonian has a 20-fold degenerate ground-state manifold, and we will show later that it is this degeneracy that leads to NFL excitations. The ground-states are

$$\{|S = S^z = 0\rangle, |\uparrow, 0\rangle, |\uparrow, 2\rangle, |\downarrow, 0\rangle, |\downarrow, 2\rangle\} \otimes \{|0\rangle, |\uparrow\rangle, |\downarrow\rangle, |\uparrow\downarrow\rangle\}. \quad (14.2)$$

The ket to the left of $\otimes$ describes the configuration of the impurity and zeroth sites; $S = S^z = 0$ represents the singlet, while the remaining four states $|\sigma, 0\rangle, |\sigma, 2\rangle$ are local moment states. The ket to the right of $\otimes$ represents the configuration of the first site. Eight of these states have zero matrix elements within the ground-state subspace, even in the presence of the perturbation, so we drop these states. Since the full Hamiltonian conserves the total number of particles $N_{\text{tot}} = \sum_{\sigma} (\hat{n}_{d\sigma} + \hat{n}_{0\sigma} + \hat{n}_{1\sigma})$ and the total spin $S_{\text{tot}}^z = S_d^z + S_0^z + S_1^z$, the remaining 12 states can be split into decoupled blocks based on the values of these two operators:

$$\begin{aligned} N_{\text{tot}} = c + 2, S_{\text{tot}}^z = 0 & : |S = S^z = 0\rangle |c\rangle, |\uparrow, c, \downarrow\rangle, |\downarrow, c, \uparrow\rangle; c = 0, 2, \\ N_{\text{tot}} = 3, S_{\text{tot}}^z = \pm \frac{1}{2} & : |S = S^z = 0\rangle |\sigma\rangle, |\sigma, 0, 2\rangle, |\sigma, 2, 0\rangle; \sigma = \uparrow, \downarrow. \end{aligned} \quad (14.3)$$

The perturbation V will lift the degeneracy in each block; to calculate the modified spectrum, we diagonalise each block in the presence of V . We will first write down the action of the perturbation on the singlet ground-states: this will make it easier to identify the true eigenstates.

$$\begin{aligned} N_{\text{tot}} = c + 2, S_{\text{tot}}^z = 0 & : V |\text{SS}_c\rangle = \frac{t}{\sqrt{2}} (|\downarrow, c, \uparrow\rangle - |\uparrow, c, \downarrow\rangle), \quad c = 0, 2, \\ N_{\text{tot}} = 3, S_{\text{tot}}^z = \frac{\sigma}{2} & : V |\text{SS}_\sigma\rangle = \frac{t}{\sqrt{2}} |\sigma\rangle \otimes (|0, 2\rangle + |2, 0\rangle), \quad \sigma = \uparrow (1), \downarrow (-1), \end{aligned} \quad (14.4)$$

where we have introduced the notation $|\text{SS}_\alpha\rangle = |S = S^z = 0\rangle \otimes |\alpha\rangle$, $\alpha = 0, \uparrow, \downarrow, 2$. This shows that the action of the perturbation on the singlet state is to create linear combinations of the local moment states in the respective blocks. We therefore define the two possible linear combinations $|\text{LM}_\pm^{N_{\text{tot}}, S_{\text{tot}}^z}\rangle$ within each block:

$$\begin{aligned} N_{\text{tot}} = c + 2, S_{\text{tot}}^z = 0 & : |\text{LM}_\pm^{c+2,0}\rangle = \frac{1}{\sqrt{2}} (|\uparrow, c, \downarrow\rangle \pm |\downarrow, c, \uparrow\rangle), \quad c = 0, 2, \\ N_{\text{tot}} = 3, S_{\text{tot}}^z = \frac{\sigma}{2} & : |\text{LM}_\sigma^{3, \frac{1}{2}}\rangle = \frac{1}{\sqrt{2}} (|\sigma, 0, 2\rangle \pm |\sigma, 2, 0\rangle), \quad \sigma = \uparrow (1), \downarrow (-1). \end{aligned} \quad (14.5)$$

In terms of these linear combinations, we then get

$$\begin{aligned} V |\text{SS}_0\rangle &= t |\text{LM}_-^{2,0}\rangle, \quad V |\text{LM}_-^{2,0}\rangle = t |\text{SS}_0\rangle, \quad V |\text{LM}_+^{2,0}\rangle = 0; \quad V |\text{SS}_\uparrow\rangle = t |\text{LM}_+^{3,\uparrow}\rangle, \\ V |\text{LM}_+^{3,\uparrow}\rangle &= t |\text{SS}_\uparrow\rangle, \quad V |\text{LM}_-^{3,\uparrow}\rangle = 0, \quad V |\text{SS}_\downarrow\rangle = t |\text{LM}_+^{3,\downarrow}\rangle, \quad V |\text{LM}_+^{3,\downarrow}\rangle = t |\text{SS}_\downarrow\rangle, \\ V |\text{LM}_-^{3,\downarrow}\rangle &= 0; \quad V |\text{SS}_2\rangle = t |\text{LM}_-^{4,0}\rangle, \quad V |\text{LM}_-^{4,0}\rangle = t |\text{SS}_2\rangle, \quad V |\text{LM}_+^{4,0}\rangle = 0. \end{aligned} \quad (14.6)$$

The zero matrix elements at the far right of each line should be understood as the fact that V takes those states out of the ground-state subspace. For e.g., the state $V |\text{LM}_-^{2,0}\rangle$ is not really zero, but the overlap of $V |\text{LM}_-^{2,0}\rangle$ with any other state in the ground-state subspace spanned by the 20 degenerate states is zero. We now see that each of the 3×3 blocks decouples into a 2×2 block $\begin{pmatrix} 0 & t \\ t & 0 \end{pmatrix}$ and a 1×1 block (0) when written in terms of the linear combinations $|\text{LM}_\pm\rangle$. The state in the final block is therefore already an eigenstate, but with zero energy, so that will drop out from the effective Hamiltonian. The remaining 2×2 block has eigenstates $|\pm^{N_{\text{tot}}, S_{\text{tot}}^z}\rangle$ that are linear combinations of the singlet states and the surviving local moment combination, with eigenvalues $\pm t$:

$$\begin{aligned} N_{\text{tot}} = 2, S_{\text{tot}}^z = 0 & : |\pm^{2,0}\rangle = \frac{1}{\sqrt{2}} (|\text{SS}_0\rangle \pm |\text{LM}_-^{2,0}\rangle), \\ N_{\text{tot}} = 3, S_{\text{tot}}^z = \frac{1}{2} & : |\pm^{3, \frac{1}{2}}\rangle = \frac{1}{\sqrt{2}} (|\text{SS}_\uparrow\rangle \pm |\text{LM}_+^{3, \frac{1}{2}}\rangle), \\ N_{\text{tot}} = 3, S_{\text{tot}}^z = -\frac{1}{2} & : |\pm^{3, -\frac{1}{2}}\rangle = \frac{1}{\sqrt{2}} (|\text{SS}_\downarrow\rangle \pm |\text{LM}_+^{3, -\frac{1}{2}}\rangle), \\ N_{\text{tot}} = 4, S_{\text{tot}}^z = 0 & : |\pm^{4,0}\rangle = \frac{1}{\sqrt{2}} (|\text{SS}_2\rangle \pm |\text{LM}_-^{4,0}\rangle). \end{aligned} \quad (14.7)$$

At this point, it is worth noting that the polarised ground-states $|S_{\text{tot}}^z = \sigma \frac{1}{2}\rangle$, $\sigma = \pm 1$ can be written as an equal superposition of the singlet states $|\text{SS}\rangle_{d0} \otimes |\sigma\rangle_1$ and the local moment states $\frac{1}{\sqrt{2}} (|\sigma, 0, 2\rangle - |\sigma, 2, 0\rangle)$.

The polarised symmetry-broken states therefore act as a bridge between the ground states of the two phases and are important in displaying the breakdown of the Kondo cloud at the QCP and the consequences arising from it (like inexact screening of the impurity and fractional magnetisation and entanglement entropy, which will be discussed in the next section).

By combining the two eigenstates $|\pm\rangle$ for each block, one can then reconstruct the effective Hamiltonian $\mathcal{H}_{\text{eff}}^{N_{\text{tot}}, S_{\text{tot}}^z}$ that describes the dynamics of the first site from within the ground-state subspace of the zeroth Hamiltonian. Formally, this is written as

$$\begin{aligned} \mathcal{H}_{\text{eff}}^{N_{\text{tot}}, S_{\text{tot}}^z} &= t |+\rangle^{N_{\text{tot}}, S_{\text{tot}}^z} \langle +^{N_{\text{tot}}, S_{\text{tot}}^z} | - t |-\rangle^{N_{\text{tot}}, S_{\text{tot}}^z} \langle -^{N_{\text{tot}}, S_{\text{tot}}^z} |, \\ &= \begin{cases} t \left[|\text{SS}_\alpha\rangle \langle \text{LM}_-^{N_{\text{tot}}, S_{\text{tot}}^z} | + |\text{LM}_-^{N_{\text{tot}}, S_{\text{tot}}^z}\rangle \langle \text{SS}_\alpha| \right], & \alpha = 0, 2, \\ t \left[|\text{SS}_\alpha\rangle \langle \text{LM}_+^{N_{\text{tot}}, S_{\text{tot}}^z} | + |\text{LM}_+^{N_{\text{tot}}, S_{\text{tot}}^z}\rangle \langle \text{SS}_\alpha| \right], & \alpha = \uparrow, \downarrow. \end{cases} \end{aligned} \quad (14.8)$$

We first write down the effective Hamiltonian for $\alpha = 0$ which corresponds to $N_{\text{tot}} = 2, S_{\text{tot}}^z = 0$:

$$\mathcal{H}_{\text{eff}}^{2,0} = t \left[|\text{SS}_0\rangle \langle \text{LM}_-^{2,0} | + |\text{LM}_-^{2,0}\rangle \langle \text{SS}_0| \right]. \quad (14.9)$$

In order to convert this into a second-quantised form, we will use the following identities:

$$|\text{SS}_0\rangle = \frac{1}{\sqrt{2}} \left(P_d^\uparrow c_{0\downarrow}^\dagger - P_d^\downarrow c_{0\uparrow}^\dagger \right) P_0^{(0)} P_1^{(0)} |0\rangle; \quad |\text{LM}_-^{2,0}\rangle = \frac{1}{\sqrt{2}} \left(P_d^\uparrow c_{1\downarrow}^\dagger - P_d^\downarrow c_{1\uparrow}^\dagger \right) P_0^{(0)} P_1^{(0)} |0\rangle, \quad (14.10)$$

where $P_i^{(m)}$ ($m = 0, 1, 2$) projects onto the $\hat{n}_i = m$ subspace on the i^{th} site, and P_d^σ projects the impurity spin into $|\sigma\rangle$; $\sigma = \uparrow, \downarrow$ configuration. We then have

$$\begin{aligned} |\text{SS}_0\rangle \langle \text{LM}_-^{2,0} | &= \frac{1}{2} \left(P_d^\uparrow c_{0\downarrow}^\dagger - P_d^\downarrow c_{0\uparrow}^\dagger \right) P_0^{(0)} P_1^{(0)} \left(P_d^\uparrow c_{1\downarrow} - P_d^\downarrow c_{1\uparrow} \right) \\ &= P_1^{(0)} \frac{1}{2} \left(P_d^\uparrow c_{0\downarrow}^\dagger c_{1\downarrow} + P_d^\downarrow c_{0\uparrow}^\dagger c_{1\uparrow} - S_d^+ c_{0\downarrow}^\dagger c_{1\uparrow} - S_d^- c_{0\uparrow}^\dagger c_{1\downarrow} \right) P_0^{(0)}, \end{aligned} \quad (14.11)$$

where we have used $P^2 = P$ if P is a projector, and that $P_d^\uparrow P_d^\downarrow = S_d^+$. Substituting these into the effective Hamiltonian and recognising that $P_d^\uparrow = \frac{1}{2} + S_d^z, P_d^\downarrow = \frac{1}{2} - S_d^z$ gives

$$\begin{aligned} \mathcal{H}_{\text{eff}}^{2,0} &= \frac{t}{2} \left[\left(\frac{1}{2} + S_d^z \right) c_{0\downarrow}^\dagger P_0^{(0)} P_1^{(0)} c_{1\downarrow} - S_d^+ c_{0\downarrow}^\dagger P_0^{(0)} P_1^{(0)} c_{1\uparrow} - S_d^- c_{0\uparrow}^\dagger P_0^{(0)} P_1^{(0)} c_{1\downarrow} \right. \\ &\quad \left. + \left(\frac{1}{2} - S_d^z \right) c_{0\uparrow}^\dagger P_0^{(0)} P_1^{(0)} c_{1\uparrow} \right] + \text{h.c.} . \end{aligned} \quad (14.12)$$

In order to make this more illuminating, we define the holon excitation operators $h_{i\sigma}^\dagger = c_{i\sigma}^\dagger P_i^{(0)}$. In terms of these operators, the effective Hamiltonian becomes

$$\begin{aligned} \mathcal{H}_{\text{eff}}^{2,0} &= \frac{t}{2} \left[\left(\frac{1}{2} + S_d^z \right) \left(h_{0\downarrow}^\dagger h_{1\downarrow} + h_{1\downarrow}^\dagger h_{0\downarrow} \right) - S_d^+ \left(h_{0\downarrow}^\dagger h_{1\uparrow} + h_{1\downarrow}^\dagger h_{0\uparrow} \right) - S_d^- \left(h_{0\uparrow}^\dagger h_{1\downarrow} + h_{1\uparrow}^\dagger h_{0\downarrow} \right) \right. \\ &\quad \left. + \left(\frac{1}{2} - S_d^z \right) \left(h_{0\uparrow}^\dagger h_{1\uparrow} + h_{1\uparrow}^\dagger h_{0\uparrow} \right) \right] \\ &= -t \left[S_d^z \frac{1}{2} \left(h_{0\uparrow}^\dagger h_{1\uparrow} + h_{1\uparrow}^\dagger h_{0\uparrow} - h_{0\downarrow}^\dagger h_{1\downarrow} - h_{1\downarrow}^\dagger h_{0\downarrow} \right) + \frac{1}{2} S_d^+ \left(h_{0\downarrow}^\dagger h_{1\uparrow} + h_{1\downarrow}^\dagger h_{0\uparrow} \right) \right. \\ &\quad \left. + \frac{1}{2} S_d^- \left(h_{0\uparrow}^\dagger h_{1\downarrow} + h_{1\uparrow}^\dagger h_{0\downarrow} \right) \right] + \frac{t}{4} \left(h_{0\uparrow}^\dagger h_{1\uparrow} + h_{1\uparrow}^\dagger h_{0\uparrow} + h_{0\downarrow}^\dagger h_{1\downarrow} + h_{1\downarrow}^\dagger h_{0\downarrow} \right). \end{aligned} \quad (14.13)$$

The first part of the Hamiltonian (with a $-t$ factor in front) is reminiscent of a two-spin Heisenberg Hamiltonian. To investigate this, we write down the potential second spin:

$$\mathcal{S}_1^z \equiv \frac{1}{2} \left(h_{0\uparrow}^\dagger h_{1\uparrow} + h_{1\uparrow}^\dagger h_{0\uparrow} - h_{0\downarrow}^\dagger h_{1\downarrow} - h_{1\downarrow}^\dagger h_{0\downarrow} \right). \quad (14.14)$$

Note that each term that comprises the full operator $\mathcal{S}_1^z$ has a combined projector $P_{01}^{(0)} = P_0^{(0)} P_1^{(0)} = P_1^{(0)} P_0^{(0)}$ in the middle with a single annihilation operator to the right. This means that $\mathcal{S}^z$ projects onto the smaller set of only those states that satisfy $\hat{n}_0 + \hat{n}_1 = 1$. That is also the meaning of the subscript 1 in $\mathcal{S}_1^z$. There are four such states: $|\uparrow, 0\rangle, |\downarrow, 0\rangle, |0, \uparrow\rangle, |0, \downarrow\rangle$. In fact, it is easy to see that the eigenstates of $\mathcal{S}^z$ are

$$|\uparrow\rangle_{1,\pm} = \frac{1}{\sqrt{2}} (|\uparrow, 0\rangle \pm |0, \uparrow\rangle), \quad \mathcal{S}_1^z |\uparrow\rangle_{1,\pm} = \pm \frac{1}{2} |\uparrow\rangle_{1,\pm}, \quad (14.15)$$

$$|\downarrow\rangle_{1,\pm} = \frac{1}{\sqrt{2}} (|\downarrow, 0\rangle \pm |0, \downarrow\rangle), \quad \mathcal{S}_1^z |\downarrow\rangle_{1,\pm} = \mp \frac{1}{2} |\downarrow\rangle_{1,\pm}. \quad (14.16)$$

The label $\pm$ on top of the eigenstates indicates that these are also eigenstates of the parity operator $\mathcal{P}_{01}$ that switches the site labels 0 and 1:

$$\mathcal{P}_{01} |\alpha, \alpha'\rangle = |\alpha', \alpha\rangle; \quad \alpha, \alpha' \in \{0, \uparrow, \downarrow, 2\}. \quad (14.17)$$

In fact, the eigenstates $|\uparrow\rangle_{1,+}$ and $|\downarrow\rangle_{1,-}$ both have $\mathcal{S}^z = \frac{1}{2}$, but are orthogonal because they have different parities. Analogous to the $\mathcal{S}_1^z$ operator, we can also read off the potential $\mathcal{S}_1^\pm$ ladder operators from the effective Hamiltonian:

$$\mathcal{S}_1^+ = h_{0\uparrow}^\dagger h_{1\downarrow} + h_{1\uparrow}^\dagger h_{0\downarrow}, \quad \mathcal{S}_1^- = (\mathcal{S}_1^+)^\dagger. \quad (14.18)$$

Applying these ladder operators on the eigenstates of $\mathcal{S}_1^z$ gives

$$\mathcal{S}_1^+ |\downarrow\rangle_{1,\pm} = \pm |\uparrow\rangle_{1,\pm}; \quad \mathcal{S}_1^- |\uparrow\rangle_{1,\pm} = \pm |\downarrow\rangle_{1,\pm}; \quad \mathcal{S}_1^+ |\uparrow\rangle_{1,\pm} = \mathcal{S}_1^- |\downarrow\rangle_{1,\pm} = 0. \quad (14.19)$$

This makes it clear that the pair of states $|\uparrow\rangle_1^+$ and $|\downarrow\rangle_1^+$ form a spin-half doublet, as does the other pair $|\uparrow\rangle_1^-$ and $|\downarrow\rangle_1^-$. We can define new spin operators that act only on a single parity subspace:

$$\mathcal{S}_{1,\pm}^a = \pm \mathcal{S}^a \frac{1}{2} (1 \pm \mathcal{P}_{01}); \quad a = x, y, z. \quad (14.20)$$

The operator $\frac{1}{2} (1 \pm \mathcal{P}_{01})$ projects on to the subspace with ± 1 parity. In the notation $\mathcal{S}_{1,\pm}^a$, the label $\pm$ indicates the parity sector to which the operator applies, while 1 indicates that we are operating in the subspace with $\hat{n}_0 + \hat{n}_1 = 1$. Two operators corresponding to different parity subspaces will commute because they act on disjoint Hilbert spaces: $[\mathcal{S}_{+1}^a, \mathcal{S}_{-1}^b] = 0$. In terms of these projected spin operators, we recover the usual spin-half SU(2) algebra:

$$\begin{aligned} \mathcal{S}_{1,\pm}^z |\uparrow\rangle_{1,\pm} &= \frac{1}{2} |\uparrow\rangle_{1,\pm}; & \mathcal{S}_{1,\pm}^z |\downarrow\rangle_{1,\pm} &= -\frac{1}{2} |\downarrow\rangle_{1,\pm}; \\ \mathcal{S}_{1,\pm}^+ |\downarrow\rangle_{1,\pm} &= |\uparrow\rangle_{1,\pm}; & \mathcal{S}_{1,\pm}^- |\uparrow\rangle_{1,\pm} &= |\downarrow\rangle_{1,\pm}. \end{aligned} \quad (14.21)$$

The part of the effective Hamiltonian that involves the impurity can now be written in terms of these new spin operators:

$$t \left(\vec{S}_d \cdot \vec{S}_{-1} - \vec{S}_d \cdot \vec{S}_{+1} \right); \quad \vec{S}_{\pm} = (S_{\pm}^x \ S_{\pm}^y \ S_{\pm}^z) . \quad (14.22)$$

The last term in eq. (14.13) (with a prefactor of $\frac{t}{4}$) can now be identified as the parity operator $\mathcal{P}_{01}$ acting within the subspace $\hat{n}_0 + \hat{n}_1 = 1$, because it has eigenstates $|\uparrow, 0\rangle \pm |0, \uparrow\rangle$ and $|\downarrow, 0\rangle \pm |0, \downarrow\rangle$ with eigenvalues ± 1 . The full effective Hamiltonian in the $N_{\text{tot}} = 2$ subspace then takes the compact form

$$\mathcal{H}_{\text{eff}}^{2,0} = t \left(\vec{S}_d \cdot \vec{S}_{1,-1} - \vec{S}_d \cdot \vec{S}_{1,+} \right) + \frac{t}{4} \mathcal{P}_{01} P_{01}^{(1)} , \quad (14.23)$$

where $P_{01}^{(1)} = P_0^{(1)} P_1^{(0)} + P_0^{(0)} P_1^{(1)}$ projects on to the $\hat{n}_0 + \hat{n}_1 = 1$ subspace. In light of these new spin operators, the original eigenstates $|\pm^{2,0}\rangle$ can be written in terms of these emergent spins $|\uparrow\rangle_{1,\pm}$ and $|\downarrow\rangle_{1,\pm}$:

$$\begin{aligned} |\pm^{2,0}\rangle &= \frac{1}{2} (|\uparrow, \downarrow, 0\rangle - |\downarrow, \uparrow, 0\rangle \pm |\uparrow, 0, \downarrow\rangle \mp |\downarrow, 0, \uparrow\rangle) = \frac{1}{2} [|\uparrow\rangle (|\downarrow, 0\rangle \pm |0, \downarrow\rangle) - |\downarrow\rangle (|\uparrow, 0\rangle \pm |0, \uparrow\rangle)] \\ &= \frac{1}{\sqrt{2}} (|\uparrow\rangle |\downarrow\rangle_{1,\pm} - |\downarrow\rangle |\uparrow\rangle_{1,\pm}) . \end{aligned} \quad (14.24)$$

Both the ground and excited states are therefore spin-singlet states, but they belong to the negative and positive parity sectors respectively.

We now proceed to the $N_{\text{tot}} = 4$ sector, which is equivalent to $\alpha = 2$ and $\hat{n}_0 + \hat{n}_1 = 3$. From eq. (14.8), we note that the effective Hamiltonian of the $\alpha = 2$ sector is obtained from that of the $\alpha = 0$ sector by replacing the holons with doublons, $|0\rangle \rightarrow |2\rangle$. This allows us to use the results of the $N_{\text{tot}} = 2$ sector simply by making the above-mentioned holon-doublon transformation. Accordingly, the composite up and down spins of the $0 + 1$ -site system become

$$|\uparrow\rangle_{3,\pm} = \frac{1}{\sqrt{2}} (|\uparrow, 2\rangle \pm |2, \uparrow\rangle), \quad |\downarrow\rangle_{3,\pm} = \frac{1}{\sqrt{2}} (|\downarrow, 2\rangle \pm |2, \downarrow\rangle) , \quad (14.25)$$

which then allow us to define spin operators for the positive and negative parity sectors:

$$S_{\pm}^z = \frac{1}{2} (|\uparrow\rangle_{3,\pm} \langle\uparrow|_{3,\pm} - |\downarrow\rangle_{3,\pm} \langle\downarrow|_{3,\pm}) , \quad S_{\pm}^{\pm} = |\uparrow\rangle_{3,\pm} \langle\downarrow|_{3,\pm} , \quad (14.26)$$

analogous to those of the $N_{\text{tot}} = 2$ sector in eqs. (14.15) and (14.18). These definitions then allow us to rewrite the effective Hamiltonian in terms of the eigenstates of $S_{\pm}^z$, and hence in terms of the spin operators $\vec{S}_{3,\pm}$. The eigenstates $|\pm^{4,0}\rangle$ and the effective Hamiltonian $H_{\text{eff}}^{4,0}$ for this sector can be written as

$$|\pm^{4,0}\rangle = \frac{1}{\sqrt{2}} (|\uparrow\rangle_d |\downarrow\rangle_{3,\pm} - |\downarrow\rangle_d |\uparrow\rangle_{3,\pm}) . \quad (14.27)$$

The effective Hamiltonian can then be written in terms of these emergent spins

$$\begin{aligned} H_{\text{eff}}^{4,0} = \frac{t}{2} & \left[(|\uparrow\rangle_d \langle\downarrow|_{3,+} - |\downarrow\rangle_d \langle\uparrow|_{3,+}) (\langle\uparrow|_d \langle\downarrow|_{3,+} - \langle\downarrow|_d \langle\uparrow|_{3,+}) \right. \\ & \left. - (|\uparrow\rangle_d \langle\downarrow|_{3,-} - |\downarrow\rangle_d \langle\uparrow|_{3,-}) (\langle\uparrow|_d \langle\downarrow|_{3,-} - \langle\downarrow|_d \langle\uparrow|_{3,-}) \right] . \end{aligned} \quad (14.28)$$

By replacing the emergent spins with the spin operators $\vec{S}_{3,\pm}$, we get the final form of the Hamiltonian

$$H_{\text{eff}}^{4,0} = t \left(\vec{S}_d \cdot \vec{S}_{3,-} - \vec{S}_d \cdot \vec{S}_{3,+} \right) + \frac{t}{4} \mathcal{P}_{01} P_{01}^{(3)}, \quad (14.29)$$

where $\mathcal{P}_{01}$ is the parity transformation operator defined in eq. (14.17) and $P_{01}^{(3)}$ projects on to the $\hat{n}_0 + \hat{n}_1 = 3$ subspace. This completes the effective Hamiltonian for the $S_{\text{tot}}^z = 0$ sector.

We now come to the $S_{\text{tot}}^z = +1/2$ sector. From eq. (14.8), the effective Hamiltonian for this sector is

$$H_{\text{eff}}^{3,\frac{1}{2}} = t \left[|\text{SS}\uparrow\rangle \langle \text{LM}^{-3,\frac{1}{2}}| + |\text{LM}^{-3,\frac{1}{2}}\rangle \langle \text{SS}\uparrow| \right], \quad (14.30)$$

where

$$|\text{SS}\uparrow\rangle = \frac{1}{\sqrt{2}} \left(P_d^\uparrow c_{0\downarrow}^\dagger - P_d^\downarrow c_{0\uparrow}^\dagger \right) c_{1\uparrow}^\dagger |0\rangle; \quad |\text{LM}^{-3,\frac{1}{2}}\rangle = \frac{1}{\sqrt{2}} P_d^\uparrow \left(c_{0\uparrow}^\dagger c_{0\downarrow}^\dagger + c_{1\uparrow}^\dagger c_{1\downarrow}^\dagger \right) |0\rangle, \quad (14.31)$$

such that

$$|\text{SS}\uparrow\rangle \langle \text{LAM}^{-3,\frac{1}{2}}| = \frac{1}{2} \left(P_d^\uparrow c_{0\downarrow}^\dagger - P_d^\downarrow c_{0\uparrow}^\dagger \right) c_{1\uparrow}^\dagger (c_{0\downarrow} c_{0\uparrow} + c_{1\downarrow} c_{1\uparrow}) P_d^\uparrow \quad (14.32)$$

$$= -\frac{1}{2} \left[P_d^\uparrow \left(c_{1\uparrow}^\dagger c_{0\uparrow} \hat{n}_{0\downarrow} + c_{0\downarrow}^\dagger c_{1\downarrow} \hat{n}_{1\uparrow} \right) + S_d^- \left(c_{1\uparrow}^\dagger c_{0\downarrow} \hat{n}_{0\uparrow} - c_{0\uparrow}^\dagger c_{1\downarrow} \hat{n}_{1\uparrow} \right) \right] \quad (14.33)$$

The full effective Hamiltonian then becomes

$$\begin{aligned} H_{\text{eff}}^{3,\frac{1}{2}} &= -\frac{t}{2} \left[P_d^\uparrow \left\{ \left(c_{1\uparrow}^\dagger c_{0\uparrow} + c_{0\downarrow}^\dagger c_{1\downarrow} \right) \left(P_0^{(2)} P_1^{(0)} + P_0^{(0)} P_1^{(2)} \right) + \left(c_{0\uparrow}^\dagger c_{1\uparrow} + c_{1\downarrow}^\dagger c_{0\downarrow} \right) P_0^{(1)} P_1^{(1)} \right\} \right. \\ &\quad \left. + S_d^- \left(c_{1\uparrow}^\dagger c_{0\downarrow} - c_{0\uparrow}^\dagger c_{1\downarrow} \right) \left(P_0^{(2)} P_1^{(0)} + P_0^{(0)} P_1^{(2)} \right) + S_d^+ \left(c_{0\downarrow}^\dagger c_{1\uparrow} - c_{1\downarrow}^\dagger c_{0\uparrow} \right) P_0^{(1)} P_1^{(1)} \right] \\ &= t \left[P_d^\uparrow \sqrt{2} \mathcal{A}_{\frac{1}{2}}^z - \frac{1}{2} S_d^+ \mathcal{B}_{\frac{1}{2}}^- - \frac{1}{2} S_d^- \mathcal{B}_{\frac{1}{2}}^+ \right], \end{aligned} \quad (14.34)$$

where we have defined

$$\mathcal{A}_{\frac{1}{2}}^z = -\frac{1}{2\sqrt{2}} \left[\left(c_{1\uparrow}^\dagger c_{0\uparrow} + c_{0\downarrow}^\dagger c_{1\downarrow} \right) \left(P_0^{(2)} P_1^{(0)} + P_0^{(0)} P_1^{(2)} \right) + \left(c_{0\uparrow}^\dagger c_{1\uparrow} + c_{1\downarrow}^\dagger c_{0\downarrow} \right) P_0^{(1)} P_1^{(1)} \right], \quad (14.35)$$

and

$$\mathcal{B}_{\frac{1}{2}}^+ = \left(c_{1\uparrow}^\dagger c_{0\downarrow} - c_{0\uparrow}^\dagger c_{1\downarrow} \right) \left(P_0^{(2)} P_1^{(0)} + P_0^{(0)} P_1^{(2)} \right); \quad \mathcal{B}_{\frac{1}{2}}^- = \left(c_{0\downarrow}^\dagger c_{1\uparrow} - c_{1\downarrow}^\dagger c_{0\uparrow} \right) P_0^{(1)} P_1^{(1)} \quad (14.36)$$

The effective Hamiltonian for the $S_{\text{tot}}^z = -1/2$ sector is obtained by applying the transformations $P_d^\sigma \rightarrow P_d^{\bar{\sigma}}, c_{i\sigma} \rightarrow c_{i\bar{\sigma}}$ on the Hamiltonian. This is motivated by the fact that applying the same transformations on the $S_{\text{tot}}^z = 1/2$ eigenstates give the $S_{\text{tot}}^z = -1/2$ eigenstates.

$$H_{\text{eff}}^{3,-\frac{1}{2}} = t \left[P_d^\uparrow \sqrt{2} \mathcal{A}_{-\frac{1}{2}}^z - \frac{1}{2} S_d^+ \mathcal{B}_{-\frac{1}{2}}^- - \frac{1}{2} S_d^- \mathcal{B}_{-\frac{1}{2}}^+ \right]. \quad (14.37)$$

The operators $\mathcal{A}_{-\frac{1}{2}}^z$ and $\mathcal{B}_{-\frac{1}{2}}^\pm$ are obtained by applying the above mentioned transformation on their $S_{\text{tot}}^z = +1/2$ counterparts.

2.15 Appendix: Non-Fermi liquid exponents in self-energy and two-particle correlations

The one-particle self-energies relating to the impurity and zeroth sites, as well as some relevant two-particle correlation functions, can be shown to follow power-law behaviour close to the Brinkman-Rice transition at r_{c2} , by mapping the impurity model to a classical Coulomb gas, similar to the work of Anderson, Yuval and Hamman for the Kondo model [465] and that of Si and Kotliar for a periodic Anderson model [460]. Indeed, the latter work uses an auxiliary that is similar to our extended Anderson impurity model. In the following, we adapt their results to our model. These non-universal power-laws are signatures of the non-Fermi liquid excitations that emerge at r_{c2} .

Near r_{c2} , extended SIAM can be reduced to a $J-U_b$ model (shown in subsection V.A of the main manuscript). The conduction electrons scattering off the impurity suffer phase shifts because of the presence of the potentials J and U_b . These phase shifts are given by $\delta_{\text{sp}} = \arctan(\pi\rho_0 J/2)$ and $\delta_{\text{ch}} = \arctan(\pi\rho_0 U_b/2)$ [460, 534], where ρ_0 is the electronic density of states in the conduction bath. The low-energy behaviour of the local impurity and bath self-energies can be expressed in terms of these phase shifts [460]:

$$\begin{aligned} \Sigma_{dd}(\omega) &\sim \omega^{\gamma_{dd}}, \quad \Sigma_{d0}(\omega) \sim \omega^{\gamma_{d0}}, \quad \Sigma_{00}(\omega) \sim \omega^{\gamma_{00}}, \\ \gamma_{dd} &= \frac{5}{4} - \left(\frac{\delta_{\text{tot}}^*}{\pi}\right)^2 - \left(\frac{\delta_{\text{ch}}^*}{\pi}\right)^2, \quad \gamma_{d0} = 2 \left[\left(\frac{\delta_{\text{ch}}^*}{\pi}\right)^2 + \left(\frac{\delta_{\text{tot}}^*}{\pi}\right)^2 - \frac{1}{4} \right], \\ \gamma_{00} &= \left(\frac{\delta_{\text{tot}}^*}{\pi} - \frac{1}{2}\right)^2 + \left(\frac{\delta_{\text{ch}}^*}{\pi}\right)^2 - 2, \end{aligned} \quad (15.1)$$

where we have defined a total phase shift parameter $\frac{\delta_{\text{tot}}}{\pi} \equiv \frac{\delta_{\text{sp}} + \delta_{\text{ch}}}{\pi} - \frac{1}{2}$. This can be extended to certain correlation functions as well. The impurity spin-flip correlation, the impurity-bath excitonic correlation and the zeroth-site pairing correlation [460]:

$$\begin{aligned} \langle S_d^+(\tau) S_d^-(0) \rangle &\sim \tau^{-\alpha_1} \implies \langle S_d^+ \rangle(\omega) \sim \omega^{(\alpha_1-1)/2}, \\ \langle (c_{d\uparrow}^\dagger c_{0\uparrow})(\tau) (c_{0\uparrow}^\dagger c_{d\uparrow})(0) \rangle &\sim \tau^{-\alpha_2} \implies \langle c_{d\uparrow}^\dagger c_{0\uparrow} c_{0\uparrow}^\dagger c_{d\uparrow} \rangle(\omega) \sim \omega^{(\alpha_2-1)/2}, \\ \langle (c_{0\uparrow}^\dagger c_{0\downarrow}^\dagger)(\tau) (c_{0\downarrow} c_{0\uparrow})(0) \rangle &\sim \tau^{-\alpha_3} \implies \langle c_{0\uparrow}^\dagger c_{0\downarrow}^\dagger \rangle(\omega) \sim \omega^{(\alpha_3-1)/2}, \end{aligned} \quad (15.2)$$

where the exponents are given by

$$\alpha_1 = 2 \left(1 - \frac{\delta_{\text{sp}}^*}{\pi}\right)^2, \quad \alpha_2 = \left(\frac{\delta_{\text{tot}}^*}{\pi} - \frac{1}{2}\right)^2 + \left(\frac{\delta_{\text{ch}}^*}{\pi}\right)^2, \quad \alpha_3 = \left(\frac{\delta_{\text{tot}}^*}{\pi} + \frac{1}{2}\right)^2 + \left(1 + \frac{\delta_{\text{ch}}^*}{\pi}\right)^2. \quad (15.3)$$

2.16 Appendix: Entanglement entropy, magnetisation and phase shift, at the QCP

As mentioned in the previous section, the polarised ground-states $|- \rangle^{3, \pm \frac{1}{2}}$ of eq. (14.7) are special because they involve the strong-coupling ground-state as well as the local moment ground-state.

The overall ground states, therefore, contain both entangled and non-entangled impurity contributions, leading to a departure from maximal entanglement. We will now calculate the impurity entanglement entropy $S_{\text{EE}}(d)$ in this polarised ground-state(s) that emerges exactly at the QCP, and show that $S_{\text{EE}}(d)$ has a fractional value (in units of $\log 2$). This turns out to be linked to the fact that the impurity is screened partially in these states, leading to an impurity magnetisation that is non-vanishing but only half of that in the local moment phase.

The fractional entropy arises because of the singular scattering of the gapless excitations at the Fermi surface leading to the destruction of the local Fermi liquid. In order to access these processes, we will work with a *semi-continuous* conduction bath beyond the zeroth site (total number of lattice sites $N \rightarrow \infty$): $H_{\text{bath}} = \sum_{k,\sigma} \epsilon(k) \psi_{\sigma}^{\dagger}(k) \psi_{\sigma}(k)$. This conduction bath that is composed of the k -states formed from the Hilbert space of the lattice sites beyond the zeroth site will be referred to as the reduced conduction bath. This is in addition to the Hamiltonian for the impurity and zeroth sites: $H_{d0} = J \vec{S}_d \cdot \vec{S}_0 - \frac{U_b}{2} (c_{0\uparrow}^{\dagger} c_{0\uparrow} - c_{0\downarrow}^{\dagger} c_{0\downarrow})^2$, where $\vec{S}_d$ is the spin operator for the impurity site, and $c_{0\sigma}^{\dagger}$ creates an electron of spin σ at the zeroth site. The full Hamiltonian also involves a single-particle hopping term that couples these two Hamiltonians: $H = H_{d0} + H_{\text{bath}} + H_{\text{int}}$, where

$$H_{\text{int}} = -t \sum_{\sigma} (c_{0\sigma}^{\dagger} c_{1\sigma} + \text{h.c.}) = -\frac{t}{N} \sum_{k,\sigma} [c_{0\sigma}^{\dagger} \psi_{\sigma}(k) + \text{h.c.}] \quad (16.1)$$

$c_{1\sigma}^{\dagger} = \frac{1}{N} \sum_k \psi_{\sigma}^{\dagger}(k)$ is the creation operator for the first site of the conduction bath of N sites (but effectively the zeroth site for the reduced conduction bath defined by H_{bath}). We will treat H_{int} perturbatively in t/J , in order to obtain the set of renormalised g -fold degenerate ground-states $\{|\psi_{\text{gs}}^{(n)}\rangle; n = 1, \dots, g\}$.

As mentioned before, the intention here is to calculate magnetisation and entanglement entropy in the non-Fermi liquid ground states. In the case of a degenerate subspace with $S_{\text{tot}}^z = \pm 1/2$ where $S_{\text{tot}}^z = S_d^z + S_0^z + S_1^z$ is the total spin operator, the magnetisation m_d^z must be calculated in any *one* of the degenerate states for our problem. This is accomplished by solving the problem in the presence of an infinitesimal magnetic field term $-h S_{\text{tot}}^z$ that lifts the degeneracy and chooses one of the symmetry-broken ground-states. This is similar to the "two self-energy description" introduced by David Logan for computing quantities in the local moment phase/Mott insulating phase in terms of different self-energies obtained by inserting positive and negative magnetic fields [161, 535, 536]. Due to the symmetry of the Hamiltonian under $S_{\text{tot}}^z \rightarrow -S_{\text{tot}}^z$, both the limits $h = 0^+$ and $h = 0^-$ give the same results for the density matrix and the entanglement entropy. We will work with $h = 0^+$. Since the total Hamiltonian H and the perturbation H_{int} preserve the total spin $S_{\text{tot}}^z = S_d^z + S_0^z + \int dk S_k^z$, it is sufficient to solve only for the sector with the most positive value of S_{tot}^z , since those are the states that will be selected by the positive magnetic field $h = 0^+$.

In the absence of H_{int} , the set of ground-states with a positive S_{tot}^z will contain, to begin with, the local moment states $|\uparrow\rangle_d |0\rangle_0 |\phi\rangle$ and $|\uparrow\rangle_d |2\rangle_0 |\phi\rangle$, where $|\phi\rangle$ represents the filled Fermi sea of the reduced conduction bath, acting as its spinless ground-state. These states have $S_{\text{tot}}^z = 1/2$. We can label these states as $|n_{\text{tot}}, \mu_d^z\rangle$, using the total number of particles $n_{\text{tot}} = n_d + n_0 + n_{\text{bath}}$ (which is also conserved, and where we set $n_{\text{bath}} = 1$ for the set $|\phi\rangle$), and the average magnetisation μ_d^z of the impurity state: $|2, \frac{1}{2}\rangle \equiv |\uparrow\rangle_d |0\rangle_0 |\phi\rangle$, $|4, \frac{1}{2}\rangle \equiv |\uparrow\rangle_d |2\rangle_0 |\phi\rangle$. Since the singlet state $|SS\rangle_{d0}$ (on the

d and zeroth sites) is degenerate with the local moment states $|\uparrow\rangle_d |0(2)\rangle_0$, one can construct other states with $S_{\text{tot}}^z = 1/2$ by creating magnetic excitations at the Fermi surface. The fact that we are working with a thermodynamically large bath ($N \rightarrow \infty$) means that there will be gapless excitations vanishingly close to the Fermi surface. Such states are of the form $|\text{SS}\rangle_{d0} |e_\uparrow\rangle$ (labeled as $|4, 0\rangle$ in the $|n_{\text{tot}}, \mu_d^z\rangle$ notation) and $|\text{SS}\rangle_{d0} |h_\downarrow\rangle$ (labeled as $|2, 0\rangle$), where $|e_\sigma\rangle$ and $|h_\downarrow\rangle$ are electron and hole states respectively:

$$|e_\sigma\rangle = \sum_k^{\text{FS}} \psi_\sigma^\dagger(k) |\phi\rangle, \quad |h_\sigma\rangle = \sum_k^{\text{FS}} \psi_\sigma(k) |\phi\rangle. \quad (16.2)$$

The sum is over k -states on the Fermi surface. The singlet state was set to have zero spin on the impurity site in the state notation, because $\langle S_d^z \rangle$ vanishes in the singlet state. The full set of relevant degenerate ground-states, classified by the total number of particles, is

$$n_{\text{tot}} = 4 : |4, 0\rangle, |4, \frac{1}{2}\rangle; \quad n_{\text{tot}} = 2 : |2, 0\rangle, |2, \frac{1}{2}\rangle. \quad (16.3)$$

Within each 2×2 block, the interaction matrix H_{int} takes the form (in the basis of the states in eq. (16.3))

$$(H_{\text{int}})_{\text{gs}}^{n_{\text{tot}}=2} = \frac{t}{N} \times \begin{pmatrix} 0 & 1/\sqrt{2} \\ 1/\sqrt{2} & 0 \end{pmatrix} = - (H_{\text{int}})_{\text{gs}}^{n_{\text{tot}}=4}, \quad (16.4)$$

and the $S_{\text{tot}}^z = +\frac{1}{2}$ ground-states are then easily obtained to be of the form $|\psi_{\text{gs}}\rangle^{(n_{\text{tot}})} = \frac{1}{\sqrt{2}} \left(|n_{\text{tot}}, 0\rangle - |n_{\text{tot}}, \frac{1}{2}\rangle \right)$, $n_{\text{tot}} = 2$ & 4 , independent of N . Both ground-states involve an equal superposition of a $\mu_d^z = 1/2$ state and a $\mu_d^z = 0$ state, such that the net impurity magnetisation in either of the renormalised polarised states is

$$m_d^z = \frac{1}{2} \left(0 + \frac{1}{2} \right) = \frac{1}{4}. \quad (16.5)$$

The reduced density matrix $\rho_{\text{imp}} = \text{Tr}' \left(|\psi_{\text{gs}}\rangle^{(n_{\text{tot}})} \langle \psi_{\text{gs}}|^{(n_{\text{tot}})} \right)$ (where Tr' indicates partial trace over all states apart from those of the impurity) for the impurity spin in the polarised states can now be immediately obtained. It is independent of n_{tot} , and actually turns out to be halfway between the maximally mixed density matrix $\rho_{\text{MM}} = \frac{1}{2}I$ and the non-entangled density matrix $\rho_{\text{NE}} = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}$:

$$\rho_{\text{imp}} = \begin{pmatrix} 3/4 & 0 \\ 0 & 1/4 \end{pmatrix} = \frac{1}{2} (\rho_{\text{MM}} + \rho_{\text{NE}}), \quad S_{\text{EE}}(d) = -\text{Tr} (\rho_{\text{imp}} \ln \rho_{\text{imp}}) \simeq 0.81 \log 2. \quad (16.6)$$

This fractional value of the entropy in units of $\log 2$ is another signature of the non-Fermi liquid. We can express this in terms of an effective degeneracy $\tilde{g}$ of the impurity degree of freedom. We define

this effective degeneracy as $\tilde{g} = 1 + 2|m_d^z|$, in order to recover the unique ground-state in the local Fermi liquid phase (that has $m_d^z = 0, \tilde{g} = 1$) and the magnetic doublet in the local moment phase (that has $m_d^z = 1/2, \tilde{g} = 2$). The proposed form also results in a non-integer impurity degeneracy for the NFL at the QCP, $\tilde{g}^* = 3/2$. Using the effective degeneracy $\tilde{g}$, one can also write down a general ground-state

$$|\psi_{\text{gs}}(g)\rangle = \frac{1}{\sqrt{(2-\tilde{g})^2 + (\tilde{g}-1)^2}} [(2-\tilde{g}) |n_{\text{tot}}, 0\rangle - (\tilde{g}-1) |n_{\text{tot}}, 1/2\rangle] , \quad (16.7)$$

that interpolates between the strong-coupling ground-state for $r < r_{c2}$ (therefore $\tilde{g} = 1$) and the weak coupling ground-state $r > r_{c2}$ (therefore $\tilde{g} = 2$), also describing the QCP in between ($\tilde{g} = 3/2$). The entanglement entropy, when written in terms of $\tilde{g}$, takes the form

$$\rho_{\text{imp}} = \frac{1}{2} \begin{pmatrix} 2-\tilde{g} & 0 \\ 0 & \tilde{g} \end{pmatrix} = \frac{1}{2} + \begin{pmatrix} m_d^z & 0 \\ 0 & -m_d^z \end{pmatrix}, \quad (16.8)$$

$$S_{\text{EE}}(d) = - \left(1 - \frac{\tilde{g}}{2}\right) \ln \left(1 - \frac{\tilde{g}}{2}\right) - \frac{\tilde{g}}{2} \ln \frac{\tilde{g}}{2}. \quad (16.9)$$

The relation between the reduced density matrix ρ_{imp} and the impurity magnetisation m_d^z is consistent with that obtained by Kopp et al., [537].

The general form $|\psi_{\text{gs}}(g)\rangle$ of polarised interpolating ground-state also allows us to calculate the excess charge (n_{exc}) added, by the impurity, to the conduction bath Fermi surface (through gapless excitations). This then allows the scattering phase shift of the conduction electrons (obtained via the Friedel sum rule, [48, 517, 518]) to be linked with the impurity magnetisation m_d^z . From the definitions of the states $|n_{\text{tot}}, 0\rangle$ and $|n_{\text{tot}}, 1/2\rangle$, we note that the former state involves an impurity-bath singlet while the latter involves a decoupled local moment. The singlet state contributes an excess charge of unity to the conduction bath: $n_{\text{exc}}^{\text{SS}} = 1$, because of the impurity-bath entanglement within it. The local state moment does not contribute any excess charge, because the impurity spin is no longer hybridising with the bath in that state: $n_{\text{exc}}^{\text{LM}} = 0$. Putting these together, the net excess charge contributed by the state in eq. (16.7) is

$$n_{\text{exc}} = \frac{1}{(2-\tilde{g})^2 + (\tilde{g}-1)^2} [(2-\tilde{g})^2 n_{\text{exc}}^{\text{SS}} + (\tilde{g}-1)^2 n_{\text{exc}}^{\text{LM}}] = \left[1 + \left(\frac{2-\tilde{g}}{\tilde{g}-1} \right)^2 \right]^{-1}. \quad (16.10)$$

At the QCP, the excess charge acquires a fractional value of 1/2, indicating once more that only half of the impurity remains coupled with the bath at the transition. The Friedel sum rule can be used to equate this excess charge with the phase shift δ^* suffered by conduction electrons as they scatter off the impurity, at the fixed point:

$$\delta^* = \pi n_{\text{exc}} = \left[1 + \left(\frac{2-\tilde{g}}{\tilde{g}-1} \right)^2 \right]^{-1} = \frac{4m_d^{z2}}{1 - 4|m_d^z| + 8m_d^{z2}}. \quad (16.11)$$

Equation (16.11) unifies the effect of several important quantities that lead to the frustration of the impurity and the destruction of the Kondo effect. The presence of U_b introduces local moment states

into the spectrum and leads to states with non-vanishing impurity magnetisation m_d^z at the QCP; this non-zero m_d^z can be interpreted as an effective impurity degeneracy $\tilde{g}$ that is only partially screened, resulting in only half the excess charge n_{exc} being contributed to the conduction bath. This reduction in the excess charge acts as a shift in the boundary conditions felt by the conduction electrons and manifests as a phase shift that is less than the unitarity limit of π .

2.17 Appendix: Additional correlations near the ESQPT and the QCP

We have included some additional correlations for the extended SIAM here to highlight the features that are mentioned in the main manuscript. Left panel of Fig. (2.15) shows the impurity and zeroth site double occupancy. Both vanish at r_{c1} , indicating the emergence of the Kondo model. Very close to r_{c2} , the zeroth site doublon occupancy picks up (see inset), showing the destruction of the Kondo cloud. Right panel of Fig. (2.15) shows various measures of entanglement close to r_{c2} . The impurity-bath mutual information (blue) decreases as the transition is approached, showing the breakdown of impurity screening. The impurity and first site MI (violet), the zeroth site and first site MI (red) and the impurity, zeroth site and first site tripartite information (grey) increase, showing the emergence of long-ranged correlations near the transition. Fig. (2.16) shows the emergence of long-ranged intra-bath mutual information near r_{c1} and r_{c2} , signalling critical behaviour near the ESQPT and QCP respectively. Intra-bath CDW correlations and impurity-bath Ising correlations in Fig. (2.17) also show the previously-mentioned emergence of long-ranged correlations.

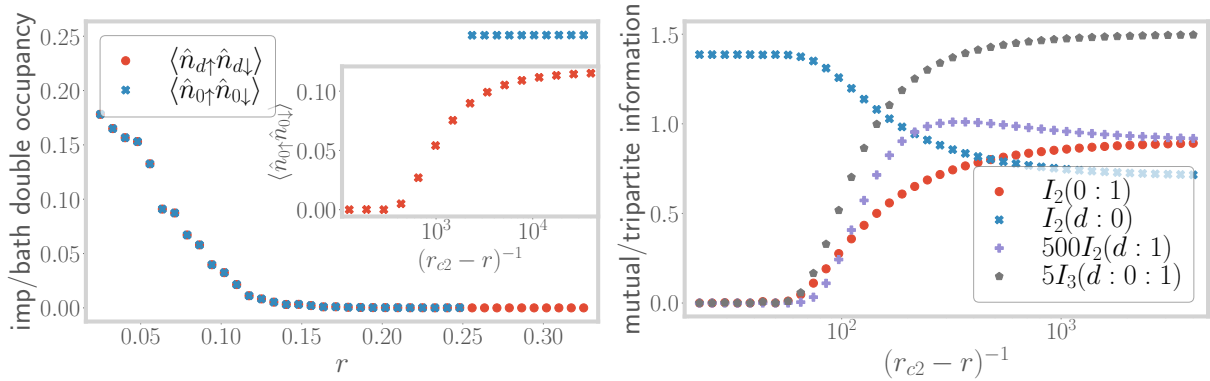


Figure 2.15: Left: Evolution of the double occupancy of impurity (red) and zeroth sites (blue) ground-state double occupancy, with r . Right: Evolution of various quantum information-theoretic measures, very close to the transition.

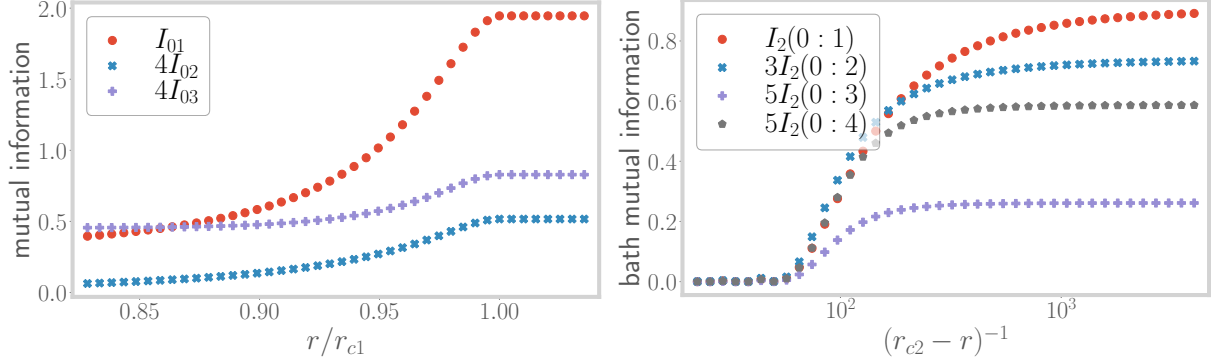


Figure 2.16: Variation of intra-bath mutual information between zeroth site and sites 1, 2 and 3, near r_{c1} (left) and r_{c2} (right).

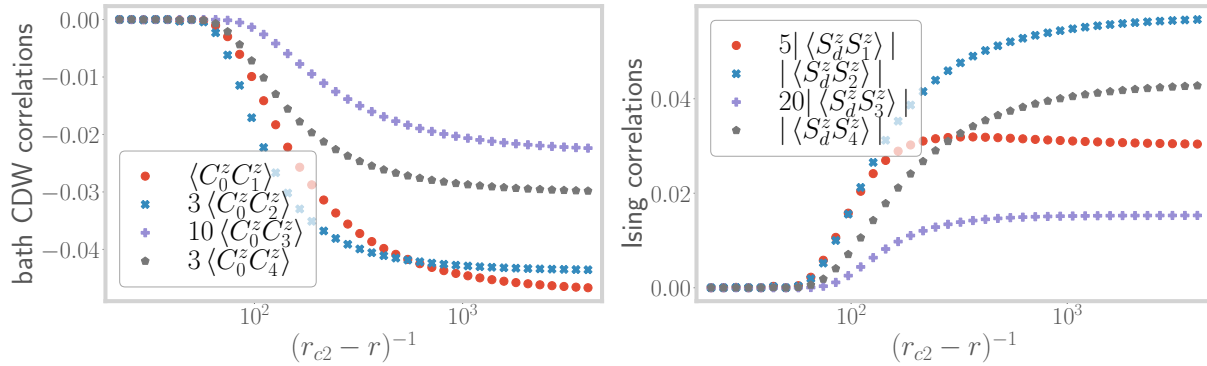


Figure 2.17: Left: Variation of charge isospin Ising correlations between the zeroth site and bath sites further down the chain. Right: Ising correlations between the impurity site and bath sites beyond the zeroth site. Both sets of correlations show an overall increase as $r \rightarrow r_{c2}$.

Chapter 3

Summary, Conclusions and Future Work

In the final chapter, we will summarise the results obtained from the research carried out in the thesis, clarify the major conclusions derived from them, and point out some avenues for further work. In a nutshell, we have applied a two-pronged approach to metal-insulator transitions through this body of work. The first approach tackles the effects of strong local correlation through the development of a new method that maps interacting fermionic models to quantum impurity models. The other complimentary approach involves studying the effects of metal-insulator transition on entanglement renormalisation, and characterising phases through the emergent geometry resulting from such flows.

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